

Import Libraries

```
In [1]: # Data manipulation
import pandas as pd
import numpy as np

# Data visualization
import matplotlib.pyplot as plt
import seaborn as sns

# Statistical analysis
from scipy import stats

# Machine learning preprocessing
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.model_selection import train_test_split

# Ignore warnings
import warnings
warnings.filterwarnings("ignore")

# Display settings
pd.set_option("display.max_columns", None)
pd.set_option("display.max_rows", 100)
pd.set_option("display.float_format", "{:.2f}".format)

# Plot settings
plt.style.use("ggplot")
sns.set_theme(style="whitegrid")

print("Libraries imported successfully!")
```

Libraries imported successfully!

Loading Dataset

```
In [2]: home_team = pd.read_csv("train_home_team_statistics_df.csv")
home_team.head()
```

```
Out[2]:
```

	ID	LEAGUE	TEAM_NAME	TEAM_SHOTS_TOTAL_season_sum	TEAM_SHOTS_INSID
0	0	Ligue 1	Toulouse	3.00	
1	1	Ligue 2	Brest	6.00	
2	2	Serie A	Sampdoria	4.00	
3	3	League One	Coventry City	7.00	
4	4	Premier League	Wolverhampton Wanderers	3.00	

```
In [3]: away_team = pd.read_csv("train_away_team_statistics_df.csv")
away_team.head()
```

```
Out[3]:
```

	ID	LEAGUE	TEAM_NAME	TEAM_SHOTS_TOTAL_season_sum	TEAM_SHOTS_INSIDEB
--	----	--------	-----------	-----------------------------	--------------------

0	0	Ligue 1	Olympique Marseille	4.00	
---	---	---------	------------------------	------	--

1	1	Ligue 2	Red Star	4.00	
---	---	---------	----------	------	--

2	2	Serie A	Bologna	4.00	
---	---	---------	---------	------	--

3	3	League One	Bolton Wanderers	0.00	
---	---	---------------	---------------------	------	--

4	4	Premier League	Leicester City	5.00	
---	---	-------------------	-------------------	------	--

```
In [4]: home_player = pd.read_csv("train_home_player_statistics_df.csv")
home_player.head()
```

```
Out[4]:
```

	ID	LEAGUE	TEAM_NAME	POSITION	PLAYER_NAME	PLAYER_ACCURATE_CROSSES
--	----	--------	-----------	----------	-------------	-------------------------

0	0	Ligue 1	Toulouse	defender	Agustín Rogel	
---	---	---------	----------	----------	---------------	--

1	0	Ligue 1	Toulouse	defender	Mathieu Goncalves	
---	---	---------	----------	----------	----------------------	--

2	0	Ligue 1	Toulouse	goalkeeper	Baptiste Reynet	
---	---	---------	----------	------------	--------------------	--

3	0	Ligue 1	Toulouse	midfielder	Jean-Victor Makengo	
---	---	---------	----------	------------	------------------------	--

4	0	Ligue 1	Toulouse	NaN	Efthymios Koulouris	
---	---	---------	----------	-----	------------------------	--

```
In [5]: away_player = pd.read_csv("train_away_player_statistics_df.csv")
away_player.head()
```

```
Out[5]:
```

	ID	LEAGUE	TEAM_NAME	POSITION	PLAYER_NAME	PLAYER_ACCURATE_CROSSES
--	----	--------	-----------	----------	-------------	-------------------------

0	0	Ligue 1	Olympique Marseille	defender	Lucas Perrin	
---	---	---------	------------------------	----------	--------------	--

1	0	Ligue 1	Olympique Marseille	midfielder	Kevin Strootman	
---	---	---------	------------------------	------------	--------------------	--

2	0	Ligue 1	Olympique Marseille	goalkeeper	Yohann Pelé	
---	---	---------	------------------------	------------	-------------	--

3	0	Ligue 1	Olympique Marseille	defender	Abdallah Ali Mohamed	
---	---	---------	------------------------	----------	-------------------------	--

4	0	Ligue 1	Olympique Marseille	NaN	Nemanja Radonjic	
---	---	---------	------------------------	-----	---------------------	--

```
In [6]: y_train = pd.read_csv("Y_train_lrknArQ.csv")
y_train.head()
```

```
Out[6]:
```

	ID	HOME_WINS	DRAW	AWAY_WINS
0	0	0	0	1
1	1	0	1	0
2	2	0	0	1
3	3	1	0	0
4	4	0	1	0

```
In [7]: y_train_supp = pd.read_csv("Y_train_supp.csv")
y_train_supp.head()
```

```
Out[7]:
```

	ID	GOAL_DIFF_HOME_AWAY
0	0	-2.00
1	1	0.00
2	2	-1.00
3	3	1.00
4	4	0.00

```
In [8]: home_team["LEAGUE"].nunique()
```

```
Out[8]: 13
```

```
In [9]: home_team["TEAM_NAME"].nunique()
```

```
Out[9]: 350
```

```
In [10]: # Understanding the Data
def dataset_summary(df, name):
    print("=" * 80)
    print(f"{name}")
    print("=" * 80)

    print(f"Rows      : {df.shape[0]:,}")
    print(f"Columns   : {df.shape[1]:,}")

    print("\nFirst 5 rows")
    display(df.head())

    print("\nData Types")
    display(df.dtypes.value_counts())

    print("\nMemory Usage")

    memory = df.memory_usage(deep=True).sum() / (1024**2)

    print(f"{memory:.2f} MB")
```

```
In [11]: dataset_summary(home_team, "HOME TEAM")

dataset_summary(away_team, "AWAY TEAM")

dataset_summary(home_player, "HOME PLAYER")

dataset_summary(away_player, "AWAY PLAYER")

dataset_summary(y_train, "Y TRAIN")

dataset_summary(y_train_supp, "Y TRAIN SUPPLEMENT")
```

```
=====
=====
HOME TEAM
=====
=====

Rows      : 12,303
Columns   : 143
```

First 5 rows

	ID	LEAGUE	TEAM_NAME	TEAM_SHOTS_TOTAL_season_sum	TEAM_SHOTS_INSIDE
0	0	Ligue 1	Toulouse	3.00	
1	1	Ligue 2	Brest	6.00	
2	2	Serie A	Sampdoria	4.00	
3	3	League One	Coventry City	7.00	
4	4	Premier League	Wolverhampton Wanderers	3.00	

Data Types

```
float64    140
object      2
int64       1
Name: count, dtype: int64
Memory Usage
14.65 MB
```

AWAY TEAM

```
Rows      : 12,303
Columns   : 143
```

First 5 rows

	ID	LEAGUE	TEAM_NAME	TEAM_SHOTS_TOTAL_season_sum	TEAM_SHOTS_INSIDEBO
0	0	Ligue 1	Olympique Marseille	4.00	
1	1	Ligue 2	Red Star	4.00	
2	2	Serie A	Bologna	4.00	
3	3	League One	Bolton Wanderers	0.00	
4	4	Premier League	Leicester City	5.00	

```
Data Types
float64    140
object      2
int64       1
Name: count, dtype: int64
Memory Usage
14.64 MB
```

HOME PLAYER

```
Rows      : 237,079
Columns   : 307
```

First 5 rows

	ID	LEAGUE	TEAM_NAME	POSITION	PLAYER_NAME	PLAYER_ACCURATE_CROSSES_s
0	0	Ligue 1	Toulouse	defender	Agustín Rogel	
1	0	Ligue 1	Toulouse	defender	Mathieu Goncalves	
2	0	Ligue 1	Toulouse	goalkeeper	Baptiste Reynet	
3	0	Ligue 1	Toulouse	midfielder	Jean-Victor Makengo	
4	0	Ligue 1	Toulouse	NaN	Efthymios Koulouris	

Data Types

float64 302

object 4

int64 1

Name: count, dtype: int64

Memory Usage

601.93 MB

=====

=====

AWAY PLAYER

=====

=====

Rows : 236,132

Columns : 307

First 5 rows

	ID	LEAGUE	TEAM_NAME	POSITION	PLAYER_NAME	PLAYER_ACCURATE_CROSSES_s
0	0	Ligue 1	Olympique Marseille	defender	Lucas Perrin	
1	0	Ligue 1	Olympique Marseille	midfielder	Kevin Strootman	
2	0	Ligue 1	Olympique Marseille	goalkeeper	Yohann Pelé	
3	0	Ligue 1	Olympique Marseille	defender	Abdallah Ali Mohamed	
4	0	Ligue 1	Olympique Marseille	NaN	Nemanja Radonjic	

Data Types

float64 302

object 4

int64 1

Name: count, dtype: int64

Memory Usage
599.54 MB

=====

Y TRAIN

=====

=====

Rows : 12,303
Columns : 4

First 5 rows

	ID	HOME_WINS	DRAW	AWAY_WINS
0	0	0	0	1
1	1	0	1	0
2	2	0	0	1
3	3	1	0	0
4	4	0	1	0

Data Types
int64 4
Name: count, dtype: int64
Memory Usage
0.38 MB

=====

Y TRAIN SUPPLEMENT

=====

=====

Rows : 12,303
Columns : 2

First 5 rows

	ID	GOAL_DIFF_HOME_AWAY
0	0	-2.00
1	1	0.00
2	2	-1.00
3	3	1.00
4	4	0.00

Data Types
int64 1
float64 1
Name: count, dtype: int64
Memory Usage
0.19 MB

```
In [12]: # Column Inspection
for col in home_team.columns:
    print(col)
```

ID
LEAGUE
TEAM_NAME
TEAM_SHOTS_TOTAL_season_sum
TEAM_SHOTS_INSIDEBOX_season_sum
TEAM_SHOTS_OFF_TARGET_season_sum
TEAM_SHOTS_ON_TARGET_season_sum
TEAM_SHOTS_OUTSIDEBOX_season_sum
TEAM_PASSES_season_sum
TEAM_SUCCESSFUL_PASSES_season_sum
TEAM_SAVES_season_sum
TEAM_CORNERS_season_sum
TEAM_FOULS_season_sum
TEAM_YELLOWCARDS_season_sum
TEAM_REDCARDS_season_sum
TEAM_OFFSIDES_season_sum
TEAM_ATTACKS_season_sum
TEAM_PENALTIES_season_sum
TEAM_SUBSTITUTIONS_season_sum
TEAM_BALL_SAFE_season_sum
TEAM_DANGEROUS_ATTACKS_season_sum
TEAM_INJURIES_season_sum
TEAM_GOALS_season_sum
TEAM_GAME_WON_season_sum
TEAM_GAME_DRAW_season_sum
TEAM_GAME_LOST_season_sum
TEAM_SHOTS_TOTAL_season_average
TEAM_SHOTS_INSIDEBOX_season_average
TEAM_SHOTS_OFF_TARGET_season_average
TEAM_SHOTS_ON_TARGET_season_average
TEAM_SHOTS_OUTSIDEBOX_season_average
TEAM_PASSES_season_average
TEAM_SUCCESSFUL_PASSES_season_average
TEAM_SUCCESSFUL_PASSES_PERCENTAGE_season_average
TEAM_SAVES_season_average
TEAM_CORNERS_season_average
TEAM_BALL_POSSESSION_season_average
TEAM_FOULS_season_average
TEAM_YELLOWCARDS_season_average
TEAM_REDCARDS_season_average
TEAM_OFFSIDES_season_average
TEAM_ATTACKS_season_average
TEAM_PENALTIES_season_average
TEAM_SUBSTITUTIONS_season_average
TEAM_BALL_SAFE_season_average
TEAM_DANGEROUS_ATTACKS_season_average
TEAM_INJURIES_season_average
TEAM_GOALS_season_average
TEAM_GAME_WON_season_average
TEAM_GAME_DRAW_season_average
TEAM_GAME_LOST_season_average
TEAM_SHOTS_TOTAL_season_std
TEAM_SHOTS_INSIDEBOX_season_std
TEAM_SHOTS_OFF_TARGET_season_std
TEAM_SHOTS_ON_TARGET_season_std
TEAM_SHOTS_OUTSIDEBOX_season_std
TEAM_PASSES_season_std
TEAM_SUCCESSFUL_PASSES_season_std
TEAM_SUCCESSFUL_PASSES_PERCENTAGE_season_std
TEAM_SAVES_season_std

TEAM_CORNERS_season_std
TEAM_BALL_POSSESSION_season_std
TEAM_FOULS_season_std
TEAM_YELLOWCARDS_season_std
TEAM_REDCARDS_season_std
TEAM_OFFSIDES_season_std
TEAM_ATTACKS_season_std
TEAM_PENALTIES_season_std
TEAM_SUBSTITUTIONS_season_std
TEAM_BALL_SAFE_season_std
TEAM_DANGEROUS_ATTACKS_season_std
TEAM_INJURIES_season_std
TEAM_GOALS_season_std
TEAM_SHOTS_TOTAL_5_last_match_sum
TEAM_SHOTS_INSIDEBOX_5_last_match_sum
TEAM_SHOTS_OFF_TARGET_5_last_match_sum
TEAM_SHOTS_ON_TARGET_5_last_match_sum
TEAM_SHOTS_OUTSIDEBOX_5_last_match_sum
TEAM_PASSES_5_last_match_sum
TEAM_SUCCESSFUL_PASSES_5_last_match_sum
TEAM_SAVES_5_last_match_sum
TEAM_CORNERS_5_last_match_sum
TEAM_FOULS_5_last_match_sum
TEAM_YELLOWCARDS_5_last_match_sum
TEAM_REDCARDS_5_last_match_sum
TEAM_OFFSIDES_5_last_match_sum
TEAM_ATTACKS_5_last_match_sum
TEAM_PENALTIES_5_last_match_sum
TEAM_SUBSTITUTIONS_5_last_match_sum
TEAM_BALL_SAFE_5_last_match_sum
TEAM_DANGEROUS_ATTACKS_5_last_match_sum
TEAM_INJURIES_5_last_match_sum
TEAM_GOALS_5_last_match_sum
TEAM_GAME_WON_5_last_match_sum
TEAM_GAME_DRAW_5_last_match_sum
TEAM_GAME_LOST_5_last_match_sum
TEAM_SHOTS_TOTAL_5_last_match_average
TEAM_SHOTS_INSIDEBOX_5_last_match_average
TEAM_SHOTS_OFF_TARGET_5_last_match_average
TEAM_SHOTS_ON_TARGET_5_last_match_average
TEAM_SHOTS_OUTSIDEBOX_5_last_match_average
TEAM_PASSES_5_last_match_average
TEAM_SUCCESSFUL_PASSES_5_last_match_average
TEAM_SUCCESSFUL_PASSES_PERCENTAGE_5_last_match_average
TEAM_SAVES_5_last_match_average
TEAM_CORNERS_5_last_match_average
TEAM_BALL_POSSESSION_5_last_match_average
TEAM_FOULS_5_last_match_average
TEAM_YELLOWCARDS_5_last_match_average
TEAM_REDCARDS_5_last_match_average
TEAM_OFFSIDES_5_last_match_average
TEAM_ATTACKS_5_last_match_average
TEAM_PENALTIES_5_last_match_average
TEAM_SUBSTITUTIONS_5_last_match_average
TEAM_BALL_SAFE_5_last_match_average
TEAM_DANGEROUS_ATTACKS_5_last_match_average
TEAM_INJURIES_5_last_match_average
TEAM_GOALS_5_last_match_average
TEAM_GAME_WON_5_last_match_average
TEAM_GAME_DRAW_5_last_match_average

TEAM_GAME_LOST_5_last_match_average
TEAM_SHOTS_TOTAL_5_last_match_std
TEAM_SHOTS_INSIDEBOX_5_last_match_std
TEAM_SHOTS_OFF_TARGET_5_last_match_std
TEAM_SHOTS_ON_TARGET_5_last_match_std
TEAM_SHOTS_OUTSIDEBOX_5_last_match_std
TEAM_PASSES_5_last_match_std
TEAM_SUCCESSFUL_PASSES_5_last_match_std
TEAM_SUCCESSFUL_PASSES_PERCENTAGE_5_last_match_std
TEAM_SAVES_5_last_match_std
TEAM_CORNERS_5_last_match_std
TEAM_BALL_POSSESSION_5_last_match_std
TEAM_FOULS_5_last_match_std
TEAM_YELLOWCARDS_5_last_match_std
TEAM_REDCARDS_5_last_match_std
TEAM_OFFSIDES_5_last_match_std
TEAM_ATTACKS_5_last_match_std
TEAM_PENALTIES_5_last_match_std
TEAM_SUBSTITUTIONS_5_last_match_std
TEAM_BALL_SAFE_5_last_match_std
TEAM_DANGEROUS_ATTACKS_5_last_match_std
TEAM_INJURIES_5_last_match_std
TEAM_GOALS_5_last_match_std

```
In [13]: for col in away_team.columns:  
         print(col)
```

ID
LEAGUE
TEAM_NAME
TEAM_SHOTS_TOTAL_season_sum
TEAM_SHOTS_INSIDEBOX_season_sum
TEAM_SHOTS_OFF_TARGET_season_sum
TEAM_SHOTS_ON_TARGET_season_sum
TEAM_SHOTS_OUTSIDEBOX_season_sum
TEAM_PASSES_season_sum
TEAM_SUCCESSFUL_PASSES_season_sum
TEAM_SAVES_season_sum
TEAM_CORNERS_season_sum
TEAM_FOULS_season_sum
TEAM_YELLOWCARDS_season_sum
TEAM_REDCARDS_season_sum
TEAM_OFFSIDES_season_sum
TEAM_ATTACKS_season_sum
TEAM_PENALTIES_season_sum
TEAM_SUBSTITUTIONS_season_sum
TEAM_BALL_SAFE_season_sum
TEAM_DANGEROUS_ATTACKS_season_sum
TEAM_INJURIES_season_sum
TEAM_GOALS_season_sum
TEAM_GAME_WON_season_sum
TEAM_GAME_DRAW_season_sum
TEAM_GAME_LOST_season_sum
TEAM_SHOTS_TOTAL_season_average
TEAM_SHOTS_INSIDEBOX_season_average
TEAM_SHOTS_OFF_TARGET_season_average
TEAM_SHOTS_ON_TARGET_season_average
TEAM_SHOTS_OUTSIDEBOX_season_average
TEAM_PASSES_season_average
TEAM_SUCCESSFUL_PASSES_season_average
TEAM_SUCCESSFUL_PASSES_PERCENTAGE_season_average
TEAM_SAVES_season_average
TEAM_CORNERS_season_average
TEAM_BALL_POSSESSION_season_average
TEAM_FOULS_season_average
TEAM_YELLOWCARDS_season_average
TEAM_REDCARDS_season_average
TEAM_OFFSIDES_season_average
TEAM_ATTACKS_season_average
TEAM_PENALTIES_season_average
TEAM_SUBSTITUTIONS_season_average
TEAM_BALL_SAFE_season_average
TEAM_DANGEROUS_ATTACKS_season_average
TEAM_INJURIES_season_average
TEAM_GOALS_season_average
TEAM_GAME_WON_season_average
TEAM_GAME_DRAW_season_average
TEAM_GAME_LOST_season_average
TEAM_SHOTS_TOTAL_season_std
TEAM_SHOTS_INSIDEBOX_season_std
TEAM_SHOTS_OFF_TARGET_season_std
TEAM_SHOTS_ON_TARGET_season_std
TEAM_SHOTS_OUTSIDEBOX_season_std
TEAM_PASSES_season_std
TEAM_SUCCESSFUL_PASSES_season_std
TEAM_SUCCESSFUL_PASSES_PERCENTAGE_season_std
TEAM_SAVES_season_std

TEAM_CORNERS_season_std
TEAM_BALL_POSSESSION_season_std
TEAM_FOULS_season_std
TEAM_YELLOWCARDS_season_std
TEAM_REDCARDS_season_std
TEAM_OFFSIDES_season_std
TEAM_ATTACKS_season_std
TEAM_PENALTIES_season_std
TEAM_SUBSTITUTIONS_season_std
TEAM_BALL_SAFE_season_std
TEAM_DANGEROUS_ATTACKS_season_std
TEAM_INJURIES_season_std
TEAM_GOALS_season_std
TEAM_SHOTS_TOTAL_5_last_match_sum
TEAM_SHOTS_INSIDEBOX_5_last_match_sum
TEAM_SHOTS_OFF_TARGET_5_last_match_sum
TEAM_SHOTS_ON_TARGET_5_last_match_sum
TEAM_SHOTS_OUTSIDEBOX_5_last_match_sum
TEAM_PASSES_5_last_match_sum
TEAM_SUCCESSFUL_PASSES_5_last_match_sum
TEAM_SAVES_5_last_match_sum
TEAM_CORNERS_5_last_match_sum
TEAM_FOULS_5_last_match_sum
TEAM_YELLOWCARDS_5_last_match_sum
TEAM_REDCARDS_5_last_match_sum
TEAM_OFFSIDES_5_last_match_sum
TEAM_ATTACKS_5_last_match_sum
TEAM_PENALTIES_5_last_match_sum
TEAM_SUBSTITUTIONS_5_last_match_sum
TEAM_BALL_SAFE_5_last_match_sum
TEAM_DANGEROUS_ATTACKS_5_last_match_sum
TEAM_INJURIES_5_last_match_sum
TEAM_GOALS_5_last_match_sum
TEAM_GAME_WON_5_last_match_sum
TEAM_GAME_DRAW_5_last_match_sum
TEAM_GAME_LOST_5_last_match_sum
TEAM_SHOTS_TOTAL_5_last_match_average
TEAM_SHOTS_INSIDEBOX_5_last_match_average
TEAM_SHOTS_OFF_TARGET_5_last_match_average
TEAM_SHOTS_ON_TARGET_5_last_match_average
TEAM_SHOTS_OUTSIDEBOX_5_last_match_average
TEAM_PASSES_5_last_match_average
TEAM_SUCCESSFUL_PASSES_5_last_match_average
TEAM_SUCCESSFUL_PASSES_PERCENTAGE_5_last_match_average
TEAM_SAVES_5_last_match_average
TEAM_CORNERS_5_last_match_average
TEAM_BALL_POSSESSION_5_last_match_average
TEAM_FOULS_5_last_match_average
TEAM_YELLOWCARDS_5_last_match_average
TEAM_REDCARDS_5_last_match_average
TEAM_OFFSIDES_5_last_match_average
TEAM_ATTACKS_5_last_match_average
TEAM_PENALTIES_5_last_match_average
TEAM_SUBSTITUTIONS_5_last_match_average
TEAM_BALL_SAFE_5_last_match_average
TEAM_DANGEROUS_ATTACKS_5_last_match_average
TEAM_INJURIES_5_last_match_average
TEAM_GOALS_5_last_match_average
TEAM_GAME_WON_5_last_match_average
TEAM_GAME_DRAW_5_last_match_average

```
TEAM_GAME_LOST_5_last_match_average
TEAM_SHOTS_TOTAL_5_last_match_std
TEAM_SHOTS_INSIDEBOX_5_last_match_std
TEAM_SHOTS_OFF_TARGET_5_last_match_std
TEAM_SHOTS_ON_TARGET_5_last_match_std
TEAM_SHOTS_OUTSIDEBOX_5_last_match_std
TEAM_PASSES_5_last_match_std
TEAM_SUCCESSFUL_PASSES_5_last_match_std
TEAM_SUCCESSFUL_PASSES_PERCENTAGE_5_last_match_std
TEAM_SAVES_5_last_match_std
TEAM_CORNERS_5_last_match_std
TEAM_BALL_POSSESSION_5_last_match_std
TEAM_FOULS_5_last_match_std
TEAM_YELLOWCARDS_5_last_match_std
TEAM_REDCARDS_5_last_match_std
TEAM_OFFSIDES_5_last_match_std
TEAM_ATTACKS_5_last_match_std
TEAM_PENALTIES_5_last_match_std
TEAM_SUBSTITUTIONS_5_last_match_std
TEAM_BALL_SAFE_5_last_match_std
TEAM_DANGEROUS_ATTACKS_5_last_match_std
TEAM_INJURIES_5_last_match_std
TEAM_GOALS_5_last_match_std
```

```
In [14]: for col in home_player.columns:
          print(col)
```

ID
LEAGUE
TEAM_NAME
POSITION
PLAYER_NAME
PLAYER_ACCURATE_CROSSES_season_sum
PLAYER_ACCURATE_PASSES_season_sum
PLAYER_AERIALS_WON_season_sum
PLAYER_ASSISTS_season_sum
PLAYER_BIG_CHANCES_CREATED_season_sum
PLAYER_BIG_CHANCES_MISSED_season_sum
PLAYER_BLOCKED_SHOTS_season_sum
PLAYER_CAPTAIN_season_sum
PLAYER_CLEARANCES_season_sum
PLAYER_CLEARANCE_OFFLINE_season_sum
PLAYER_DISPOSSESSED_season_sum
PLAYER_DRIBBLED_ATTEMPTS_season_sum
PLAYER_DRIBBLED_PAST_season_sum
PLAYER_DUELS_LOST_season_sum
PLAYER_DUELS_WON_season_sum
PLAYER_ERROR_LEAD_TO_GOAL_season_sum
PLAYER_FOULS_season_sum
PLAYER_FOULS_DRAWN_season_sum
PLAYER_GOALKEEPER_GOALS_CONCEDED_season_sum
PLAYER_GOALS_season_sum
PLAYER_GOALS_CONCEDED_season_sum
PLAYER_HIT_WOODWORK_season_sum
PLAYER_INTERCEPTIONS_season_sum
PLAYER_KEY_PASSES_season_sum
PLAYER_MINUTES_PLAYED_season_sum
PLAYER_OFFSIDES_season_sum
PLAYER_OWN_GOALS_season_sum
PLAYER_PASSES_season_sum
PLAYER_PENALTIES_COMMITTED_season_sum
PLAYER_PENALTIES_MISSES_season_sum
PLAYER_PENALTIES_SAVED_season_sum
PLAYER_PENALTIES_SCORED_season_sum
PLAYER_PENALTIES_WON_season_sum
PLAYER_REDCARDS_season_sum
PLAYER_SAVES_season_sum
PLAYER_SAVES_INSIDE_BOX_season_sum
PLAYER_SHOTS_BLOCKED_season_sum
PLAYER_SHOTS_ON_TARGET_season_sum
PLAYER_SHOTS_TOTAL_season_sum
PLAYER_STARTING_LINEUP_season_sum
PLAYER_SUCCESSFUL_DRIBBLES_season_sum
PLAYER_TACKLES_season_sum
PLAYER_TOTAL_CROSSES_season_sum
PLAYER_TOTAL_DUELS_season_sum
PLAYER_YELLOWCARDS_season_sum
PLAYER_PUNCHES_season_sum
PLAYER_LONG_BALLS_season_sum
PLAYER_LONG_BALLS_WON_season_sum
PLAYER_SHOTS_OFF_TARGET_season_sum
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PLAYER_ACCURATE_PASSES_season_average
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average
PLAYER_AERIALS_WON_season_average
PLAYER_ASSISTS_season_average
PLAYER_BIG_CHANCES_CREATED_season_average

PLAYER_BIG_CHANCES_MISSED_season_average
PLAYER_BLOCKED_SHOTS_season_average
PLAYER_CAPTAIN_season_average
PLAYER_CLEARANCES_season_average
PLAYER_CLEARANCE_OFFLINE_season_average
PLAYER_DISPOSSESSED_season_average
PLAYER_DRIBBLED_ATTEMPTS_season_average
PLAYER_DRIBBLED_PAST_season_average
PLAYER_DUELS_LOST_season_average
PLAYER_DUELS_WON_season_average
PLAYER_ERROR_LEAD_TO_GOAL_season_average
PLAYER_FOULS_season_average
PLAYER_FOULS_DRAWN_season_average
PLAYER_GOALKEEPER_GOALS_CONCEDED_season_average
PLAYER_GOALS_season_average
PLAYER_GOALS_CONCEDED_season_average
PLAYER_HIT_WOODWORK_season_average
PLAYER_INTERCEPTIONS_season_average
PLAYER_KEY_PASSES_season_average
PLAYER_MINUTES_PLAYED_season_average
PLAYER_OFFSIDES_season_average
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PLAYER_PENALTIES_MISSES_season_average
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PLAYER_PENALTIES_SCORED_season_average
PLAYER_PENALTIES_WON_season_average
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PLAYER_ACCURATE_PASSES_season_std
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std
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PLAYER_DUELS_LOST_season_std

PLAYER_DUELS_WON_season_std
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PLAYER_FOULS_DRAWN_season_std
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PLAYER_GOALS_season_std
PLAYER_GOALS_CONCEDED_season_std
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PLAYER_PENALTIES_MISSES_season_std
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PLAYER_PENALTIES_WON_season_std
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PLAYER_SAVES_INSIDE_BOX_season_std
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PLAYER_TOTAL_DUELS_season_std
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PLAYER_LONG_BALLS_WON_season_std
PLAYER_SHOTS_OFF_TARGET_season_std
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PLAYER_DUELS_WON_5_last_match_sum
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PLAYER_FOULS_DRAWN_5_last_match_sum
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PLAYER_GOALS_CONCEDED_5_last_match_sum
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PLAYER_INTERCEPTIONS_5_last_match_sum
PLAYER_KEY_PASSES_5_last_match_sum

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PLAYER_OFFSIDES_5_last_match_sum
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PLAYER_PENALTIES_SCORED_5_last_match_sum
PLAYER_PENALTIES_WON_5_last_match_sum
PLAYER_REDCARDS_5_last_match_sum
PLAYER_SAVES_5_last_match_sum
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PLAYER_SHOTS_ON_TARGET_5_last_match_sum
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PLAYER_TOTAL_DUELS_5_last_match_sum
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PLAYER_LONG_BALLS_WON_5_last_match_sum
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PLAYER_ACCURATE_PASSES_5_last_match_average
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average
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PLAYER_BIG_CHANCES_MISSED_5_last_match_average
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PLAYER_CLEARANCE_OFFLINE_5_last_match_average
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PLAYER_DRIBBLED_PAST_5_last_match_average
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PLAYER_DUELS_WON_5_last_match_average
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PLAYER_FOULS_DRAWN_5_last_match_average
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PLAYER_GOALS_CONCEDED_5_last_match_average
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PLAYER_PASSES_5_last_match_average
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PLAYER_PENALTIES_MISSES_5_last_match_average
PLAYER_PENALTIES_SAVED_5_last_match_average
PLAYER_PENALTIES_SCORED_5_last_match_average
PLAYER_PENALTIES_WON_5_last_match_average
PLAYER_RATING_5_last_match_average

PLAYER_REDCARDS_5_last_match_average
PLAYER_SAVES_5_last_match_average
PLAYER_SAVES_INSIDE_BOX_5_last_match_average
PLAYER_SHOTS_BLOCKED_5_last_match_average
PLAYER_SHOTS_ON_TARGET_5_last_match_average
PLAYER_SHOTS_TOTAL_5_last_match_average
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PLAYER_TACKLES_5_last_match_average
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PLAYER_TOTAL_DUELS_5_last_match_average
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PLAYER_PUNCHES_5_last_match_average
PLAYER_LONG_BALLS_5_last_match_average
PLAYER_LONG_BALLS_WON_5_last_match_average
PLAYER_SHOTS_OFF_TARGET_5_last_match_average
PLAYER_ACCURATE_CROSSES_5_last_match_std
PLAYER_ACCURATE_PASSES_5_last_match_std
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std
PLAYER_AERIALS_WON_5_last_match_std
PLAYER_ASSISTS_5_last_match_std
PLAYER_BIG_CHANCES_CREATED_5_last_match_std
PLAYER_BIG_CHANCES_MISSED_5_last_match_std
PLAYER_BLOCKED_SHOTS_5_last_match_std
PLAYER_CAPTAIN_5_last_match_std
PLAYER_CLEARANCES_5_last_match_std
PLAYER_CLEARANCE_OFFLINE_5_last_match_std
PLAYER_DISPOSSESSED_5_last_match_std
PLAYER_DRIBBLED_ATTEMPTS_5_last_match_std
PLAYER_DRIBBLED_PAST_5_last_match_std
PLAYER_DUELS_LOST_5_last_match_std
PLAYER_DUELS_WON_5_last_match_std
PLAYER_ERROR_LEAD_TO_GOAL_5_last_match_std
PLAYER_FOULS_5_last_match_std
PLAYER_FOULS_DRAWN_5_last_match_std
PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_std
PLAYER_GOALS_5_last_match_std
PLAYER_GOALS_CONCEDED_5_last_match_std
PLAYER_HIT_WOODWORK_5_last_match_std
PLAYER_INTERCEPTIONS_5_last_match_std
PLAYER_KEY_PASSES_5_last_match_std
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PLAYER_OFFSIDES_5_last_match_std
PLAYER_OWN_GOALS_5_last_match_std
PLAYER_PASSES_5_last_match_std
PLAYER_PENALTIES_COMMITTED_5_last_match_std
PLAYER_PENALTIES_MISSES_5_last_match_std
PLAYER_PENALTIES_SAVED_5_last_match_std
PLAYER_PENALTIES_SCORED_5_last_match_std
PLAYER_PENALTIES_WON_5_last_match_std
PLAYER_RATING_5_last_match_std
PLAYER_REDCARDS_5_last_match_std
PLAYER_SAVES_5_last_match_std
PLAYER_SAVES_INSIDE_BOX_5_last_match_std
PLAYER_SHOTS_BLOCKED_5_last_match_std
PLAYER_SHOTS_ON_TARGET_5_last_match_std
PLAYER_SHOTS_TOTAL_5_last_match_std
PLAYER_STARTING_LINEUP_5_last_match_std
PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_std
PLAYER_TACKLES_5_last_match_std

```
PLAYER_TOTAL_CROSSES_5_last_match_std  
PLAYER_TOTAL_DUELS_5_last_match_std  
PLAYER_YELLOWCARDS_5_last_match_std  
PLAYER_PUNCHES_5_last_match_std  
PLAYER_LONG_BALLS_5_last_match_std  
PLAYER_LONG_BALLS_WON_5_last_match_std  
PLAYER_SHOTS_OFF_TARGET_5_last_match_std
```

```
In [15]: for col in away_player.columns:  
         print(col)
```

ID
LEAGUE
TEAM_NAME
POSITION
PLAYER_NAME
PLAYER_ACCURATE_CROSSES_season_sum
PLAYER_ACCURATE_PASSES_season_sum
PLAYER_AERIALS_WON_season_sum
PLAYER_ASSISTS_season_sum
PLAYER_BIG_CHANCES_CREATED_season_sum
PLAYER_BIG_CHANCES_MISSED_season_sum
PLAYER_BLOCKED_SHOTS_season_sum
PLAYER_CAPTAIN_season_sum
PLAYER_CLEARANCES_season_sum
PLAYER_CLEARANCE_OFFLINE_season_sum
PLAYER_DISPOSSESSED_season_sum
PLAYER_DRIBBLED_ATTEMPTS_season_sum
PLAYER_DRIBBLED_PAST_season_sum
PLAYER_DUELS_LOST_season_sum
PLAYER_DUELS_WON_season_sum
PLAYER_ERROR_LEAD_TO_GOAL_season_sum
PLAYER_FOULS_season_sum
PLAYER_FOULS_DRAWN_season_sum
PLAYER_GOALKEEPER_GOALS_CONCEDED_season_sum
PLAYER_GOALS_season_sum
PLAYER_GOALS_CONCEDED_season_sum
PLAYER_HIT_WOODWORK_season_sum
PLAYER_INTERCEPTIONS_season_sum
PLAYER_KEY_PASSES_season_sum
PLAYER_MINUTES_PLAYED_season_sum
PLAYER_OFFSIDES_season_sum
PLAYER_OWN_GOALS_season_sum
PLAYER_PASSES_season_sum
PLAYER_PENALTIES_COMMITTED_season_sum
PLAYER_PENALTIES_MISSES_season_sum
PLAYER_PENALTIES_SAVED_season_sum
PLAYER_PENALTIES_SCORED_season_sum
PLAYER_PENALTIES_WON_season_sum
PLAYER_REDCARDS_season_sum
PLAYER_SAVES_season_sum
PLAYER_SAVES_INSIDE_BOX_season_sum
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PLAYER_SHOTS_ON_TARGET_season_sum
PLAYER_SHOTS_TOTAL_season_sum
PLAYER_STARTING_LINEUP_season_sum
PLAYER_SUCCESSFUL_DRIBBLES_season_sum
PLAYER_TACKLES_season_sum
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PLAYER_TOTAL_DUELS_season_sum
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PLAYER_PUNCHES_season_sum
PLAYER_LONG_BALLS_season_sum
PLAYER_LONG_BALLS_WON_season_sum
PLAYER_SHOTS_OFF_TARGET_season_sum
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PLAYER_ACCURATE_PASSES_season_average
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average
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PLAYER_ASSISTS_season_average
PLAYER_BIG_CHANCES_CREATED_season_average

PLAYER_BIG_CHANCES_MISSED_season_average
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PLAYER_CAPTAIN_season_average
PLAYER_CLEARANCES_season_average
PLAYER_CLEARANCE_OFFLINE_season_average
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PLAYER_DRIBBLED_ATTEMPTS_season_average
PLAYER_DRIBBLED_PAST_season_average
PLAYER_DUELS_LOST_season_average
PLAYER_DUELS_WON_season_average
PLAYER_ERROR_LEAD_TO_GOAL_season_average
PLAYER_FOULS_season_average
PLAYER_FOULS_DRAWN_season_average
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PLAYER_GOALS_season_average
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PLAYER_INTERCEPTIONS_season_average
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PLAYER_PASSES_season_average
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PLAYER_LONG_BALLS_WON_season_average
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PLAYER_DUELS_WON_season_std
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PLAYER_LONG_BALLS_WON_season_std
PLAYER_SHOTS_OFF_TARGET_season_std
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PLAYER_ACCURATE_PASSES_5_last_match_sum
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PLAYER_DRIBBLED_PAST_5_last_match_sum
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PLAYER_DUELS_WON_5_last_match_sum
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PLAYER_FOULS_DRAWN_5_last_match_sum
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PLAYER_GOALS_CONCEDED_5_last_match_sum
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PLAYER_INTERCEPTIONS_5_last_match_sum
PLAYER_KEY_PASSES_5_last_match_sum

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PLAYER_DUELS_WON_5_last_match_average
PLAYER_ERROR_LEAD_TO_GOAL_5_last_match_average
PLAYER_FOULS_5_last_match_average
PLAYER_FOULS_DRAWN_5_last_match_average
PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_average
PLAYER_GOALS_5_last_match_average
PLAYER_GOALS_CONCEDED_5_last_match_average
PLAYER_HIT_WOODWORK_5_last_match_average
PLAYER_INTERCEPTIONS_5_last_match_average
PLAYER_KEY_PASSES_5_last_match_average
PLAYER_MINUTES_PLAYED_5_last_match_average
PLAYER_OFFSIDES_5_last_match_average
PLAYER_OWN_GOALS_5_last_match_average
PLAYER_PASSES_5_last_match_average
PLAYER_PENALTIES_COMMITTED_5_last_match_average
PLAYER_PENALTIES_MISSES_5_last_match_average
PLAYER_PENALTIES_SAVED_5_last_match_average
PLAYER_PENALTIES_SCORED_5_last_match_average
PLAYER_PENALTIES_WON_5_last_match_average
PLAYER_RATING_5_last_match_average

PLAYER_REDCARDS_5_last_match_average
PLAYER_SAVES_5_last_match_average
PLAYER_SAVES_INSIDE_BOX_5_last_match_average
PLAYER_SHOTS_BLOCKED_5_last_match_average
PLAYER_SHOTS_ON_TARGET_5_last_match_average
PLAYER_SHOTS_TOTAL_5_last_match_average
PLAYER_STARTING_LINEUP_5_last_match_average
PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_average
PLAYER_TACKLES_5_last_match_average
PLAYER_TOTAL_CROSSES_5_last_match_average
PLAYER_TOTAL_DUELS_5_last_match_average
PLAYER_YELLOWCARDS_5_last_match_average
PLAYER_PUNCHES_5_last_match_average
PLAYER_LONG_BALLS_5_last_match_average
PLAYER_LONG_BALLS_WON_5_last_match_average
PLAYER_SHOTS_OFF_TARGET_5_last_match_average
PLAYER_ACCURATE_CROSSES_5_last_match_std
PLAYER_ACCURATE_PASSES_5_last_match_std
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std
PLAYER_AERIALS_WON_5_last_match_std
PLAYER_ASSISTS_5_last_match_std
PLAYER_BIG_CHANCES_CREATED_5_last_match_std
PLAYER_BIG_CHANCES_MISSED_5_last_match_std
PLAYER_BLOCKED_SHOTS_5_last_match_std
PLAYER_CAPTAIN_5_last_match_std
PLAYER_CLEARANCES_5_last_match_std
PLAYER_CLEARANCE_OFFLINE_5_last_match_std
PLAYER_DISPOSSESSED_5_last_match_std
PLAYER_DRIBBLED_ATTEMPTS_5_last_match_std
PLAYER_DRIBBLED_PAST_5_last_match_std
PLAYER_DUELS_LOST_5_last_match_std
PLAYER_DUELS_WON_5_last_match_std
PLAYER_ERROR_LEAD_TO_GOAL_5_last_match_std
PLAYER_FOULS_5_last_match_std
PLAYER_FOULS_DRAWN_5_last_match_std
PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_std
PLAYER_GOALS_5_last_match_std
PLAYER_GOALS_CONCEDED_5_last_match_std
PLAYER_HIT_WOODWORK_5_last_match_std
PLAYER_INTERCEPTIONS_5_last_match_std
PLAYER_KEY_PASSES_5_last_match_std
PLAYER_MINUTES_PLAYED_5_last_match_std
PLAYER_OFFSIDES_5_last_match_std
PLAYER_OWN_GOALS_5_last_match_std
PLAYER_PASSES_5_last_match_std
PLAYER_PENALTIES_COMMITTED_5_last_match_std
PLAYER_PENALTIES_MISSES_5_last_match_std
PLAYER_PENALTIES_SAVED_5_last_match_std
PLAYER_PENALTIES_SCORED_5_last_match_std
PLAYER_PENALTIES_WON_5_last_match_std
PLAYER_RATING_5_last_match_std
PLAYER_REDCARDS_5_last_match_std
PLAYER_SAVES_5_last_match_std
PLAYER_SAVES_INSIDE_BOX_5_last_match_std
PLAYER_SHOTS_BLOCKED_5_last_match_std
PLAYER_SHOTS_ON_TARGET_5_last_match_std
PLAYER_SHOTS_TOTAL_5_last_match_std
PLAYER_STARTING_LINEUP_5_last_match_std
PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_std
PLAYER_TACKLES_5_last_match_std


```

PLAYER_TOTAL_CROSSES_5_last_match_std
PLAYER_TOTAL_DUELS_5_last_match_std
PLAYER_YELLOWCARDS_5_last_match_std
PLAYER_PUNCHES_5_last_match_std
PLAYER_LONG_BALLS_5_last_match_std
PLAYER_LONG_BALLS_WON_5_last_match_std
PLAYER_SHOTS_OFF_TARGET_5_last_match_std

```

```

In [16]: for col in y_train.columns:
          print(col)

```

```

ID
HOME_WINS
DRAW
AWAY_WINS

```

```

In [17]: for col in y_train_supp.columns:
          print(col)

```

```

ID
GOAL_DIFF_HOME_AWAY

```

```

In [18]: # Checking Data Types
def datatype_summary(df):

    summary = pd.DataFrame({
        "Data Type": df.dtypes,
        "Missing": df.isna().sum(),
        "Unique Values": df.nunique()
    })

    return summary

datatype_summary(home_team)

```

Out[18]:

	Data Type	Missing	Unique Values
ID	int64	0	12303
LEAGUE	object	0	13
TEAM_NAME	object	0	350
TEAM_SHOTS_TOTAL_season_sum	float64	260	11
TEAM_SHOTS_INSIDEBOX_season_sum	float64	1385	11
...	...	...	...
TEAM_SUBSTITUTIONS_5_last_match_std	float64	91	11
TEAM_BALL_SAFE_5_last_match_std	float64	1795	11
TEAM_DANGEROUS_ATTACKS_5_last_match_std	float64	0	11
TEAM_INJURIES_5_last_match_std	float64	3318	11
TEAM_GOALS_5_last_match_std	float64	2	11

143 rows × 3 columns

```
In [19]: def datatype_summary(df):

    summary = pd.DataFrame({
        "Data Type": df.dtypes,
        "Missing": df.isna().sum(),
        "Unique Values": df.nunique()
    })

    return summary

datatype_summary(away_team)
```

Out[19]:

	Data Type	Missing	Unique Values
ID	int64	0	12303
LEAGUE	object	0	13
TEAM_NAME	object	0	350
TEAM_SHOTS_TOTAL_season_sum	float64	259	11
TEAM_SHOTS_INSIDEBOX_season_sum	float64	1386	11
...	...	...	...
TEAM_SUBSTITUTIONS_5_last_match_std	float64	91	11
TEAM_BALL_SAFE_5_last_match_std	float64	1836	11
TEAM_DANGEROUS_ATTACKS_5_last_match_std	float64	0	11
TEAM_INJURIES_5_last_match_std	float64	3313	11
TEAM_GOALS_5_last_match_std	float64	2	11

143 rows × 3 columns

```
In [20]: def datatype_summary(df):

    summary = pd.DataFrame({
        "Data Type": df.dtypes,
        "Missing": df.isna().sum(),
        "Unique Values": df.nunique()
    })

    return summary

datatype_summary(home_player)
```

Out[20]:

	Data Type	Missing	Unique Values
ID	int64	0	12303
LEAGUE	object	0	13
TEAM_NAME	object	0	350
POSITION	object	42820	4
PLAYER_NAME	object	0	13551
...	...	...	...
PLAYER_YELLOWCARDS_5_last_match_std	float64	7443	35
PLAYER_PUNCHES_5_last_match_std	float64	170579	75
PLAYER_LONG_BALLS_5_last_match_std	float64	237079	0
PLAYER_LONG_BALLS_WON_5_last_match_std	float64	237079	0
PLAYER_SHOTS_OFF_TARGET_5_last_match_std	float64	237079	0

307 rows × 3 columns

```
In [21]: def datatype_summary(df):  
  
    summary = pd.DataFrame({  
        "Data Type": df.dtypes,  
        "Missing": df.isna().sum(),  
        "Unique Values": df.nunique()  
    })  
  
    return summary  
  
datatype_summary(away_player)
```

Out[21]:

	Data Type	Missing	Unique Values
ID	int64	0	12303
LEAGUE	object	0	13
TEAM_NAME	object	0	350
POSITION	object	42255	4
PLAYER_NAME	object	0	13673
...	...	...	...
PLAYER_YELLOWCARDS_5_last_match_std	float64	7687	35
PLAYER_PUNCHES_5_last_match_std	float64	170078	74
PLAYER_LONG_BALLS_5_last_match_std	float64	236132	0
PLAYER_LONG_BALLS_WON_5_last_match_std	float64	236132	0
PLAYER_SHOTS_OFF_TARGET_5_last_match_std	float64	236132	0

307 rows × 3 columns

In [22]: **def** datatype_summary(df):

```

    summary = pd.DataFrame({
        "Data Type": df.dtypes,
        "Missing": df.isna().sum(),
        "Unique Values": df.nunique()
    })

```

```

    return summary

```

```

datatype_summary(y_train)

```

Out[22]:

	Data Type	Missing	Unique Values
ID	int64	0	12303
HOME_WINS	int64	0	2
DRAW	int64	0	2
AWAY_WINS	int64	0	2

In [23]: **def** datatype_summary(df):

```

    summary = pd.DataFrame({
        "Data Type": df.dtypes,
        "Missing": df.isna().sum(),
        "Unique Values": df.nunique()
    })

```

```

    return summary

```

```

datatype_summary(y_train_supp)

```

Out[23]:

	Data Type	Missing	Unique Values
ID	int64	0	12303
GOAL_DIFF_HOME_AWAY	float64	0	21

In [24]: *# Duplicate Analysis*

```
def check_duplicates(df, name):  
  
    duplicate_rows = df.duplicated().sum()  
  
    print("="*70)  
    print(name)  
    print("="*70)  
  
    print(f"Duplicate Rows : {duplicate_rows:,}")  
  
    if duplicate_rows > 0:  
        print("\nExample Duplicate Rows")  
        display(df[df.duplicated()].head())
```

```
In [25]: check_duplicates(home_team,"HOME TEAM")  
  
check_duplicates(away_team,"AWAY TEAM")  
  
check_duplicates(home_player,"HOME PLAYER")  
  
check_duplicates(away_player,"AWAY PLAYER")  
  
check_duplicates(y_train,"Y TRAIN")  
  
check_duplicates(y_train_supp,"Y TRAIN SUPP")
```

```
=====
HOME TEAM
```

```
=====
Duplicate Rows : 0
```

```
=====
AWAY TEAM
```

```
=====
Duplicate Rows : 0
```

```
=====
HOME PLAYER
```

```
=====
Duplicate Rows : 0
```

```
=====
AWAY PLAYER
```

```
=====
Duplicate Rows : 0
```

```
=====
Y TRAIN
```

```
=====
Duplicate Rows : 0
```

```
=====
Y TRAIN SUPP
```

```
=====
Duplicate Rows : 0
```

```
In [26]: # ID Consistency
def check_id(df,name):

    print("="*70)
    print(name)
    print("="*70)

    print("Total Rows :",len(df))
    print("Unique IDs :",df["ID"].nunique())

    print("Duplicate IDs :",df["ID"].duplicated().sum())
```

```
In [27]: check_id(home_team,"HOME TEAM")

check_id(away_team,"AWAY TEAM")

check_id(home_player,"HOME PLAYER")

check_id(away_player,"AWAY PLAYER")

check_id(y_train,"Y TRAIN")

check_id(y_train_supp,"Y TRAIN SUPP")
```

```
=====
HOME TEAM
=====
Total Rows : 12303
Unique IDs : 12303
Duplicate IDs : 0
=====
AWAY TEAM
=====
Total Rows : 12303
Unique IDs : 12303
Duplicate IDs : 0
=====
HOME PLAYER
=====
Total Rows : 237079
Unique IDs : 12303
Duplicate IDs : 224776
=====
AWAY PLAYER
=====
Total Rows : 236132
Unique IDs : 12303
Duplicate IDs : 223829
=====
Y TRAIN
=====
Total Rows : 12303
Unique IDs : 12303
Duplicate IDs : 0
=====
Y TRAIN SUPP
=====
Total Rows : 12303
Unique IDs : 12303
Duplicate IDs : 0
=====
```

For Home team, Away Team, Y Train, Y Train Supp Duplicate IDs will be 0, whereas for player datasets duplicate IDs should be very large as one match contains many players.

```
In [28]: # Missing Value Summary

def missing_summary(df):

    missing = pd.DataFrame({

        "Missing Count":df.isna().sum(),

        "Missing %":round(df.isna().mean()*100,2),

        "Data Type":df.dtypes

    })

    missing = missing[missing["Missing Count"]>0]

    return missing.sort_values(
        by="Missing %",
```

```
) ascending=False
```

```
In [29]: missing_summary(home_team)
```

```
Out[29]:
```

	Missing Count	Missing %	Data Type
TEAM_INJURIES_5_last_match_std	3318	26.97	float64
TEAM_INJURIES_5_last_match_sum	2932	23.83	float64
TEAM_INJURIES_5_last_match_average	2931	23.82	float64
TEAM_INJURIES_season_std	2924	23.77	float64
TEAM_INJURIES_season_sum	1960	15.93	float64
...	...	...	...
TEAM_YELLOWCARDS_5_last_match_std	1	0.01	float64
TEAM_CORNERS_5_last_match_average	1	0.01	float64
TEAM_YELLOWCARDS_season_std	1	0.01	float64
TEAM_CORNERS_season_sum	1	0.01	float64
TEAM_CORNERS_5_last_match_sum	1	0.01	float64

118 rows × 3 columns

```
In [30]: missing_summary(away_team)
```

```
Out[30]:
```

	Missing Count	Missing %	Data Type
TEAM_INJURIES_5_last_match_std	3313	26.93	float64
TEAM_INJURIES_5_last_match_sum	2933	23.84	float64
TEAM_INJURIES_5_last_match_average	2932	23.83	float64
TEAM_INJURIES_season_std	2926	23.78	float64
TEAM_INJURIES_season_sum	1961	15.94	float64
...	...	...	...
TEAM_YELLOWCARDS_5_last_match_std	1	0.01	float64
TEAM_CORNERS_5_last_match_average	1	0.01	float64
TEAM_YELLOWCARDS_season_std	1	0.01	float64
TEAM_CORNERS_season_sum	1	0.01	float64
TEAM_CORNERS_5_last_match_sum	1	0.01	float64

118 rows × 3 columns

```
In [31]: missing_summary(home_player)
```


Out[31]:

	Missing Count	Missing %	Data Type
PLAYER_SHOTS_OFF_TARGET_season_std	237079	100.00	float64
PLAYER_LONG_BALLS_5_last_match_sum	237079	100.00	float64
PLAYER_SHOTS_OFF_TARGET_season_sum	237079	100.00	float64
PLAYER_CAPTAIN_season_average	237079	100.00	float64
PLAYER_LONG_BALLS_season_average	237079	100.00	float64
...	...	...	...
PLAYER_ASSISTS_season_sum	3597	1.52	float64
PLAYER_ASSISTS_season_average	3597	1.52	float64
PLAYER_MINUTES_PLAYED_5_last_match_sum	3597	1.52	float64
PLAYER_GOALS_season_sum	3597	1.52	float64
PLAYER_GOALS_season_average	3597	1.52	float64

303 rows × 3 columns

In [32]: `missing_summary(away_player)`

Out[32]:

	Missing Count	Missing %	Data Type
PLAYER_SHOTS_OFF_TARGET_season_std	236132	100.00	float64
PLAYER_LONG_BALLS_5_last_match_sum	236132	100.00	float64
PLAYER_SHOTS_OFF_TARGET_season_sum	236132	100.00	float64
PLAYER_CAPTAIN_season_average	236132	100.00	float64
PLAYER_LONG_BALLS_season_average	236132	100.00	float64
...	...	...	...
PLAYER_ASSISTS_season_sum	3829	1.62	float64
PLAYER_ASSISTS_season_average	3829	1.62	float64
PLAYER_MINUTES_PLAYED_5_last_match_sum	3829	1.62	float64
PLAYER_GOALS_season_sum	3829	1.62	float64
PLAYER_GOALS_season_average	3829	1.62	float64

303 rows × 3 columns

In [33]: `# Categorizing Missing Values`

In [34]: `def classify_missing(df):`

```

missing = (
    df.isna().mean()*100
)

summary = pd.DataFrame({

    "Missing %":missing

})

summary["Category"] = pd.cut(

    summary["Missing %"],

    bins=[-1,0,5,20,50,80,100],

    labels=[
        "No Missing",
        "Very Low",
        "Low",
        "Moderate",
        "High",
        "Very High"
    ]

)

return summary.sort_values(
    by="Missing %",
    ascending=False
)

```

In [35]: `classify_missing(home_team)`

Out[35]:

	Missing %	Category
TEAM_INJURIES_5_last_match_std	26.97	Moderate
TEAM_INJURIES_5_last_match_sum	23.83	Moderate
TEAM_INJURIES_5_last_match_average	23.82	Moderate
TEAM_INJURIES_season_std	23.77	Moderate
TEAM_INJURIES_season_sum	15.93	Low
...	...	...
TEAM_ATTACKS_5_last_match_average	0.00	No Missing
TEAM_ATTACKS_5_last_match_sum	0.00	No Missing
TEAM_DANGEROUS_ATTACKS_5_last_match_sum	0.00	No Missing
TEAM_SHOTS_OFF_TARGET_5_last_match_average	0.00	No Missing
ID	0.00	No Missing

143 rows × 2 columns

```
In [36]: classify_missing(away_team)
```

```
Out[36]:
```

	Missing %	Category
TEAM_INJURIES_5_last_match_std	26.93	Moderate
TEAM_INJURIES_5_last_match_sum	23.84	Moderate
TEAM_INJURIES_5_last_match_average	23.83	Moderate
TEAM_INJURIES_season_std	23.78	Moderate
TEAM_INJURIES_season_sum	15.94	Low
...	...	...
TEAM_ATTACKS_5_last_match_average	0.00	No Missing
TEAM_ATTACKS_5_last_match_sum	0.00	No Missing
TEAM_DANGEROUS_ATTACKS_5_last_match_sum	0.00	No Missing
TEAM_SHOTS_OFF_TARGET_5_last_match_average	0.00	No Missing
ID	0.00	No Missing

143 rows × 2 columns

```
In [37]: classify_missing(home_player)
```

```
Out[37]:
```

	Missing %	Category
PLAYER_LONG_BALLS_season_std	100.00	Very High
PLAYER_LONG_BALLS_5_last_match_sum	100.00	Very High
PLAYER_SHOTS_OFF_TARGET_season_sum	100.00	Very High
PLAYER_CAPTAIN_season_average	100.00	Very High
PLAYER_LONG_BALLS_season_average	100.00	Very High
...	...	...
PLAYER_GOALS_5_last_match_sum	1.52	Very Low
LEAGUE	0.00	No Missing
PLAYER_NAME	0.00	No Missing
TEAM_NAME	0.00	No Missing
ID	0.00	No Missing

307 rows × 2 columns

```
In [38]: classify_missing(away_player)
```

Out[38]:

	Missing %	Category
PLAYER_LONG_BALLS_season_std	100.00	Very High
PLAYER_LONG_BALLS_5_last_match_sum	100.00	Very High
PLAYER_SHOTS_OFF_TARGET_season_sum	100.00	Very High
PLAYER_CAPTAIN_season_average	100.00	Very High
PLAYER_LONG_BALLS_season_average	100.00	Very High
...	...	...
PLAYER_GOALS_5_last_match_sum	1.62	Very Low
LEAGUE	0.00	No Missing
PLAYER_NAME	0.00	No Missing
TEAM_NAME	0.00	No Missing
ID	0.00	No Missing

307 rows × 2 columns

```
In [39]: # Constant Columns
def constant_columns(df):

    constant=[]

    for col in df.columns:

        if df[col].nunique()==1:

            constant.append(col)

    return constant
```

```
In [40]: constant_columns(home_team)
```

Out[40]: []

```
In [41]: constant_columns(away_team)
```

Out[41]: []

```
In [42]: constant_columns(home_player)
```

Out[42]: []

```
In [43]: constant_columns(away_player)
```

Out[43]: []

[] is the result for every dataset which means no useless columns, therefore nothing to drop.

```
In [44]: # Numerical Summary
home_team.describe().T
```

```
Out[44]:
```

	count	mean	std	min	2
ID	12303.00	6151.00	3551.71	0.00	3075
TEAM_SHOTS_TOTAL_season_sum	12043.00	4.39	2.87	0.00	2
TEAM_SHOTS_INSIDEBOX_season_sum	10918.00	4.30	2.93	0.00	2
TEAM_SHOTS_OFF_TARGET_season_sum	12303.00	4.51	2.81	0.00	2
TEAM_SHOTS_ON_TARGET_season_sum	12301.00	4.15	2.86	0.00	2
...	...	...	...	...	...
TEAM_SUBSTITUTIONS_5_last_match_std	12212.00	3.30	3.35	0.00	0
TEAM_BALL_SAFE_5_last_match_std	10508.00	3.80	2.87	0.00	1
TEAM_DANGEROUS_ATTACKS_5_last_match_std	12303.00	3.69	2.78	0.00	2
TEAM_INJURIES_5_last_match_std	8985.00	3.87	2.79	0.00	2
TEAM_GOALS_5_last_match_std	12301.00	3.63	2.85	0.00	1

141 rows × 8 columns

```
In [45]: away_team.describe().T
```

```
Out[45]:
```

	count	mean	std	min	2
ID	12303.00	6151.00	3551.71	0.00	3075
TEAM_SHOTS_TOTAL_season_sum	12044.00	4.47	2.89	0.00	2
TEAM_SHOTS_INSIDEBOX_season_sum	10917.00	4.38	2.93	0.00	2
TEAM_SHOTS_OFF_TARGET_season_sum	12303.00	4.59	2.81	0.00	2
TEAM_SHOTS_ON_TARGET_season_sum	12301.00	4.22	2.87	0.00	2
...	...	...	...	...	...
TEAM_SUBSTITUTIONS_5_last_match_std	12212.00	3.27	3.34	0.00	0
TEAM_BALL_SAFE_5_last_match_std	10467.00	3.80	2.86	0.00	2
TEAM_DANGEROUS_ATTACKS_5_last_match_std	12303.00	3.76	2.81	0.00	2
TEAM_INJURIES_5_last_match_std	8990.00	3.89	2.80	0.00	2
TEAM_GOALS_5_last_match_std	12301.00	3.69	2.86	0.00	2

141 rows × 8 columns

For team dataset the highest missing percentage is 26.9% and it comes team injuries which makes sense as not every team has injuries recorded so we will not drop these columns

```
In [46]: home_player.describe().T
```

```
Out[46]:
```

	count	mean	std	min	2
ID	237079.00	6151.42	3552.22	0.00	3080
PLAYER_ACCURATE_CROSSES_season_sum	208790.00	7.93	15.10	0.00	0
PLAYER_ACCURATE_PASSES_season_sum	208790.00	22.72	20.85	0.00	6
PLAYER_AERIALS_WON_season_sum	208790.00	12.72	16.20	0.00	1
PLAYER_ASSISTS_season_sum	233482.00	9.30	16.38	0.00	0
...	...	...	...	...	...
PLAYER_YELLOWCARDS_5_last_match_std	229636.00	25.54	33.78	0.00	0
PLAYER_PUNCHES_5_last_match_std	66500.00	2.12	11.19	0.00	0
PLAYER_LONG_BALLS_5_last_match_std	0.00	NaN	NaN	NaN	0
PLAYER_LONG_BALLS_WON_5_last_match_std	0.00	NaN	NaN	NaN	0
PLAYER_SHOTS_OFF_TARGET_5_last_match_std	0.00	NaN	NaN	NaN	0

303 rows × 8 columns

```
In [47]: away_player.describe().T
```

```
Out[47]:
```

	count	mean	std	min	2
ID	236132.00	6151.38	3551.67	0.00	3079
PLAYER_ACCURATE_CROSSES_season_sum	207662.00	7.94	15.15	0.00	0
PLAYER_ACCURATE_PASSES_season_sum	207662.00	22.73	20.89	0.00	6
PLAYER_AERIALS_WON_season_sum	207662.00	12.70	16.17	0.00	1
PLAYER_ASSISTS_season_sum	232303.00	9.28	16.39	0.00	0
...	...	...	...	...	...
PLAYER_YELLOWCARDS_5_last_match_std	228445.00	25.17	33.65	0.00	0
PLAYER_PUNCHES_5_last_match_std	66054.00	2.10	11.14	0.00	0
PLAYER_LONG_BALLS_5_last_match_std	0.00	NaN	NaN	NaN	0
PLAYER_LONG_BALLS_WON_5_last_match_std	0.00	NaN	NaN	NaN	0
PLAYER_SHOTS_OFF_TARGET_5_last_match_std	0.00	NaN	NaN	NaN	0

303 rows × 8 columns

For player dataset many variables have 100% missing, eg player long balls, player captain, player shots off target, etc. and from numerical summary these columns contain absolutely no information, therefore these columns must be removed

```
In [48]: # Data Cleaning
def remove_all_missing_columns(df):

    cols_to_drop = df.columns[df.isnull().all()]

    print(f"Columns removed : {len(cols_to_drop)}")

    return df.drop(columns=cols_to_drop), list(cols_to_drop)
```

```
In [49]: home_player, dropped_home = remove_all_missing_columns(home_player)

        away_player, dropped_away = remove_all_missing_columns(away_player)
```

```
Columns removed : 22
Columns removed : 22
```

```
In [50]: home_player.shape

        away_player.shape
```

```
Out[50]: (236132, 285)
```

```
In [51]: missing_summary(home_player).head(20)
```

Out[51]:

	Missing Count	Missing %	Da Ty
PLAYER_CAPTAIN_5_last_match_sum	219006	92.38	float
PLAYER_CAPTAIN_season_sum	211789	89.33	float
PLAYER_PUNCHES_season_std	170579	71.95	float
PLAYER_PUNCHES_5_last_match_std	170579	71.95	float
PLAYER_PUNCHES_season_sum	169599	71.54	float
PLAYER_PUNCHES_season_average	169599	71.54	float
PLAYER_PUNCHES_5_last_match_average	169599	71.54	float
PLAYER_PUNCHES_5_last_match_sum	169599	71.54	float
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std	95161	40.14	float
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std	81891	34.54	float
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average	77848	32.84	float
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average	70539	29.75	float
PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_std	56915	24.01	float
PLAYER_DRIBBLED_ATTEMPTS_5_last_match_std	56915	24.01	float
PLAYER_SUCCESSFUL_DRIBBLES_season_std	56915	24.01	float
PLAYER_DRIBBLED_PAST_5_last_match_std	56915	24.01	float
PLAYER_DRIBBLED_PAST_season_std	56915	24.01	float
PLAYER_DRIBBLED_ATTEMPTS_season_std	56915	24.01	float
PLAYER_SUCCESSFUL_DRIBBLES_season_average	54107	22.82	float
PLAYER_DRIBBLED_ATTEMPTS_season_sum	54107	22.82	float

In [52]: `missing_summary(away_player).head(20)`

Out[52]:

	Missing Count	Missing %	Da Ty
PLAYER_CAPTAIN_5_last_match_sum	218254	92.43	float
PLAYER_CAPTAIN_season_sum	211147	89.42	float
PLAYER_PUNCHES_season_std	170078	72.03	float
PLAYER_PUNCHES_5_last_match_std	170078	72.03	float
PLAYER_PUNCHES_season_sum	169063	71.60	float
PLAYER_PUNCHES_season_average	169063	71.60	float
PLAYER_PUNCHES_5_last_match_average	169063	71.60	float
PLAYER_PUNCHES_5_last_match_sum	169063	71.60	float
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std	95523	40.45	float
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std	82102	34.77	float
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average	78109	33.08	float
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average	70770	29.97	float
PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_std	57084	24.17	float
PLAYER_DRIBBLED_ATTEMPTS_5_last_match_std	57084	24.17	float
PLAYER_SUCCESSFUL_DRIBBLES_season_std	57084	24.17	float
PLAYER_DRIBBLED_PAST_5_last_match_std	57084	24.17	float
PLAYER_DRIBBLED_PAST_season_std	57084	24.17	float
PLAYER_DRIBBLED_ATTEMPTS_season_std	57084	24.17	float
PLAYER_SUCCESSFUL_DRIBBLES_season_average	54161	22.94	float
PLAYER_DRIBBLED_ATTEMPTS_season_sum	54161	22.94	float

We notice that the football specific variables for eg captain - only one player per team is captain so this variable is naturally sparse, punches - only goalkeepers perform punches so outfield players legitimately have missing values; so we don't drop these columns yet we make a more informed decision after we aggregate player data to match level.

```
In [53]: home_team_clean = home_team.copy()
         away_team_clean = away_team.copy()
         home_player_clean = home_player.copy()
         away_player_clean = away_player.copy()
```

```
In [54]: # Remove columns that are 100% missing
         home_team_clean, dropped_home_team = remove_all_missing_columns(home_team
```

```

away_team_clean, dropped_away_team = remove_all_missing_columns(away_team
home_player_clean, dropped_home_player = remove_all_missing_columns(home_
away_player_clean, dropped_away_player = remove_all_missing_columns(away_

```

```

Columns removed : 0
Columns removed : 0
Columns removed : 0
Columns removed : 0

```

6. Treating Missing Values

6.1.1 Home Team Dataset

```
In [55]: # CREATE A COPY OF THE HOME TEAM DATASET
```

```
home_team_processed = home_team_clean.copy()
```

```
In [56]: # Select numerical columns
```

```
numerical_columns = home_team_processed.select_dtypes(include=['number'])
```

```
print(f"Total numerical columns: {len(numerical_columns)}")
```

Total numerical columns: 141

```
In [57]: # Missing values before imputation
```

```
missing_before = home_team_processed[numerical_columns].isnull().sum()
```

```
# Display only columns with missing values
```

```
missing_before = missing_before[missing_before > 0].sort_values(ascending
```

```
print(f"Numerical columns with missing values: {len(missing_before)}")
```

```
missing_before.head(10)
```

Numerical columns with missing values: 118

```

Out[57]: TEAM_INJURIES_5_last_match_std          3318
         TEAM_INJURIES_5_last_match_sum          2932
         TEAM_INJURIES_5_last_match_average       2931
         TEAM_INJURIES_season_std                2924
         TEAM_INJURIES_season_sum                1960
         TEAM_INJURIES_season_average            1959
         TEAM_BALL_SAFE_5_last_match_std          1795
         TEAM_SUCCESSFUL_PASSES_PERCENTAGE_5_last_match_std 1589
         TEAM_PASSES_5_last_match_std            1589
         TEAM_SUCCESSFUL_PASSES_PERCENTAGE_season_std 1529
         dtype: int64

```

```
In [58]: from sklearn.impute import SimpleImputer

# Create the imputer
median_imputer = SimpleImputer(strategy='median')

# Apply median imputation only to numerical columns
home_team_processed[numerical_columns] = median_imputer.fit_transform(
    home_team_processed[numerical_columns]
)

In [59]: # Check remaining missing values
remaining_missing = (
    home_team_processed[numerical_columns]
    .isnull()
    .sum()
    .sum()
)

print(f"Remaining missing values in numerical columns: {remaining_missing}")
Remaining missing values in numerical columns: 0

In [60]: home_team_processed["TEAM_INJURIES_5_last_match_std"].isna().sum()

Out[60]: np.int64(0)

In [61]: home_team_processed["TEAM_INJURIES_5_last_match_std"].head(10)

Out[61]: 0    4.00
1    8.00
2    2.00
3    3.00
4    4.00
5    2.00
6    4.00
7    5.00
8    3.00
9    7.00
Name: TEAM_INJURIES_5_last_match_std, dtype: float64

In [62]: # Check for missing values in non-numerical (categorical/object) columns
categorical_columns = home_team_processed.select_dtypes(include=['object'])

home_team_processed[categorical_columns].isna().sum()

Out[62]: LEAGUE      0
TEAM_NAME    0
dtype: int64

In [63]: home_team_processed.isnull().sum().sum()

Out[63]: np.int64(0)
```

6.1.2 Away Team Dataset

```
In [64]: # Create a copy of the cleaned Away Team dataset
away_team_processed = away_team_clean.copy()
```

```
In [65]: # Select numerical columns
numerical_columns = away_team_processed.select_dtypes(include=['number'])

print(f"Total numerical columns: {len(numerical_columns)}")
```

Total numerical columns: 141

```
In [66]: # Missing values before imputation
missing_before = away_team_processed[numerical_columns].isnull().sum()

# Display only columns with missing values
missing_before = missing_before[missing_before > 0].sort_values(ascending

print(f"Numerical columns with missing values: {len(missing_before)}")
missing_before.head(10)
```

Numerical columns with missing values: 118

```
Out[66]: TEAM_INJURIES_5_last_match_std      3313
TEAM_INJURIES_5_last_match_sum      2933
TEAM_INJURIES_5_last_match_average      2932
TEAM_INJURIES_season_std      2926
TEAM_INJURIES_season_sum      1961
TEAM_INJURIES_season_average      1960
TEAM_BALL_SAFE_5_last_match_std      1836
TEAM_SUCCESSFUL_PASSES_PERCENTAGE_5_last_match_std      1588
TEAM_PASSES_5_last_match_std      1588
TEAM_SUCCESSFUL_PASSES_PERCENTAGE_season_std      1529
dtype: int64
```

```
In [67]: # Create the median imputer
median_imputer = SimpleImputer(strategy='median')

# Apply median imputation
away_team_processed[numerical_columns] = median_imputer.fit_transform(
    away_team_processed[numerical_columns]
)
```

```
In [68]: remaining_missing = (
    away_team_processed[numerical_columns]
    .isnull()
    .sum()
    .sum()
)

print(f"Remaining numerical missing values: {remaining_missing}")
```

Remaining numerical missing values: 0

```
In [69]: categorical_columns = away_team_processed.select_dtypes(
    include=['object', 'category']
).columns

print(away_team_processed[categorical_columns].isnull().sum())
```

```
LEAGUE      0
TEAM_NAME    0
dtype: int64
```

```
In [70]: print("Total missing values:", away_team_processed.isnull().sum().sum())
```

Total missing values: 0

6.1.3 Home Player Dataset

```
In [71]: # Percentage of missing values
missing_percent = (home_player.isnull().mean() * 100).sort_values(ascending=True)

# Show columns with more than 50% missing values
missing_percent[missing_percent > 50]
```

```
Out[71]: PLAYER_CAPTAIN_5_last_match_sum      92.38
PLAYER_CAPTAIN_season_sum      89.33
PLAYER_PUNCHES_5_last_match_std    71.95
PLAYER_PUNCHES_season_std    71.95
PLAYER_PUNCHES_5_last_match_sum    71.54
PLAYER_PUNCHES_season_sum    71.54
PLAYER_PUNCHES_5_last_match_average    71.54
PLAYER_PUNCHES_season_average    71.54
dtype: float64
```

```
In [72]: print(home_player["POSITION"].value_counts(dropna=False))
```

```
POSITION
midfielder    78524
defender      67746
NaN           42820
goalkeeper    24298
attacker      23691
Name: count, dtype: int64
```

```
In [73]: home_player[home_player["POSITION"].isnull()][
        ["PLAYER_NAME", "TEAM_NAME", "LEAGUE"]
        ].head(20)
```

Out[73]:

	PLAYER_NAME	TEAM_NAME	LEAGUE
4	Efthymios Koulouris	Toulouse	Ligue 1
9	Aaron Leya Iseka	Toulouse	Ligue 1
10	Yaya Sanogo	Toulouse	Ligue 1
17	Wesley Saïd	Toulouse	Ligue 1
21	Édouard Butin	Brest	Ligue 2
27	Valentin Henry	Brest	Ligue 2
31	Gaëtan Charbonnier	Brest	Ligue 2
97	Gonzalo Plata	Sporting CP	Liga Portugal
100	Jovane Cabral	Sporting CP	Liga Portugal
108	Andraž Šporar	Sporting CP	Liga Portugal
110	Luciano Vietto	Sporting CP	Liga Portugal
117	Bryan Gil	Sevilla	La Liga
119	Quincy Promes	Sevilla	La Liga
121	Munir El Haddadi	Sevilla	La Liga
126	André Silva	Sevilla	La Liga
128	Wissam Ben Yedder	Sevilla	La Liga
132	Josh Maja	Bordeaux	Ligue 1
134	Yann Karamoh	Bordeaux	Ligue 1
137	Nicolas de Préville	Bordeaux	Ligue 1
143	Jimmy Briand	Bordeaux	Ligue 1

```
In [74]: home_player.groupby("POSITION")[\n    "PLAYER_PUNCHES_season_sum"\n].apply(lambda x: x.notna().sum())
```

```
Out[74]: POSITION\nattacker      14097\ndefender       21141\ngoalkeeper     7298\nmidfielder    23908\nName: PLAYER_PUNCHES_season_sum, dtype: int64
```

```
In [75]: home_player.groupby("POSITION")[\n    "PLAYER_CAPTAIN_season_sum"\n].apply(lambda x: x.notna().sum())
```

```
Out[75]: POSITION
attacker      1605
defender      9516
goalkeeper    2105
midfielder    8124
Name: PLAYER_CAPTAIN_season_sum, dtype: int64
```

```
In [76]: home_player["PLAYER_PUNCHES_season_sum"].describe()
```

```
Out[76]: count    67480.00
mean         2.35
std         11.51
min          0.00
25%          0.00
50%          0.00
75%          0.00
max         100.00
Name: PLAYER_PUNCHES_season_sum, dtype: float64
```

```
In [77]: home_player["PLAYER_CAPTAIN_season_sum"].describe()
```

```
Out[77]: count    25290.00
mean         34.37
std         35.10
min          0.00
25%          2.00
50%         20.00
75%         66.00
max         100.00
Name: PLAYER_CAPTAIN_season_sum, dtype: float64
```

```
In [78]: home_player["PLAYER_CAPTAIN_5_last_match_sum"].describe()
```

```
Out[78]: count    18073.00
mean         52.43
std         41.23
min          0.00
25%          0.00
50%         50.00
75%        100.00
max        100.00
Name: PLAYER_CAPTAIN_5_last_match_sum, dtype: float64
```

```
In [79]: home_player.groupby("POSITION")["
        "PLAYER_CAPTAIN_5_last_match_sum"
        ].apply(lambda x: x.notna().sum())
```

```
Out[79]: POSITION
attacker      1038
defender      6922
goalkeeper    1483
midfielder    5805
Name: PLAYER_CAPTAIN_5_last_match_sum, dtype: int64
```

```
In [80]: home_player["PLAYER_PUNCHES_5_last_match_std"].describe()
```

```
Out[80]: count    66500.00
         mean       2.12
         std       11.19
         min        0.00
         25%        0.00
         50%        0.00
         75%        0.00
         max       100.00
         Name: PLAYER_PUNCHES_5_last_match_std, dtype: float64
```

```
In [81]: home_player.groupby("POSITION")["PLAYER_PUNCHES_5_last_match_std"]
         .apply(lambda x: x.notna().sum())
```

```
Out[81]: POSITION
         attacker    13868
         defender    20867
         goalkeeper    7200
         midfielder  23555
         Name: PLAYER_PUNCHES_5_last_match_std, dtype: int64
```

```
In [82]: home_player["PLAYER_PUNCHES_season_std"].describe()
```

```
Out[82]: count    66500.00
         mean       3.05
         std       13.32
         min        0.00
         25%        0.00
         50%        0.00
         75%        0.00
         max       100.00
         Name: PLAYER_PUNCHES_season_std, dtype: float64
```

```
In [83]: home_player.groupby("POSITION")["PLAYER_PUNCHES_season_std"]
         .apply(lambda x: x.notna().sum())
```

```
Out[83]: POSITION
         attacker    13868
         defender    20867
         goalkeeper    7200
         midfielder  23555
         Name: PLAYER_PUNCHES_season_std, dtype: int64
```

```
In [84]: home_player["PLAYER_PUNCHES_5_last_match_average"].describe()
```

```
Out[84]: count    67480.00
         mean       1.73
         std        9.85
         min        0.00
         25%        0.00
         50%        0.00
         75%        0.00
         max       100.00
         Name: PLAYER_PUNCHES_5_last_match_average, dtype: float64
```

```
In [85]: home_player.groupby("POSITION")["PLAYER_PUNCHES_5_last_match_average"]
         .apply(lambda x: x.notna().sum())
```



```
Out[85]: POSITION
attacker      14097
defender       21141
goalkeeper     7298
midfielder    23908
Name: PLAYER_PUNCHES_5_last_match_average, dtype: int64
```

```
In [86]: home_player["PLAYER_PUNCHES_season_average"].describe()
```

```
Out[86]: count      67480.00
mean           2.18
std            10.71
min             0.00
25%             0.00
50%             0.00
75%             0.00
max            100.00
Name: PLAYER_PUNCHES_season_average, dtype: float64
```

```
In [87]: home_player.groupby("POSITION")["
        "PLAYER_PUNCHES_season_average"
        ].apply(lambda x: x.notna().sum())
```

```
Out[87]: POSITION
attacker      14097
defender       21141
goalkeeper     7298
midfielder    23908
Name: PLAYER_PUNCHES_season_average, dtype: int64
```

Eight player-level variables with more than 70% missing values were removed prior to imputation. Two captain-related variables contained values only for players who served as team captains, meaning the missingness reflected the absence of the role rather than missing observations. Similarly, six goalkeeper-specific punching statistics exhibited more than 70% missing values and highly zero-inflated distributions (median = 0, 75th percentile = 0), indicating that the statistics were only relevant for a small subset of players. Imputing these variables would introduce artificial information and provide little predictive value. Consequently, these variables were excluded before median imputation was applied to the remaining numerical features.

```
In [88]: # Create a copy of the Home Player dataset
home_player_processed = home_player.copy()
```

```
In [89]: high_missing_cols = [
        "PLAYER_CAPTAIN_5_last_match_sum",
        "PLAYER_CAPTAIN_season_sum",
        "PLAYER_PUNCHES_5_last_match_std",
        "PLAYER_PUNCHES_season_std",
        "PLAYER_PUNCHES_5_last_match_sum",
        "PLAYER_PUNCHES_season_sum",
        "PLAYER_PUNCHES_5_last_match_average",
        "PLAYER_PUNCHES_season_average"
    ]

home_player_processed.drop(columns=high_missing_cols, inplace=True)
```

```
In [90]: print(f"Columns remaining: {home_player_processed.shape[1]}")

missing_percent = (
    home_player_processed.isnull().mean() * 100
).sort_values(ascending=False)

missing_percent.head(20)
```

Columns remaining: 277

```
Out[90]: PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std      40.14
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std      34.54
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average  32.84
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average  29.75
PLAYER_DRIBBLED_ATTEMPTS_5_last_match_std      24.01
PLAYER_DRIBBLED_ATTEMPTS_season_std      24.01
PLAYER_DRIBBLED_PAST_season_std      24.01
PLAYER_SUCCESSFUL_DRIBBLES_season_std      24.01
PLAYER_DRIBBLED_PAST_5_last_match_std      24.01
PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_std      24.01
PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_average  22.82
PLAYER_DRIBBLED_PAST_season_average      22.82
PLAYER_DRIBBLED_ATTEMPTS_season_average      22.82
PLAYER_SUCCESSFUL_DRIBBLES_season_sum      22.82
PLAYER_DRIBBLED_ATTEMPTS_5_last_match_sum      22.82
PLAYER_DRIBBLED_PAST_5_last_match_sum      22.82
PLAYER_SUCCESSFUL_DRIBBLES_season_average      22.82
PLAYER_DRIBBLED_ATTEMPTS_5_last_match_average  22.82
PLAYER_DRIBBLED_PAST_season_sum      22.82
PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_sum      22.82
dtype: float64
```

```
In [91]: missing_percent[missing_percent > 40]
```

```
Out[91]: PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std      40.14
dtype: float64
```

```
In [92]: # Columns with missing values between 13% and 22.82%
missing_13_22 = missing_percent[
    (missing_percent >= 13) & (missing_percent < 22.82)
]

print(f"Number of columns: {len(missing_13_22)}")

missing_13_22
```

Number of columns: 73

```

Out[92]:  PLAYER_PENALTIES_SAVED_5_last_match_std      22.05
          PLAYER_PENALTIES_SAVED_5_last_match_sum    20.79
          PLAYER_PENALTIES_SAVED_5_last_match_average 20.79
          POSITION                                     18.06
          PLAYER_PENALTIES_SAVED_season_std          16.57
          PLAYER_PENALTIES_SAVED_season_average      15.23
          PLAYER_PENALTIES_SAVED_season_sum          15.23
          PLAYER_CLEARANCE_OFFLINE_5_last_match_std  15.04
          PLAYER_RATING_5_last_match_std             14.55
          PLAYER_RATING_season_std                   14.55
          PLAYER_ERROR_LEAD_TO_GOAL_5_last_match_std 14.41
          PLAYER_PENALTIES_WON_5_last_match_std       14.29
          PLAYER_PENALTIES_COMMITTED_5_last_match_std 14.17
          PLAYER_ERROR_LEAD_TO_GOAL_season_std        14.06
          PLAYER_CLEARANCE_OFFLINE_season_std         14.05
          PLAYER_DUELS_LOST_season_std               14.02
          PLAYER_BLOCKED_SHOTS_5_last_match_std       14.02
          PLAYER_BLOCKED_SHOTS_season_std            14.02
          PLAYER_BIG_CHANCES_CREATED_season_std       14.02
          PLAYER_BIG_CHANCES_CREATED_5_last_match_std 14.02
          PLAYER_BIG_CHANCES_MISSED_5_last_match_std  14.02
          PLAYER_DUELS_LOST_5_last_match_std          14.02
          PLAYER_BIG_CHANCES_MISSED_season_std        14.02
          PLAYER_OFFSIDES_season_std                 13.96
          PLAYER_OFFSIDES_5_last_match_std           13.96
          PLAYER_PENALTIES_WON_season_std            13.78
          PLAYER_PENALTIES_COMMITTED_season_std      13.77
          PLAYER_CLEARANCE_OFFLINE_5_last_match_sum   13.64
          PLAYER_CLEARANCE_OFFLINE_5_last_match_average 13.64
          PLAYER_HIT_WOODWORK_5_last_match_std       13.51
          PLAYER_HIT_WOODWORK_season_std             13.38
          PLAYER_ACCURATE_PASSES_season_std          13.37
          PLAYER_ACCURATE_CROSSES_season_std         13.37
          PLAYER_KEY_PASSES_5_last_match_std         13.37
          PLAYER_DISPOSSESSED_5_last_match_std        13.37
          PLAYER_ACCURATE_CROSSES_5_last_match_std    13.37
          PLAYER_AERIALS_WON_season_std              13.37
          PLAYER_INTERCEPTIONS_5_last_match_std      13.37
          PLAYER_DUELS_WON_5_last_match_std          13.37
          PLAYER_TOTAL_DUELS_season_std              13.37
          PLAYER_FOULS_5_last_match_std              13.37
          PLAYER_TACKLES_season_std                  13.37
          PLAYER_CLEARANCES_season_std               13.37
          PLAYER_SHOTS_TOTAL_season_std               13.37
          PLAYER_DISPOSSESSED_season_std              13.37
          PLAYER_FOULS_DRAWN_5_last_match_std        13.37
          PLAYER_PASSES_season_std                   13.37
          PLAYER_SHOTS_ON_TARGET_season_std           13.37
          PLAYER_ACCURATE_PASSES_5_last_match_std    13.37
          PLAYER_AERIALS_WON_5_last_match_std        13.37
          PLAYER_DUELS_WON_season_std                 13.37
          PLAYER_SHOTS_BLOCKED_season_std             13.37
          PLAYER_FOULS_season_std                    13.37
          PLAYER_FOULS_DRAWN_season_std              13.37
          PLAYER_INTERCEPTIONS_season_std             13.37
          PLAYER_SAVES_season_std                    13.37
          PLAYER_KEY_PASSES_season_std               13.37
          PLAYER_CLEARANCES_5_last_match_std         13.37
          PLAYER_SAVES_INSIDE_BOX_season_std         13.37
          PLAYER_TOTAL_CROSSES_season_std            13.37

```

PLAYER_TACKLES_5_last_match_std	13.37
PLAYER_SAVES_5_last_match_std	13.37
PLAYER_SAVES_INSIDE_BOX_5_last_match_std	13.37
PLAYER_SHOTS_TOTAL_5_last_match_std	13.37
PLAYER_SHOTS_BLOCKED_5_last_match_std	13.37
PLAYER_TOTAL_DUELS_5_last_match_std	13.37
PLAYER_PASSES_5_last_match_std	13.37
PLAYER_SHOTS_ON_TARGET_5_last_match_std	13.37
PLAYER_TOTAL_CROSSES_5_last_match_std	13.37
PLAYER_RATING_season_average	13.15
PLAYER_RATING_5_last_match_average	13.15
PLAYER_ERROR_LEAD_TO_GOAL_5_last_match_average	13.00
PLAYER_ERROR_LEAD_TO_GOAL_5_last_match_sum	13.00

dtype: float64

```
In [93]: # Columns with missing values between 12% and 13%
missing_12_13 = missing_percent[
    (missing_percent >= 12) & (missing_percent < 13)
]

print(f"Number of columns: {len(missing_12_13)}")

missing_12_13
```

Number of columns: 34

```
Out[93]:
```

PLAYER_PENALTIES_WON_5_last_match_sum	12.88
PLAYER_PENALTIES_WON_5_last_match_average	12.88
PLAYER_PENALTIES_COMMITTED_5_last_match_average	12.76
PLAYER_PENALTIES_COMMITTED_5_last_match_sum	12.76
PLAYER_ERROR_LEAD_TO_GOAL_season_average	12.65
PLAYER_ERROR_LEAD_TO_GOAL_season_sum	12.65
PLAYER_CLEARANCE_OFFLINE_season_sum	12.63
PLAYER_CLEARANCE_OFFLINE_season_average	12.63
PLAYER_BIG_CHANCES_MISSED_5_last_match_average	12.61
PLAYER_DUELS_LOST_5_last_match_average	12.61
PLAYER_BIG_CHANCES_CREATED_season_sum	12.61
PLAYER_BIG_CHANCES_MISSED_season_sum	12.61
PLAYER_BIG_CHANCES_CREATED_5_last_match_sum	12.61
PLAYER_BIG_CHANCES_CREATED_5_last_match_average	12.61
PLAYER_DUELS_LOST_season_sum	12.61
PLAYER_BLOCKED_SHOTS_season_sum	12.61
PLAYER_DUELS_LOST_5_last_match_sum	12.61
PLAYER_BLOCKED_SHOTS_5_last_match_sum	12.61
PLAYER_BIG_CHANCES_MISSED_5_last_match_sum	12.61
PLAYER_BIG_CHANCES_CREATED_season_average	12.61
PLAYER_BLOCKED_SHOTS_5_last_match_average	12.61
PLAYER_BIG_CHANCES_MISSED_season_average	12.61
PLAYER_DUELS_LOST_season_average	12.61
PLAYER_BLOCKED_SHOTS_season_average	12.61
PLAYER_OFFSIDES_season_average	12.55
PLAYER_OFFSIDES_season_sum	12.55
PLAYER_OFFSIDES_5_last_match_average	12.55
PLAYER_OFFSIDES_5_last_match_sum	12.55
PLAYER_PENALTIES_WON_season_sum	12.36
PLAYER_PENALTIES_WON_season_average	12.36
PLAYER_PENALTIES_COMMITTED_season_sum	12.35
PLAYER_PENALTIES_COMMITTED_season_average	12.35
PLAYER_HIT_WOODWORK_5_last_match_average	12.08
PLAYER_HIT_WOODWORK_5_last_match_sum	12.08

dtype: float64

```
In [94]: # Columns with missing values between 10% and 12%
missing_10_12 = missing_percent[
    (missing_percent >= 10) & (missing_percent < 12)
]

print(f"Number of columns: {len(missing_10_12)}")

missing_10_12
```

Number of columns: 80

```

Out[94]:  PLAYER_HIT_WOODWORK_season_average 11.95
          PLAYER_HIT_WOODWORK_season_sum 11.95
          PLAYER_SHOTS_TOTAL_5_last_match_sum 11.93
          PLAYER_SAVES_5_last_match_average 11.93
          PLAYER_SHOTS_ON_TARGET_5_last_match_sum 11.93
          PLAYER_SHOTS_BLOCKED_5_last_match_sum 11.93
          PLAYER_SAVES_INSIDE_BOX_5_last_match_sum 11.93
          PLAYER_SAVES_INSIDE_BOX_5_last_match_average 11.93
          PLAYER_PASSES_5_last_match_sum 11.93
          PLAYER_TOTAL_CROSSES_5_last_match_average 11.93
          PLAYER_FOULS_season_sum 11.93
          PLAYER_SHOTS_BLOCKED_5_last_match_average 11.93
          PLAYER_TOTAL_DUELS_5_last_match_average 11.93
          PLAYER_SAVES_5_last_match_sum 11.93
          PLAYER_DUELS_WON_season_sum 11.93
          PLAYER_SHOTS_ON_TARGET_5_last_match_average 11.93
          PLAYER_SHOTS_TOTAL_5_last_match_average 11.93
          PLAYER_CLEARANCES_5_last_match_average 11.93
          PLAYER_FOULS_DRAWN_5_last_match_average 11.93
          PLAYER_FOULS_5_last_match_average 11.93
          PLAYER_FOULS_DRAWN_season_sum 11.93
          PLAYER_AERIALS_WON_season_sum 11.93
          PLAYER_INTERCEPTIONS_5_last_match_average 11.93
          PLAYER_AERIALS_WON_5_last_match_average 11.93
          PLAYER_CLEARANCES_season_sum 11.93
          PLAYER_ACCURATE_PASSES_5_last_match_average 11.93
          PLAYER_ACCURATE_CROSSES_5_last_match_average 11.93
          PLAYER_KEY_PASSES_5_last_match_average 11.93
          PLAYER_TOTAL_DUELS_5_last_match_sum 11.93
          PLAYER_ACCURATE_PASSES_season_sum 11.93
          PLAYER_DISPOSSESSED_5_last_match_average 11.93
          PLAYER_TOTAL_CROSSES_5_last_match_sum 11.93
          PLAYER_ACCURATE_CROSSES_season_sum 11.93
          PLAYER_DUELS_WON_5_last_match_average 11.93
          PLAYER_DISPOSSESSED_season_sum 11.93
          PLAYER_PASSES_5_last_match_average 11.93
          PLAYER_TACKLES_5_last_match_sum 11.93
          PLAYER_DUELS_WON_5_last_match_sum 11.93
          PLAYER_SHOTS_TOTAL_season_sum 11.93
          PLAYER_INTERCEPTIONS_season_average 11.93
          PLAYER_TACKLES_season_sum 11.93
          PLAYER_TACKLES_season_average 11.93
          PLAYER_TOTAL_CROSSES_season_sum 11.93
          PLAYER_TOTAL_DUELS_season_sum 11.93
          PLAYER_TOTAL_CROSSES_season_average 11.93
          PLAYER_ACCURATE_CROSSES_season_average 11.93
          PLAYER_KEY_PASSES_5_last_match_sum 11.93
          PLAYER_ACCURATE_PASSES_season_average 11.93
          PLAYER_AERIALS_WON_season_average 11.93
          PLAYER_TOTAL_DUELS_season_average 11.93
          PLAYER_TACKLES_5_last_match_average 11.93
          PLAYER_CLEARANCES_season_average 11.93
          PLAYER_PASSES_season_average 11.93
          PLAYER_KEY_PASSES_season_average 11.93
          PLAYER_DISPOSSESSED_season_average 11.93
          PLAYER_DUELS_WON_season_average 11.93
          PLAYER_FOULS_season_average 11.93
          PLAYER_SHOTS_TOTAL_season_average 11.93
          PLAYER_FOULS_DRAWN_season_average 11.93
          PLAYER_SHOTS_ON_TARGET_season_average 11.93

```

PLAYER_DISPOSSESSED_5_last_match_sum	11.93
PLAYER_SAVES_season_average	11.93
PLAYER_INTERCEPTIONS_5_last_match_sum	11.93
PLAYER_SAVES_INSIDE_BOX_season_average	11.93
PLAYER_SHOTS_BLOCKED_season_average	11.93
PLAYER_FOULS_DRAWN_5_last_match_sum	11.93
PLAYER_FOULS_5_last_match_sum	11.93
PLAYER_INTERCEPTIONS_season_sum	11.93
PLAYER_KEY_PASSES_season_sum	11.93
PLAYER_CLEARANCES_5_last_match_sum	11.93
PLAYER_SHOTS_ON_TARGET_season_sum	11.93
PLAYER_AERIALS_WON_5_last_match_sum	11.93
PLAYER_ACCURATE_PASSES_5_last_match_sum	11.93
PLAYER_ACCURATE_CROSSES_5_last_match_sum	11.93
PLAYER_PASSES_season_sum	11.93
PLAYER_SAVES_season_sum	11.93
PLAYER_SAVES_INSIDE_BOX_season_sum	11.93
PLAYER_SHOTS_BLOCKED_season_sum	11.93
PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_std	10.04
PLAYER_GOALKEEPER_GOALS_CONCEDED_season_std	10.04

dtype: float64

```
In [95]: # Columns with missing values between 0% and 10%
missing_0_10 = missing_percent[
    (missing_percent >= 0) & (missing_percent < 10)
]

print(f"Number of columns: {len(missing_0_10)}")

missing_0_10
```

Number of columns: 68

```

Out[95]:  PLAYER_GOALKEEPER_GOALS_CONCEDED_season_sum      8.57
          PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_average 8.57
          PLAYER_GOALKEEPER_GOALS_CONCEDED_season_average      8.57
          PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_sum    8.57
          PLAYER_PENALTIES_MISSES_5_last_match_std             5.65
          PLAYER_REDCARDS_5_last_match_std                     4.61
          PLAYER_PENALTIES_MISSES_5_last_match_sum             4.08
          PLAYER_PENALTIES_MISSES_5_last_match_average         4.08
          PLAYER_OWN_GOALS_5_last_match_std                     3.65
          PLAYER_PENALTIES_MISSES_season_std                   3.60
          PLAYER_REDCARDS_season_std                           3.42
          PLAYER_OWN_GOALS_season_std                           3.24
          PLAYER_STARTING_LINEUP_5_last_match_std              3.14
          PLAYER_PENALTIES_SCORED_5_last_match_std              3.14
          PLAYER_MINUTES_PLAYED_5_last_match_std                3.14
          PLAYER_GOALS_CONCEDED_5_last_match_std                3.14
          PLAYER_ASSISTS_5_last_match_std                       3.14
          PLAYER_GOALS_5_last_match_std                         3.14
          PLAYER_YELLOWCARDS_5_last_match_std                   3.14
          PLAYER_GOALS_season_std                               3.14
          PLAYER_ASSISTS_season_std                             3.14
          PLAYER_GOALS_CONCEDED_season_std                     3.14
          PLAYER_MINUTES_PLAYED_season_std                     3.14
          PLAYER_PENALTIES_SCORED_season_std                   3.14
          PLAYER_STARTING_LINEUP_season_std                     3.14
          PLAYER_YELLOWCARDS_season_std                         3.14
          PLAYER_REDCARDS_5_last_match_average                 3.04
          PLAYER_REDCARDS_5_last_match_sum                     3.04
          PLAYER_OWN_GOALS_5_last_match_sum                     2.04
          PLAYER_OWN_GOALS_5_last_match_average                 2.04
          PLAYER_PENALTIES_MISSES_season_average               1.99
          PLAYER_PENALTIES_MISSES_season_sum                   1.99
          PLAYER_REDCARDS_season_average                       1.82
          PLAYER_REDCARDS_season_sum                           1.82
          PLAYER_OWN_GOALS_season_sum                           1.63
          PLAYER_OWN_GOALS_season_average                       1.63
          PLAYER_PENALTIES_SCORED_season_average               1.52
          PLAYER_ASSISTS_season_average                         1.52
          PLAYER_GOALS_season_average                           1.52
          PLAYER_GOALS_CONCEDED_season_average                 1.52
          PLAYER_STARTING_LINEUP_season_sum                     1.52
          PLAYER_MINUTES_PLAYED_season_average                 1.52
          PLAYER_PENALTIES_SCORED_season_sum                   1.52
          PLAYER_MINUTES_PLAYED_season_sum                     1.52
          PLAYER_GOALS_CONCEDED_season_sum                     1.52
          PLAYER_GOALS_season_sum                               1.52
          PLAYER_ASSISTS_season_sum                             1.52
          PLAYER_YELLOWCARDS_season_sum                         1.52
          PLAYER_STARTING_LINEUP_5_last_match_average           1.52
          PLAYER_STARTING_LINEUP_season_average                 1.52
          PLAYER_YELLOWCARDS_season_average                     1.52
          PLAYER_YELLOWCARDS_5_last_match_average               1.52
          PLAYER_PENALTIES_SCORED_5_last_match_average           1.52
          PLAYER_MINUTES_PLAYED_5_last_match_average             1.52
          PLAYER_GOALS_CONCEDED_5_last_match_average             1.52
          PLAYER_GOALS_5_last_match_average                     1.52
          PLAYER_ASSISTS_5_last_match_average                   1.52
          PLAYER_YELLOWCARDS_5_last_match_sum                   1.52
          PLAYER_STARTING_LINEUP_5_last_match_sum               1.52
          PLAYER_PENALTIES_SCORED_5_last_match_sum               1.52

```


PLAYER_MINUTES_PLAYED_5_last_match_sum	1.52
PLAYER_GOALS_CONCEDED_5_last_match_sum	1.52
PLAYER_GOALS_5_last_match_sum	1.52
PLAYER_ASSISTS_5_last_match_sum	1.52
LEAGUE	0.00
PLAYER_NAME	0.00
TEAM_NAME	0.00
ID	0.00

dtype: float64

Investigating Goalkeeper Family Statistics

```
In [96]: goalkeeper_cols = [col for col in home_player_processed.columns
                             if "GOALKEEPER" in col]

print(goalkeeper_cols)

['PLAYER_GOALKEEPER_GOALS_CONCEDED_season_sum', 'PLAYER_GOALKEEPER_GOALS_CONCEDED_season_average', 'PLAYER_GOALKEEPER_GOALS_CONCEDED_season_std', 'PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_sum', 'PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_average', 'PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_std']
```

```
In [97]: missing_percent[goalkeeper_cols].sort_values(ascending=False)
```

```
Out[97]: PLAYER_GOALKEEPER_GOALS_CONCEDED_season_std      10.04
         PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_std  10.04
         PLAYER_GOALKEEPER_GOALS_CONCEDED_season_sum       8.57
         PLAYER_GOALKEEPER_GOALS_CONCEDED_season_average   8.57
         PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_sum  8.57
         PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_average 8.57
         dtype: float64
```

```
In [98]: goalkeeper_rows = home_player_processed[
         home_player_processed["POSITION"] == "goalkeeper"
         ]

goalkeeper_rows[goalkeeper_cols].isnull().mean() * 100
```

```
Out[98]: PLAYER_GOALKEEPER_GOALS_CONCEDED_season_sum      3.50
         PLAYER_GOALKEEPER_GOALS_CONCEDED_season_average  3.50
         PLAYER_GOALKEEPER_GOALS_CONCEDED_season_std     4.85
         PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_sum 3.50
         PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_average 3.50
         PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_std 4.85
         dtype: float64
```

```
In [99]: non_goalkeeper_rows = home_player_processed[
         home_player_processed["POSITION"] != "goalkeeper"
         ]

non_goalkeeper_rows[goalkeeper_cols].isnull().mean() * 100
```

```
Out[99]: PLAYER_GOALKEEPER_GOALS_CONCEDED_season_sum      9.15
         PLAYER_GOALKEEPER_GOALS_CONCEDED_season_average  9.15
         PLAYER_GOALKEEPER_GOALS_CONCEDED_season_std     10.63
         PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_sum 9.15
         PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_average 9.15
         PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_match_std 10.63
         dtype: float64
```

```
In [100... # Numerical columns
numerical_cols = home_player_processed.select_dtypes(include=['number']).

# Categorical columns
categorical_cols = home_player_processed.select_dtypes(include=['object',

print("Numerical columns:", len(numerical_cols))
print("Categorical columns:", len(categorical_cols))
```

Numerical columns: 273
Categorical columns: 4

```
In [101... home_player_processed[categorical_cols].isnull().sum().sort_values(ascend
```

```
Out[101... POSITION          42820
LEAGUE              0
TEAM_NAME           0
PLAYER_NAME         0
dtype: int64
```

```
In [102... missing_numeric = (
    home_player_processed[numerical_cols]
    .isnull()
    .mean() * 100
).sort_values(ascending=False)

missing_numeric.head(30)
```

```
Out[102... PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std      40.14
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std            34.54
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average  32.84
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average        29.75
PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_std              24.01
PLAYER_DRIBBLED_PAST_5_last_match_std                    24.01
PLAYER_DRIBBLED_ATTEMPTS_5_last_match_std                24.01
PLAYER_DRIBBLED_PAST_season_std                          24.01
PLAYER_SUCCESSFUL_DRIBBLES_season_std                    24.01
PLAYER_DRIBBLED_ATTEMPTS_season_std                     24.01
PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_sum              22.82
PLAYER_DRIBBLED_PAST_5_last_match_sum                    22.82
PLAYER_DRIBBLED_ATTEMPTS_5_last_match_sum                22.82
PLAYER_SUCCESSFUL_DRIBBLES_season_sum                   22.82
PLAYER_DRIBBLED_ATTEMPTS_5_last_match_average            22.82
PLAYER_DRIBBLED_PAST_season_sum                          22.82
PLAYER_DRIBBLED_PAST_5_last_match_average                22.82
PLAYER_DRIBBLED_PAST_season_average                     22.82
PLAYER_SUCCESSFUL_DRIBBLES_season_average                22.82
PLAYER_DRIBBLED_ATTEMPTS_season_average                 22.82
PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_average          22.82
PLAYER_DRIBBLED_ATTEMPTS_season_sum                     22.82
PLAYER_PENALTIES_SAVED_5_last_match_std                  22.05
PLAYER_PENALTIES_SAVED_5_last_match_sum                  20.79
PLAYER_PENALTIES_SAVED_5_last_match_average              20.79
PLAYER_PENALTIES_SAVED_season_std                        16.57
PLAYER_PENALTIES_SAVED_season_sum                        15.23
PLAYER_PENALTIES_SAVED_season_average                    15.23
PLAYER_CLEARANCE_OFFLINE_5_last_match_std                15.04
PLAYER_RATING_season_std                                 14.55
dtype: float64
```

```
In [103... # Accurate Passes
```

```
In [104... pass_cols = [col for col in home_player_processed.columns
                if "ACCURATE_PASSES_PERCENTAGE" in col]

pass_cols
```

```
Out[104... ['PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average',
            'PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std',
            'PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average',
            'PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std']
```

```
In [105... missing_numeric[pass_cols]
```

```
Out[105... PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average    29.75
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std              34.54
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average    32.84
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std        40.14
dtype: float64
```

```
In [106... home_player_processed[pass_cols].describe()
```

```
Out[106...      PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average  PLAYER_ACCURATE_PA
```

count	166540.00
mean	64.77
std	15.33
min	0.00
25%	57.00
50%	67.00
75%	75.00
max	100.00

```
In [107... for col in pass_cols:
    print(f"\n{col}")
    print(
        home_player_processed
        .groupby("POSITION")[col]
        .apply(lambda x: x.notna().sum())
    )
```

```

PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average
POSITION
attacker      19877
defender      51714
goalkeeper    13056
midfielder    61019
Name: PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average, dtype: int64

```

```

PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std
POSITION
attacker      19052
defender      48239
goalkeeper    11544
midfielder    57578
Name: PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std, dtype: int64

```

```

PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average
POSITION
attacker      19549
defender      49534
goalkeeper    10999
midfielder    59573
Name: PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average, dtype: int64

```

```

PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std
POSITION
attacker      18038
defender      43686
goalkeeper     9583
midfielder    53913
Name: PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std, dtype: int64

```

```
In [108... home_player_processed.groupby("POSITION")[pass_cols].median()
```

```
Out[108...      PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average  PLAYER_ACCURA'
```

POSITION	
attacker	62.00
defender	69.00
goalkeeper	46.00
midfielder	70.00

```
In [109... for col in pass_cols:
    home_player_processed[col] = (
        home_player_processed
        .groupby("POSITION")[col]
        .transform(lambda x: x.fillna(x.median()))
    )
```

```
In [110... home_player_processed[pass_cols].isnull().sum()
```

```
Out[110...] PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average      42820
            PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std    42820
            PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average  42820
            PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std    42820
            dtype: int64
```

```
In [111...] home_player_processed.loc[
            home_player_processed["POSITION"].isna(),
            pass_cols
        ].isnull().sum()
```

```
Out[111...] PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average      42820
            PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std    42820
            PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average  42820
            PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std    42820
            dtype: int64
```

```
In [112...] # Number of rows with missing POSITION
            home_player_processed["POSITION"].isnull().sum()
```

```
Out[112...] np.int64(42820)
```

```
In [113...] # Look at a few of those rows
            home_player_processed[
                home_player_processed["POSITION"].isnull()
            ].head(10)
```

```
Out[113...]      ID  LEAGUE  TEAM_NAME  POSITION  PLAYER_NAME  PLAYER_ACCURATE_CROSS
```

4	0	Ligue 1	Toulouse	NaN	Efthymios Koulouris	
9	0	Ligue 1	Toulouse	NaN	Aaron Leya Iseka	
10	0	Ligue 1	Toulouse	NaN	Yaya Sanogo	
17	0	Ligue 1	Toulouse	NaN	Wesley Saïd	
21	1	Ligue 2	Brest	NaN	Édouard Butin	
27	1	Ligue 2	Brest	NaN	Valentin Henry	
31	1	Ligue 2	Brest	NaN	Gaëtan Charbonnier	
97	5	Liga Portugal	Sporting CP	NaN	Gonzalo Plata	
100	5	Liga Portugal	Sporting CP	NaN	Jovane Cabral	
108	5	Liga Portugal	Sporting CP	NaN	Andraž Šporar	

```
In [114...] home_player_processed["POSITION"].value_counts(dropna=False)
```

```
Out[114...] POSITION
midfielder      78524
defender        67746
NaN             42820
goalkeeper      24298
attacker        23691
Name: count, dtype: int64
```

```
In [115...] [col for col in home_player_processed.columns if "POSITION" in col.upper()
```

```
Out[115...] ['POSITION']
```

```
In [116...] for col in pass_cols:

    # Step 1: position-wise median
    home_player_processed[col] = (
        home_player_processed
        .groupby("POSITION")[col]
        .transform(lambda x: x.fillna(x.median()))
    )

    # Step 2: remaining NaNs (missing POSITION rows)
    home_player_processed[col] = (
        home_player_processed[col]
        .fillna(home_player_processed[col].median())
    )
```

```
In [117...] home_player_processed[pass_cols].isnull().sum()
```

```
Out[117...] PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average      0
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std                0
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average      0
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std          0
dtype: int64
```

```
In [118...] # Dribble
```

```
In [119...] dribble_cols = [
    col for col in home_player_processed.columns
    if "DRIBBL" in col.upper()
]

dribble_cols
```

```
Out[119...] ['PLAYER_DRIBBLED_ATTEMPTS_season_sum',
'PLAYER_DRIBBLED_PAST_season_sum',
'PLAYER_SUCCESSFUL_DRIBBLES_season_sum',
'PLAYER_DRIBBLED_ATTEMPTS_season_average',
'PLAYER_DRIBBLED_PAST_season_average',
'PLAYER_SUCCESSFUL_DRIBBLES_season_average',
'PLAYER_DRIBBLED_ATTEMPTS_season_std',
'PLAYER_DRIBBLED_PAST_season_std',
'PLAYER_SUCCESSFUL_DRIBBLES_season_std',
'PLAYER_DRIBBLED_ATTEMPTS_5_last_match_sum',
'PLAYER_DRIBBLED_PAST_5_last_match_sum',
'PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_sum',
'PLAYER_DRIBBLED_ATTEMPTS_5_last_match_average',
'PLAYER_DRIBBLED_PAST_5_last_match_average',
'PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_average',
'PLAYER_DRIBBLED_ATTEMPTS_5_last_match_std',
'PLAYER_DRIBBLED_PAST_5_last_match_std',
'PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_std']
```

```
In [120...] missing_numeric[dribble_cols]
```

```
Out[120...] PLAYER_DRIBBLED_ATTEMPTS_season_sum      22.82
PLAYER_DRIBBLED_PAST_season_sum              22.82
PLAYER_SUCCESSFUL_DRIBBLES_season_sum         22.82
PLAYER_DRIBBLED_ATTEMPTS_season_average       22.82
PLAYER_DRIBBLED_PAST_season_average           22.82
PLAYER_SUCCESSFUL_DRIBBLES_season_average     22.82
PLAYER_DRIBBLED_ATTEMPTS_season_std           24.01
PLAYER_DRIBBLED_PAST_season_std               24.01
PLAYER_SUCCESSFUL_DRIBBLES_season_std         24.01
PLAYER_DRIBBLED_ATTEMPTS_5_last_match_sum     22.82
PLAYER_DRIBBLED_PAST_5_last_match_sum         22.82
PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_sum   22.82
PLAYER_DRIBBLED_ATTEMPTS_5_last_match_average 22.82
PLAYER_DRIBBLED_PAST_5_last_match_average     22.82
PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_average 22.82
PLAYER_DRIBBLED_ATTEMPTS_5_last_match_std     24.01
PLAYER_DRIBBLED_PAST_5_last_match_std         24.01
PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_std   24.01
dtype: float64
```

```
In [121...] home_player_processed[dribble_cols].describe()
```

```
Out[121...]      PLAYER_DRIBBLED_ATTEMPTS_season_sum  PLAYER_DRIBBLED_PAST_season_sum
```

count	182972.00	182972.00
mean	11.76	17.29
std	16.13	19.07
min	0.00	0.00
25%	0.00	1.00
50%	6.00	12.00
75%	17.00	26.00
max	100.00	100.00

```
In [122... for col in dribble_cols:
            print(f"\n{col}")
            print(
                home_player_processed
                .groupby("POSITION")[col]
                .apply(lambda x: x.notna().sum())
            )
```


PLAYER_DRIBBLED_ATTEMPTS_season_sum
POSITION
attacker 20493
defender 55405
goalkeeper 19827
midfielder 64147
Name: PLAYER_DRIBBLED_ATTEMPTS_season_sum, dtype: int64

PLAYER_DRIBBLED_PAST_season_sum
POSITION
attacker 20493
defender 55405
goalkeeper 19827
midfielder 64147
Name: PLAYER_DRIBBLED_PAST_season_sum, dtype: int64

PLAYER_SUCCESSFUL_DRIBBLES_season_sum
POSITION
attacker 20493
defender 55405
goalkeeper 19827
midfielder 64147
Name: PLAYER_SUCCESSFUL_DRIBBLES_season_sum, dtype: int64

PLAYER_DRIBBLED_ATTEMPTS_season_average
POSITION
attacker 20493
defender 55405
goalkeeper 19827
midfielder 64147
Name: PLAYER_DRIBBLED_ATTEMPTS_season_average, dtype: int64

PLAYER_DRIBBLED_PAST_season_average
POSITION
attacker 20493
defender 55405
goalkeeper 19827
midfielder 64147
Name: PLAYER_DRIBBLED_PAST_season_average, dtype: int64

PLAYER_SUCCESSFUL_DRIBBLES_season_average
POSITION
attacker 20493
defender 55405
goalkeeper 19827
midfielder 64147
Name: PLAYER_SUCCESSFUL_DRIBBLES_season_average, dtype: int64

PLAYER_DRIBBLED_ATTEMPTS_season_std
POSITION
attacker 20141
defender 54614
goalkeeper 19561
midfielder 63205
Name: PLAYER_DRIBBLED_ATTEMPTS_season_std, dtype: int64

PLAYER_DRIBBLED_PAST_season_std
POSITION
attacker 20141
defender 54614

goalkeeper 19561
midfielder 63205
Name: PLAYER_DRIBBLED_PAST_season_std, dtype: int64

PLAYER_SUCCESSFUL_DRIBBLES_season_std
POSITION
attacker 20141
defender 54614
goalkeeper 19561
midfielder 63205
Name: PLAYER_SUCCESSFUL_DRIBBLES_season_std, dtype: int64

PLAYER_DRIBBLED_ATTEMPTS_5_last_match_sum
POSITION
attacker 20493
defender 55405
goalkeeper 19827
midfielder 64147
Name: PLAYER_DRIBBLED_ATTEMPTS_5_last_match_sum, dtype: int64

PLAYER_DRIBBLED_PAST_5_last_match_sum
POSITION
attacker 20493
defender 55405
goalkeeper 19827
midfielder 64147
Name: PLAYER_DRIBBLED_PAST_5_last_match_sum, dtype: int64

PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_sum
POSITION
attacker 20493
defender 55405
goalkeeper 19827
midfielder 64147
Name: PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_sum, dtype: int64

PLAYER_DRIBBLED_ATTEMPTS_5_last_match_average
POSITION
attacker 20493
defender 55405
goalkeeper 19827
midfielder 64147
Name: PLAYER_DRIBBLED_ATTEMPTS_5_last_match_average, dtype: int64

PLAYER_DRIBBLED_PAST_5_last_match_average
POSITION
attacker 20493
defender 55405
goalkeeper 19827
midfielder 64147
Name: PLAYER_DRIBBLED_PAST_5_last_match_average, dtype: int64

PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_average
POSITION
attacker 20493
defender 55405
goalkeeper 19827
midfielder 64147
Name: PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_average, dtype: int64

```

PLAYER_DRIBBLED_ATTEMPTS_5_last_match_std
POSITION
attacker      20141
defender      54614
goalkeeper    19561
midfielder    63205
Name: PLAYER_DRIBBLED_ATTEMPTS_5_last_match_std, dtype: int64

```

```

PLAYER_DRIBBLED_PAST_5_last_match_std
POSITION
attacker      20141
defender      54614
goalkeeper    19561
midfielder    63205
Name: PLAYER_DRIBBLED_PAST_5_last_match_std, dtype: int64

```

```

PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_std
POSITION
attacker      20141
defender      54614
goalkeeper    19561
midfielder    63205
Name: PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_std, dtype: int64

```

```
In [123...] home_player_processed.groupby("POSITION")[dribble_cols].median()
```

```
Out[123...]      PLAYER_DRIBBLED_ATTEMPTS_season_sum  PLAYER_DRIBBLED_PAST_season_sum
```

POSITION	
attacker	12.00
defender	3.00
goalkeeper	0.00
midfielder	11.00

```

In [124...] for col in dribble_cols:

    # Position-wise median
    home_player_processed[col] = (
        home_player_processed
        .groupby("POSITION")[col]
        .transform(lambda x: x.fillna(x.median()))
    )

    # Remaining NaNs (missing POSITION)
    home_player_processed[col] = (
        home_player_processed[col]
        .fillna(home_player_processed[col].median())
    )

```

```
In [125...] home_player_processed[dribble_cols].isnull().sum()
```

```

Out[125...  PLAYER_DRIBBLED_ATTEMPTS_season_sum      0
           PLAYER_DRIBBLED_PAST_season_sum    0
           PLAYER_SUCCESSFUL_DRIBBLES_season_sum  0
           PLAYER_DRIBBLED_ATTEMPTS_season_average  0
           PLAYER_DRIBBLED_PAST_season_average  0
           PLAYER_SUCCESSFUL_DRIBBLES_season_average  0
           PLAYER_DRIBBLED_ATTEMPTS_season_std    0
           PLAYER_DRIBBLED_PAST_season_std      0
           PLAYER_SUCCESSFUL_DRIBBLES_season_std  0
           PLAYER_DRIBBLED_ATTEMPTS_5_last_match_sum  0
           PLAYER_DRIBBLED_PAST_5_last_match_sum    0
           PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_sum  0
           PLAYER_DRIBBLED_ATTEMPTS_5_last_match_average  0
           PLAYER_DRIBBLED_PAST_5_last_match_average  0
           PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_average  0
           PLAYER_DRIBBLED_ATTEMPTS_5_last_match_std  0
           PLAYER_DRIBBLED_PAST_5_last_match_std    0
           PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_std  0
dtype: int64

```

Missing values in player dribbling features were imputed using position-specific median imputation. Since dribbling performance differs substantially across attackers, midfielders, defenders and goalkeepers, calculating the median within each playing position preserves realistic positional behaviour while remaining robust to skewed distributions and extreme values. For observations with missing playing position information, the overall feature median was used.

```

In [126... home_player_processed.columns

```

```

Out[126... Index(['ID', 'LEAGUE', 'TEAM_NAME', 'POSITION', 'PLAYER_NAME',
           'PLAYER_ACCURATE_CROSSES_season_sum',
           'PLAYER_ACCURATE_PASSES_season_sum', 'PLAYER_AERIALS_WON_season_s
um',
           'PLAYER_ASSISTS_season_sum', 'PLAYER_BIG_CHANCES_CREATED_season_s
um',
           ...,
           'PLAYER_SAVES_INSIDE_BOX_5_last_match_std',
           'PLAYER_SHOTS_BLOCKED_5_last_match_std',
           'PLAYER_SHOTS_ON_TARGET_5_last_match_std',
           'PLAYER_SHOTS_TOTAL_5_last_match_std',
           'PLAYER_STARTING_LINEUP_5_last_match_std',
           'PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_std',
           'PLAYER_TACKLES_5_last_match_std',
           'PLAYER_TOTAL_CROSSES_5_last_match_std',
           'PLAYER_TOTAL_DUELS_5_last_match_std',
           'PLAYER_YELLOWCARDS_5_last_match_std'],
          dtype='object', length=277)

```

```

In [127... # Crossing

```

```

In [128... cross_cols = [col for col in home_player_processed.columns if "CROSS" in
cross_cols

```

```
Out[128...] ['PLAYER_ACCURATE_CROSSES_season_sum',
             'PLAYER_TOTAL_CROSSES_season_sum',
             'PLAYER_ACCURATE_CROSSES_season_average',
             'PLAYER_TOTAL_CROSSES_season_average',
             'PLAYER_ACCURATE_CROSSES_season_std',
             'PLAYER_TOTAL_CROSSES_season_std',
             'PLAYER_ACCURATE_CROSSES_5_last_match_sum',
             'PLAYER_TOTAL_CROSSES_5_last_match_sum',
             'PLAYER_ACCURATE_CROSSES_5_last_match_average',
             'PLAYER_TOTAL_CROSSES_5_last_match_average',
             'PLAYER_ACCURATE_CROSSES_5_last_match_std',
             'PLAYER_TOTAL_CROSSES_5_last_match_std']
```

```
In [129...] missing_numeric[cross_cols]
```

```
Out[129...] PLAYER_ACCURATE_CROSSES_season_sum      11.93
PLAYER_TOTAL_CROSSES_season_sum      11.93
PLAYER_ACCURATE_CROSSES_season_average 11.93
PLAYER_TOTAL_CROSSES_season_average 11.93
PLAYER_ACCURATE_CROSSES_season_std    13.37
PLAYER_TOTAL_CROSSES_season_std    13.37
PLAYER_ACCURATE_CROSSES_5_last_match_sum 11.93
PLAYER_TOTAL_CROSSES_5_last_match_sum 11.93
PLAYER_ACCURATE_CROSSES_5_last_match_average 11.93
PLAYER_TOTAL_CROSSES_5_last_match_average 11.93
PLAYER_ACCURATE_CROSSES_5_last_match_std 13.37
PLAYER_TOTAL_CROSSES_5_last_match_std 13.37
dtype: float64
```

```
In [130...] home_player_processed[cross_cols].describe()
```

```
Out[130...]      PLAYER_ACCURATE_CROSSES_season_sum  PLAYER_TOTAL_CROSSES_season_sum
```

count	208790.00	208790.00
mean	7.93	9.66
std	15.10	16.18
min	0.00	0.00
25%	0.00	0.00
50%	1.00	2.00
75%	9.00	12.00
max	100.00	100.00

```
In [131...] for col in cross_cols:
              print(f"\n{col}")
              print(
                  home_player_processed
                  .groupby("POSITION")[col]
                  .apply(lambda x: x.notna().sum())
              )
```

PLAYER_ACCURATE_CROSSES_season_sum
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_ACCURATE_CROSSES_season_sum, dtype: int64

PLAYER_TOTAL_CROSSES_season_sum
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_TOTAL_CROSSES_season_sum, dtype: int64

PLAYER_ACCURATE_CROSSES_season_average
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_ACCURATE_CROSSES_season_average, dtype: int64

PLAYER_TOTAL_CROSSES_season_average
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_TOTAL_CROSSES_season_average, dtype: int64

PLAYER_ACCURATE_CROSSES_season_std
POSITION
attacker 22263
defender 58409
goalkeeper 21000
midfielder 67969
Name: PLAYER_ACCURATE_CROSSES_season_std, dtype: int64

PLAYER_TOTAL_CROSSES_season_std
POSITION
attacker 22263
defender 58409
goalkeeper 21000
midfielder 67969
Name: PLAYER_TOTAL_CROSSES_season_std, dtype: int64

PLAYER_ACCURATE_CROSSES_5_last_match_sum
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_ACCURATE_CROSSES_5_last_match_sum, dtype: int64

PLAYER_TOTAL_CROSSES_5_last_match_sum
POSITION
attacker 22658
defender 59297

```

goalkeeper      21290
midfielder      69008
Name: PLAYER_TOTAL_CROSSES_5_last_match_sum, dtype: int64

PLAYER_ACCURATE_CROSSES_5_last_match_average
POSITION
attacker        22658
defender        59297
goalkeeper      21290
midfielder      69008
Name: PLAYER_ACCURATE_CROSSES_5_last_match_average, dtype: int64

PLAYER_TOTAL_CROSSES_5_last_match_average
POSITION
attacker        22658
defender        59297
goalkeeper      21290
midfielder      69008
Name: PLAYER_TOTAL_CROSSES_5_last_match_average, dtype: int64

PLAYER_ACCURATE_CROSSES_5_last_match_std
POSITION
attacker        22263
defender        58409
goalkeeper      21000
midfielder      67969
Name: PLAYER_ACCURATE_CROSSES_5_last_match_std, dtype: int64

PLAYER_TOTAL_CROSSES_5_last_match_std
POSITION
attacker        22263
defender        58409
goalkeeper      21000
midfielder      67969
Name: PLAYER_TOTAL_CROSSES_5_last_match_std, dtype: int64

```

```
In [132]: home_player_processed.groupby("POSITION")[cross_cols].median()
```

```
Out[132]:
```

PLAYER_ACCURATE_CROSSES_season_sum		PLAYER_TOTAL_CROSSES_season_sum	
POSITION			
attacker	2.00		
defender	1.00		
goalkeeper	0.00		
midfielder	4.00		

```
In [133]: for col in cross_cols:

# Step 1: Fill using position-specific median
home_player_processed[col] = (
    home_player_processed
    .groupby("POSITION")[col]
    .transform(lambda x: x.fillna(x.median()))
)

# Step 2: Fill remaining NaNs (missing POSITION)
```

```
home_player_processed[col] = (
    home_player_processed[col]
    .fillna(home_player_processed[col].median())
)
```

```
In [134... home_player_processed[cross_cols].isnull().sum()
```

```
Out[134... PLAYER_ACCURATE_CROSSES_season_sum      0
PLAYER_TOTAL_CROSSES_season_sum          0
PLAYER_ACCURATE_CROSSES_season_average   0
PLAYER_TOTAL_CROSSES_season_average       0
PLAYER_ACCURATE_CROSSES_season_std       0
PLAYER_TOTAL_CROSSES_season_std          0
PLAYER_ACCURATE_CROSSES_5_last_match_sum 0
PLAYER_TOTAL_CROSSES_5_last_match_sum     0
PLAYER_ACCURATE_CROSSES_5_last_match_average 0
PLAYER_TOTAL_CROSSES_5_last_match_average 0
PLAYER_ACCURATE_CROSSES_5_last_match_std  0
PLAYER_TOTAL_CROSSES_5_last_match_std     0
dtype: int64
```

```
In [135... # Shooting
```

```
In [136... shoot_cols = [col for col in home_player_processed.columns if "SHOT" in c
shoot_cols
```

```
Out[136... ['PLAYER_BLOCKED_SHOTS_season_sum',
'PLAYER_SHOTS_BLOCKED_season_sum',
'PLAYER_SHOTS_ON_TARGET_season_sum',
'PLAYER_SHOTS_TOTAL_season_sum',
'PLAYER_BLOCKED_SHOTS_season_average',
'PLAYER_SHOTS_BLOCKED_season_average',
'PLAYER_SHOTS_ON_TARGET_season_average',
'PLAYER_SHOTS_TOTAL_season_average',
'PLAYER_BLOCKED_SHOTS_season_std',
'PLAYER_SHOTS_BLOCKED_season_std',
'PLAYER_SHOTS_ON_TARGET_season_std',
'PLAYER_SHOTS_TOTAL_season_std',
'PLAYER_BLOCKED_SHOTS_5_last_match_sum',
'PLAYER_SHOTS_BLOCKED_5_last_match_sum',
'PLAYER_SHOTS_ON_TARGET_5_last_match_sum',
'PLAYER_SHOTS_TOTAL_5_last_match_sum',
'PLAYER_BLOCKED_SHOTS_5_last_match_average',
'PLAYER_SHOTS_BLOCKED_5_last_match_average',
'PLAYER_SHOTS_ON_TARGET_5_last_match_average',
'PLAYER_SHOTS_TOTAL_5_last_match_average',
'PLAYER_BLOCKED_SHOTS_5_last_match_std',
'PLAYER_SHOTS_BLOCKED_5_last_match_std',
'PLAYER_SHOTS_ON_TARGET_5_last_match_std',
'PLAYER_SHOTS_TOTAL_5_last_match_std']
```

```
In [137... missing_numeric[shoot_cols]
```



```
Out[137...] PLAYER_BLOCKED_SHOTS_season_sum      12.61
PLAYER_SHOTS_BLOCKED_season_sum      11.93
PLAYER_SHOTS_ON_TARGET_season_sum    11.93
PLAYER_SHOTS_TOTAL_season_sum        11.93
PLAYER_BLOCKED_SHOTS_season_average  12.61
PLAYER_SHOTS_BLOCKED_season_average  11.93
PLAYER_SHOTS_ON_TARGET_season_average 11.93
PLAYER_SHOTS_TOTAL_season_average    11.93
PLAYER_BLOCKED_SHOTS_season_std      14.02
PLAYER_SHOTS_BLOCKED_season_std      13.37
PLAYER_SHOTS_ON_TARGET_season_std    13.37
PLAYER_SHOTS_TOTAL_season_std        13.37
PLAYER_BLOCKED_SHOTS_5_last_match_sum 12.61
PLAYER_SHOTS_BLOCKED_5_last_match_sum 11.93
PLAYER_SHOTS_ON_TARGET_5_last_match_sum 11.93
PLAYER_SHOTS_TOTAL_5_last_match_sum   11.93
PLAYER_BLOCKED_SHOTS_5_last_match_average 12.61
PLAYER_SHOTS_BLOCKED_5_last_match_average 11.93
PLAYER_SHOTS_ON_TARGET_5_last_match_average 11.93
PLAYER_SHOTS_TOTAL_5_last_match_average 11.93
PLAYER_BLOCKED_SHOTS_5_last_match_std 14.02
PLAYER_SHOTS_BLOCKED_5_last_match_std 13.37
PLAYER_SHOTS_ON_TARGET_5_last_match_std 13.37
PLAYER_SHOTS_TOTAL_5_last_match_std   13.37
dtype: float64
```

```
In [138...] home_player_processed[shoot_cols].describe()
```

```
Out[138...]      PLAYER_BLOCKED_SHOTS_season_sum  PLAYER_SHOTS_BLOCKED_season_sum  P
```

count	207184.00	208790.00
mean	10.45	17.13
std	16.98	19.38
min	0.00	0.00
25%	0.00	0.00
50%	3.00	11.00
75%	14.00	27.00
max	100.00	100.00

```
In [139...] for col in shoot_cols:
    print(f"\n{col}")
    print(
        home_player_processed
        .groupby("POSITION")[col]
        .apply(lambda x: x.notna().sum())
    )
```

PLAYER_BLOCKED_SHOTS_season_sum
POSITION
attacker 22633
defender 59096
goalkeeper 21227
midfielder 68726
Name: PLAYER_BLOCKED_SHOTS_season_sum, dtype: int64

PLAYER_SHOTS_BLOCKED_season_sum
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_SHOTS_BLOCKED_season_sum, dtype: int64

PLAYER_SHOTS_ON_TARGET_season_sum
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_SHOTS_ON_TARGET_season_sum, dtype: int64

PLAYER_SHOTS_TOTAL_season_sum
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_SHOTS_TOTAL_season_sum, dtype: int64

PLAYER_BLOCKED_SHOTS_season_average
POSITION
attacker 22633
defender 59096
goalkeeper 21227
midfielder 68726
Name: PLAYER_BLOCKED_SHOTS_season_average, dtype: int64

PLAYER_SHOTS_BLOCKED_season_average
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_SHOTS_BLOCKED_season_average, dtype: int64

PLAYER_SHOTS_ON_TARGET_season_average
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_SHOTS_ON_TARGET_season_average, dtype: int64

PLAYER_SHOTS_TOTAL_season_average
POSITION
attacker 22658
defender 59297

goalkeeper 21290
midfielder 69008
Name: PLAYER_SHOTS_TOTAL_season_average, dtype: int64

PLAYER_BLOCKED_SHOTS_season_std
POSITION
attacker 22238
defender 58215
goalkeeper 20939
midfielder 67697
Name: PLAYER_BLOCKED_SHOTS_season_std, dtype: int64

PLAYER_SHOTS_BLOCKED_season_std
POSITION
attacker 22263
defender 58409
goalkeeper 21000
midfielder 67969
Name: PLAYER_SHOTS_BLOCKED_season_std, dtype: int64

PLAYER_SHOTS_ON_TARGET_season_std
POSITION
attacker 22263
defender 58409
goalkeeper 21000
midfielder 67969
Name: PLAYER_SHOTS_ON_TARGET_season_std, dtype: int64

PLAYER_SHOTS_TOTAL_season_std
POSITION
attacker 22263
defender 58409
goalkeeper 21000
midfielder 67969
Name: PLAYER_SHOTS_TOTAL_season_std, dtype: int64

PLAYER_BLOCKED_SHOTS_5_last_match_sum
POSITION
attacker 22633
defender 59096
goalkeeper 21227
midfielder 68726
Name: PLAYER_BLOCKED_SHOTS_5_last_match_sum, dtype: int64

PLAYER_SHOTS_BLOCKED_5_last_match_sum
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_SHOTS_BLOCKED_5_last_match_sum, dtype: int64

PLAYER_SHOTS_ON_TARGET_5_last_match_sum
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_SHOTS_ON_TARGET_5_last_match_sum, dtype: int64

PLAYER_SHOTS_TOTAL_5_last_match_sum
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_SHOTS_TOTAL_5_last_match_sum, dtype: int64

PLAYER_BLOCKED_SHOTS_5_last_match_average
POSITION
attacker 22633
defender 59096
goalkeeper 21227
midfielder 68726
Name: PLAYER_BLOCKED_SHOTS_5_last_match_average, dtype: int64

PLAYER_SHOTS_BLOCKED_5_last_match_average
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_SHOTS_BLOCKED_5_last_match_average, dtype: int64

PLAYER_SHOTS_ON_TARGET_5_last_match_average
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_SHOTS_ON_TARGET_5_last_match_average, dtype: int64

PLAYER_SHOTS_TOTAL_5_last_match_average
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_SHOTS_TOTAL_5_last_match_average, dtype: int64

PLAYER_BLOCKED_SHOTS_5_last_match_std
POSITION
attacker 22238
defender 58215
goalkeeper 20939
midfielder 67697
Name: PLAYER_BLOCKED_SHOTS_5_last_match_std, dtype: int64

PLAYER_SHOTS_BLOCKED_5_last_match_std
POSITION
attacker 22263
defender 58409
goalkeeper 21000
midfielder 67969
Name: PLAYER_SHOTS_BLOCKED_5_last_match_std, dtype: int64

PLAYER_SHOTS_ON_TARGET_5_last_match_std
POSITION
attacker 22263
defender 58409

```
goalkeeper      21000
midfielder      67969
Name: PLAYER_SHOTS_ON_TARGET_5_last_match_std, dtype: int64
```

```
PLAYER_SHOTS_TOTAL_5_last_match_std
POSITION
attacker      22263
defender      58409
goalkeeper     21000
midfielder     67969
Name: PLAYER_SHOTS_TOTAL_5_last_match_std, dtype: int64
```

```
In [140...] home_player_processed.groupby("POSITION")[shoot_cols].median()
```

```
Out[140...]      PLAYER_BLOCKED_SHOTS_season_sum  PLAYER_SHOTS_BLOCKED_season_su

POSITION
attacker      0.00                                16.
defender     15.00                                13.
goalkeeper    0.00                                 0.
midfielder    4.00                                15.
```

```
In [141...] for col in shoot_cols:
    home_player_processed[col] = (
        home_player_processed
        .groupby("POSITION")[col]
        .transform(lambda x: x.fillna(x.median()))
    )
```

```
In [142...] for col in shoot_cols:
    home_player_processed[col].fillna(
        home_player_processed[col].median(),
        inplace=True
    )
```

```
In [143...] home_player_processed[shoot_cols].isnull().sum()
```

```
Out[143...] PLAYER_BLOCKED_SHOTS_season_sum 0
PLAYER_SHOTS_BLOCKED_season_sum 0
PLAYER_SHOTS_ON_TARGET_season_sum 0
PLAYER_SHOTS_TOTAL_season_sum 0
PLAYER_BLOCKED_SHOTS_season_average 0
PLAYER_SHOTS_BLOCKED_season_average 0
PLAYER_SHOTS_ON_TARGET_season_average 0
PLAYER_SHOTS_TOTAL_season_average 0
PLAYER_BLOCKED_SHOTS_season_std 0
PLAYER_SHOTS_BLOCKED_season_std 0
PLAYER_SHOTS_ON_TARGET_season_std 0
PLAYER_SHOTS_TOTAL_season_std 0
PLAYER_BLOCKED_SHOTS_5_last_match_sum 0
PLAYER_SHOTS_BLOCKED_5_last_match_sum 0
PLAYER_SHOTS_ON_TARGET_5_last_match_sum 0
PLAYER_SHOTS_TOTAL_5_last_match_sum 0
PLAYER_BLOCKED_SHOTS_5_last_match_average 0
PLAYER_SHOTS_BLOCKED_5_last_match_average 0
PLAYER_SHOTS_ON_TARGET_5_last_match_average 0
PLAYER_SHOTS_TOTAL_5_last_match_average 0
PLAYER_BLOCKED_SHOTS_5_last_match_std 0
PLAYER_SHOTS_BLOCKED_5_last_match_std 0
PLAYER_SHOTS_ON_TARGET_5_last_match_std 0
PLAYER_SHOTS_TOTAL_5_last_match_std 0
dtype: int64
```

```
In [144...] missing_numeric = (
    home_player_processed
    .select_dtypes(include='number')
    .isnull()
    .mean() * 100
).sort_values(ascending=False)

missing_numeric.head(20)
```

```
Out[144...] PLAYER_PENALTIES_SAVED_5_last_match_std 22.05
PLAYER_PENALTIES_SAVED_5_last_match_sum 20.79
PLAYER_PENALTIES_SAVED_5_last_match_average 20.79
PLAYER_PENALTIES_SAVED_season_std 16.57
PLAYER_PENALTIES_SAVED_season_sum 15.23
PLAYER_PENALTIES_SAVED_season_average 15.23
PLAYER_CLEARANCE_OFFLINE_5_last_match_std 15.04
PLAYER_RATING_season_std 14.55
PLAYER_RATING_5_last_match_std 14.55
PLAYER_ERROR_LEAD_TO_GOAL_5_last_match_std 14.41
PLAYER_PENALTIES_WON_5_last_match_std 14.29
PLAYER_PENALTIES_COMMITTED_5_last_match_std 14.17
PLAYER_ERROR_LEAD_TO_GOAL_season_std 14.06
PLAYER_CLEARANCE_OFFLINE_season_std 14.05
PLAYER_BIG_CHANCES_MISSED_5_last_match_std 14.02
PLAYER_BIG_CHANCES_MISSED_season_std 14.02
PLAYER_DUELS_LOST_season_std 14.02
PLAYER_DUELS_LOST_5_last_match_std 14.02
PLAYER_BIG_CHANCES_CREATED_season_std 14.02
PLAYER_BIG_CHANCES_CREATED_5_last_match_std 14.02
dtype: float64
```

```
In [145...] for col in shoot_cols:
    print(f"\n{col}")
```

```
print(home_player_processed.groupby("POSITION")[col].apply(lambda x:
```

PLAYER_BLOCKED_SHOTS_season_sum
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_BLOCKED_SHOTS_season_sum, dtype: int64

PLAYER_SHOTS_BLOCKED_season_sum
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_SHOTS_BLOCKED_season_sum, dtype: int64

PLAYER_SHOTS_ON_TARGET_season_sum
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_SHOTS_ON_TARGET_season_sum, dtype: int64

PLAYER_SHOTS_TOTAL_season_sum
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_SHOTS_TOTAL_season_sum, dtype: int64

PLAYER_BLOCKED_SHOTS_season_average
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_BLOCKED_SHOTS_season_average, dtype: int64

PLAYER_SHOTS_BLOCKED_season_average
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_SHOTS_BLOCKED_season_average, dtype: int64

PLAYER_SHOTS_ON_TARGET_season_average
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_SHOTS_ON_TARGET_season_average, dtype: int64

PLAYER_SHOTS_TOTAL_season_average
POSITION
attacker 0
defender 0

goalkeeper 0
midfielder 0
Name: PLAYER_SHOTS_TOTAL_season_average, dtype: int64

PLAYER_BLOCKED_SHOTS_season_std
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_BLOCKED_SHOTS_season_std, dtype: int64

PLAYER_SHOTS_BLOCKED_season_std
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_SHOTS_BLOCKED_season_std, dtype: int64

PLAYER_SHOTS_ON_TARGET_season_std
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_SHOTS_ON_TARGET_season_std, dtype: int64

PLAYER_SHOTS_TOTAL_season_std
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_SHOTS_TOTAL_season_std, dtype: int64

PLAYER_BLOCKED_SHOTS_5_last_match_sum
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_BLOCKED_SHOTS_5_last_match_sum, dtype: int64

PLAYER_SHOTS_BLOCKED_5_last_match_sum
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_SHOTS_BLOCKED_5_last_match_sum, dtype: int64

PLAYER_SHOTS_ON_TARGET_5_last_match_sum
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_SHOTS_ON_TARGET_5_last_match_sum, dtype: int64

PLAYER_SHOTS_TOTAL_5_last_match_sum
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_SHOTS_TOTAL_5_last_match_sum, dtype: int64

PLAYER_BLOCKED_SHOTS_5_last_match_average
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_BLOCKED_SHOTS_5_last_match_average, dtype: int64

PLAYER_SHOTS_BLOCKED_5_last_match_average
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_SHOTS_BLOCKED_5_last_match_average, dtype: int64

PLAYER_SHOTS_ON_TARGET_5_last_match_average
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_SHOTS_ON_TARGET_5_last_match_average, dtype: int64

PLAYER_SHOTS_TOTAL_5_last_match_average
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_SHOTS_TOTAL_5_last_match_average, dtype: int64

PLAYER_BLOCKED_SHOTS_5_last_match_std
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_BLOCKED_SHOTS_5_last_match_std, dtype: int64

PLAYER_SHOTS_BLOCKED_5_last_match_std
POSITION
attacker 0
defender 0
goalkeeper 0
midfielder 0
Name: PLAYER_SHOTS_BLOCKED_5_last_match_std, dtype: int64

PLAYER_SHOTS_ON_TARGET_5_last_match_std
POSITION
attacker 0
defender 0

```
goalkeeper      0
midfielder      0
Name: PLAYER_SHOTS_ON_TARGET_5_last_match_std, dtype: int64
```

```
PLAYER_SHOTS_TOTAL_5_last_match_std
POSITION
attacker        0
defender        0
goalkeeper      0
midfielder      0
Name: PLAYER_SHOTS_TOTAL_5_last_match_std, dtype: int64
```

```
In [146... home_player_processed[col] = home_player_processed[col].fillna(
            home_player_processed[col].median()
        )
```

```
In [147... home_player_processed[shoot_cols].isnull().sum()
```

```
Out[147... PLAYER_BLOCKED_SHOTS_season_sum          0
PLAYER_SHOTS_BLOCKED_season_sum          0
PLAYER_SHOTS_ON_TARGET_season_sum        0
PLAYER_SHOTS_TOTAL_season_sum            0
PLAYER_BLOCKED_SHOTS_season_average       0
PLAYER_SHOTS_BLOCKED_season_average       0
PLAYER_SHOTS_ON_TARGET_season_average     0
PLAYER_SHOTS_TOTAL_season_average         0
PLAYER_BLOCKED_SHOTS_season_std           0
PLAYER_SHOTS_BLOCKED_season_std           0
PLAYER_SHOTS_ON_TARGET_season_std         0
PLAYER_SHOTS_TOTAL_season_std             0
PLAYER_BLOCKED_SHOTS_5_last_match_sum     0
PLAYER_SHOTS_BLOCKED_5_last_match_sum     0
PLAYER_SHOTS_ON_TARGET_5_last_match_sum   0
PLAYER_SHOTS_TOTAL_5_last_match_sum       0
PLAYER_BLOCKED_SHOTS_5_last_match_average 0
PLAYER_SHOTS_BLOCKED_5_last_match_average 0
PLAYER_SHOTS_ON_TARGET_5_last_match_average 0
PLAYER_SHOTS_TOTAL_5_last_match_average   0
PLAYER_BLOCKED_SHOTS_5_last_match_std     0
PLAYER_SHOTS_BLOCKED_5_last_match_std     0
PLAYER_SHOTS_ON_TARGET_5_last_match_std   0
PLAYER_SHOTS_TOTAL_5_last_match_std       0
dtype: int64
```

```
In [148... # Passing
```

```
In [149... pass_cols = [col for col in home_player_processed.columns if "PASS" in co
pass_cols
```

```
Out[149... ['PLAYER_ACCURATE_PASSES_season_sum',
            'PLAYER_KEY_PASSES_season_sum',
            'PLAYER_PASSES_season_sum',
            'PLAYER_ACCURATE_PASSES_season_average',
            'PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average',
            'PLAYER_KEY_PASSES_season_average',
            'PLAYER_PASSES_season_average',
            'PLAYER_ACCURATE_PASSES_season_std',
            'PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std',
            'PLAYER_KEY_PASSES_season_std',
            'PLAYER_PASSES_season_std',
            'PLAYER_ACCURATE_PASSES_5_last_match_sum',
            'PLAYER_KEY_PASSES_5_last_match_sum',
            'PLAYER_PASSES_5_last_match_sum',
            'PLAYER_ACCURATE_PASSES_5_last_match_average',
            'PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average',
            'PLAYER_KEY_PASSES_5_last_match_average',
            'PLAYER_PASSES_5_last_match_average',
            'PLAYER_ACCURATE_PASSES_5_last_match_std',
            'PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std',
            'PLAYER_KEY_PASSES_5_last_match_std',
            'PLAYER_PASSES_5_last_match_std']
```

```
In [150... missing_numeric[pass_cols]
```

```
Out[150... PLAYER_ACCURATE_PASSES_season_sum      11.93
PLAYER_KEY_PASSES_season_sum      11.93
PLAYER_PASSES_season_sum          11.93
PLAYER_ACCURATE_PASSES_season_average      11.93
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average      0.00
PLAYER_KEY_PASSES_season_average      11.93
PLAYER_PASSES_season_average      11.93
PLAYER_ACCURATE_PASSES_season_std      13.37
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std      0.00
PLAYER_KEY_PASSES_season_std      13.37
PLAYER_PASSES_season_std      13.37
PLAYER_ACCURATE_PASSES_5_last_match_sum      11.93
PLAYER_KEY_PASSES_5_last_match_sum      11.93
PLAYER_PASSES_5_last_match_sum      11.93
PLAYER_ACCURATE_PASSES_5_last_match_average      11.93
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average      0.00
PLAYER_KEY_PASSES_5_last_match_average      11.93
PLAYER_PASSES_5_last_match_average      11.93
PLAYER_ACCURATE_PASSES_5_last_match_std      13.37
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std      0.00
PLAYER_KEY_PASSES_5_last_match_std      13.37
PLAYER_PASSES_5_last_match_std      13.37
dtype: float64
```

```
In [151... home_player_processed[pass_cols].describe()
```

Out[151...

	PLAYER_ACCURATE_PASSES_season_sum	PLAYER_KEY_PASSES_season_sum	PLAY
count	208790.00	208790.00	
mean	22.72	13.79	
std	20.85	17.21	
min	0.00	0.00	
25%	6.00	1.00	
50%	18.00	7.00	
75%	34.00	20.00	
max	100.00	100.00	

In [152...

```
for col in pass_cols:
    print(f"\n{col}")
    print(
        home_player_processed
        .groupby("POSITION")[col]
        .apply(lambda x: x.notna().sum())
    )
```

PLAYER_ACCURATE_PASSES_season_sum
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_ACCURATE_PASSES_season_sum, dtype: int64

PLAYER_KEY_PASSES_season_sum
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_KEY_PASSES_season_sum, dtype: int64

PLAYER_PASSES_season_sum
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_PASSES_season_sum, dtype: int64

PLAYER_ACCURATE_PASSES_season_average
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_ACCURATE_PASSES_season_average, dtype: int64

PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average
POSITION
attacker 23691
defender 67746
goalkeeper 24298
midfielder 78524
Name: PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average, dtype: int64

PLAYER_KEY_PASSES_season_average
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_KEY_PASSES_season_average, dtype: int64

PLAYER_PASSES_season_average
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_PASSES_season_average, dtype: int64

PLAYER_ACCURATE_PASSES_season_std
POSITION
attacker 22263
defender 58409

goalkeeper 21000
midfielder 67969
Name: PLAYER_ACCURATE_PASSES_season_std, dtype: int64

PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std
POSITION
attacker 23691
defender 67746
goalkeeper 24298
midfielder 78524
Name: PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std, dtype: int64

PLAYER_KEY_PASSES_season_std
POSITION
attacker 22263
defender 58409
goalkeeper 21000
midfielder 67969
Name: PLAYER_KEY_PASSES_season_std, dtype: int64

PLAYER_PASSES_season_std
POSITION
attacker 22263
defender 58409
goalkeeper 21000
midfielder 67969
Name: PLAYER_PASSES_season_std, dtype: int64

PLAYER_ACCURATE_PASSES_5_last_match_sum
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_ACCURATE_PASSES_5_last_match_sum, dtype: int64

PLAYER_KEY_PASSES_5_last_match_sum
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_KEY_PASSES_5_last_match_sum, dtype: int64

PLAYER_PASSES_5_last_match_sum
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_PASSES_5_last_match_sum, dtype: int64

PLAYER_ACCURATE_PASSES_5_last_match_average
POSITION
attacker 22658
defender 59297
goalkeeper 21290
midfielder 69008
Name: PLAYER_ACCURATE_PASSES_5_last_match_average, dtype: int64

```
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average
POSITION
attacker      23691
defender      67746
goalkeeper    24298
midfielder    78524
Name: PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average, dtype: int64
```

```
PLAYER_KEY_PASSES_5_last_match_average
POSITION
attacker      22658
defender      59297
goalkeeper    21290
midfielder    69008
Name: PLAYER_KEY_PASSES_5_last_match_average, dtype: int64
```

```
PLAYER_PASSES_5_last_match_average
POSITION
attacker      22658
defender      59297
goalkeeper    21290
midfielder    69008
Name: PLAYER_PASSES_5_last_match_average, dtype: int64
```

```
PLAYER_ACCURATE_PASSES_5_last_match_std
POSITION
attacker      22263
defender      58409
goalkeeper    21000
midfielder    67969
Name: PLAYER_ACCURATE_PASSES_5_last_match_std, dtype: int64
```

```
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std
POSITION
attacker      23691
defender      67746
goalkeeper    24298
midfielder    78524
Name: PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std, dtype: int64
```

```
PLAYER_KEY_PASSES_5_last_match_std
POSITION
attacker      22263
defender      58409
goalkeeper    21000
midfielder    67969
Name: PLAYER_KEY_PASSES_5_last_match_std, dtype: int64
```

```
PLAYER_PASSES_5_last_match_std
POSITION
attacker      22263
defender      58409
goalkeeper    21000
midfielder    67969
Name: PLAYER_PASSES_5_last_match_std, dtype: int64
```

```
In [153... home_player_processed.groupby("POSITION")[pass_cols].median()
```


Out[153...

PLAYER_ACCURATE_PASSES_season_sum PLAYER_KEY_PASSES_season_sum

POSITION		
attacker	10.00	13.00
defender	28.00	5.00
goalkeeper	6.00	0.00
midfielder	21.00	14.00

In [154...

```
for col in pass_cols:
    home_player_processed[col] = (
        home_player_processed
        .groupby("POSITION")[col]
        .transform(lambda x: x.fillna(x.median()))
    )
```

In [155...

```
home_player_processed[pass_cols].isnull().sum()
```

Out[155...

```
PLAYER_ACCURATE_PASSES_season_sum      42820
PLAYER_KEY_PASSES_season_sum            42820
PLAYER_PASSES_season_sum                42820
PLAYER_ACCURATE_PASSES_season_average   42820
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average 42820
PLAYER_KEY_PASSES_season_average        42820
PLAYER_PASSES_season_average            42820
PLAYER_ACCURATE_PASSES_season_std       42820
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std 42820
PLAYER_KEY_PASSES_season_std            42820
PLAYER_PASSES_season_std                42820
PLAYER_ACCURATE_PASSES_5_last_match_sum 42820
PLAYER_KEY_PASSES_5_last_match_sum      42820
PLAYER_PASSES_5_last_match_sum          42820
PLAYER_ACCURATE_PASSES_5_last_match_average 42820
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average 42820
PLAYER_KEY_PASSES_5_last_match_average   42820
PLAYER_PASSES_5_last_match_average       42820
PLAYER_ACCURATE_PASSES_5_last_match_std   42820
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std 42820
PLAYER_KEY_PASSES_5_last_match_std       42820
PLAYER_PASSES_5_last_match_std           42820
dtype: int64
```

In [156...

```
for col in pass_cols:
    home_player_processed[col] = home_player_processed[col].fillna(
        home_player_processed[col].median()
    )
```

In [157...

```
home_player_processed[pass_cols].isnull().sum()
```

```

Out[157...] PLAYER_ACCURATE_PASSES_season_sum      0
            PLAYER_KEY_PASSES_season_sum    0
            PLAYER_PASSES_season_sum         0
            PLAYER_ACCURATE_PASSES_season_average  0
            PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average  0
            PLAYER_KEY_PASSES_season_average  0
            PLAYER_PASSES_season_average       0
            PLAYER_ACCURATE_PASSES_season_std    0
            PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std    0
            PLAYER_KEY_PASSES_season_std         0
            PLAYER_PASSES_season_std            0
            PLAYER_ACCURATE_PASSES_5_last_match_sum    0
            PLAYER_KEY_PASSES_5_last_match_sum         0
            PLAYER_PASSES_5_last_match_sum            0
            PLAYER_ACCURATE_PASSES_5_last_match_average  0
            PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average  0
            PLAYER_KEY_PASSES_5_last_match_average  0
            PLAYER_PASSES_5_last_match_average       0
            PLAYER_ACCURATE_PASSES_5_last_match_std    0
            PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std    0
            PLAYER_KEY_PASSES_5_last_match_std         0
            PLAYER_PASSES_5_last_match_std            0
            dtype: int64

```

```

In [158...] # Tackle

```

```

In [159...] tackle_cols = [col for col in home_player_processed.columns if "TACKLE" in col]
            tackle_cols

```

```

Out[159...] ['PLAYER_TACKLES_season_sum',
            'PLAYER_TACKLES_season_average',
            'PLAYER_TACKLES_season_std',
            'PLAYER_TACKLES_5_last_match_sum',
            'PLAYER_TACKLES_5_last_match_average',
            'PLAYER_TACKLES_5_last_match_std']

```

```

In [160...] missing_numeric[tackle_cols]

```

```

Out[160...] PLAYER_TACKLES_season_sum      11.93
            PLAYER_TACKLES_season_average    11.93
            PLAYER_TACKLES_season_std        13.37
            PLAYER_TACKLES_5_last_match_sum   11.93
            PLAYER_TACKLES_5_last_match_average  11.93
            PLAYER_TACKLES_5_last_match_std    13.37
            dtype: float64

```

```

In [161...] home_player_processed[tackle_cols].describe()

```

Out[161...

	PLAYER_TACKLES_season_sum	PLAYER_TACKLES_season_average	PLAYER_TACKI
count	208790.00	208790.00	
mean	19.81	19.55	
std	20.13	18.66	
min	0.00	0.00	
25%	3.00	4.00	
50%	14.00	15.00	
75%	31.00	30.00	
max	100.00	100.00	

In [162...

```
for col in tackle_cols:
    print(f"\n{col}")
    print(
        home_player_processed
        .groupby("POSITION")[col]
        .apply(lambda x: x.notna().sum())
    )
```

```
PLAYER_TACKLES_season_sum
POSITION
attacker      22658
defender      59297
goalkeeper    21290
midfielder    69008
Name: PLAYER_TACKLES_season_sum, dtype: int64
```

```
PLAYER_TACKLES_season_average
POSITION
attacker      22658
defender      59297
goalkeeper    21290
midfielder    69008
Name: PLAYER_TACKLES_season_average, dtype: int64
```

```
PLAYER_TACKLES_season_std
POSITION
attacker      22263
defender      58409
goalkeeper    21000
midfielder    67969
Name: PLAYER_TACKLES_season_std, dtype: int64
```

```
PLAYER_TACKLES_5_last_match_sum
POSITION
attacker      22658
defender      59297
goalkeeper    21290
midfielder    69008
Name: PLAYER_TACKLES_5_last_match_sum, dtype: int64
```

```
PLAYER_TACKLES_5_last_match_average
POSITION
attacker      22658
defender      59297
goalkeeper    21290
midfielder    69008
Name: PLAYER_TACKLES_5_last_match_average, dtype: int64
```

```
PLAYER_TACKLES_5_last_match_std
POSITION
attacker      22263
defender      58409
goalkeeper    21000
midfielder    67969
Name: PLAYER_TACKLES_5_last_match_std, dtype: int64
```

```
In [163... home_player_processed.groupby("POSITION")[tackle_cols].median()
```

Out[163... **PLAYER_TACKLES_season_sum** **PLAYER_TACKLES_season_average** **PLAYER_**

POSITION		
attacker	9.00	10.00
defender	22.00	23.00
goalkeeper	0.00	0.00
midfielder	21.00	22.00

```
In [164... for col in pass_cols:
    home_player_processed[col] = (
        home_player_processed
        .groupby("POSITION")[col]
        .transform(lambda x: x.fillna(x.median()))
    )
```

```
In [165... home_player_processed[pass_cols].isnull().sum()
```

```
Out[165... PLAYER_ACCURATE_PASSES_season_sum          42820
PLAYER_KEY_PASSES_season_sum                42820
PLAYER_PASSES_season_sum                   42820
PLAYER_ACCURATE_PASSES_season_average       42820
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average 42820
PLAYER_KEY_PASSES_season_average            42820
PLAYER_PASSES_season_average                42820
PLAYER_ACCURATE_PASSES_season_std           42820
PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std 42820
PLAYER_KEY_PASSES_season_std                42820
PLAYER_PASSES_season_std                   42820
PLAYER_ACCURATE_PASSES_5_last_match_sum     42820
PLAYER_KEY_PASSES_5_last_match_sum           42820
PLAYER_PASSES_5_last_match_sum              42820
PLAYER_ACCURATE_PASSES_5_last_match_average 42820
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average 42820
PLAYER_KEY_PASSES_5_last_match_average       42820
PLAYER_PASSES_5_last_match_average          42820
PLAYER_ACCURATE_PASSES_5_last_match_std      42820
PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std 42820
PLAYER_KEY_PASSES_5_last_match_std           42820
PLAYER_PASSES_5_last_match_std              42820
dtype: int64
```

```
In [166... for col in pass_cols:
    home_player_processed[col].fillna(
        home_player_processed[col].median(),
        inplace=True
    )
```

```
In [167... home_player_processed[pass_cols].isnull().sum()
```

```

Out[167...  PLAYER_ACCURATE_PASSES_season_sum      0
            PLAYER_KEY_PASSES_season_sum    0
            PLAYER_PASSES_season_sum         0
            PLAYER_ACCURATE_PASSES_season_average  0
            PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average  0
            PLAYER_KEY_PASSES_season_average  0
            PLAYER_PASSES_season_average      0
            PLAYER_ACCURATE_PASSES_season_std  0
            PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std  0
            PLAYER_KEY_PASSES_season_std      0
            PLAYER_PASSES_season_std         0
            PLAYER_ACCURATE_PASSES_5_last_match_sum  0
            PLAYER_KEY_PASSES_5_last_match_sum    0
            PLAYER_PASSES_5_last_match_sum       0
            PLAYER_ACCURATE_PASSES_5_last_match_average  0
            PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_average  0
            PLAYER_KEY_PASSES_5_last_match_average  0
            PLAYER_PASSES_5_last_match_average    0
            PLAYER_ACCURATE_PASSES_5_last_match_std  0
            PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_match_std  0
            PLAYER_KEY_PASSES_5_last_match_std    0
            PLAYER_PASSES_5_last_match_std       0
dtype: int64

```

```

In [168... # Tackles

```

```

In [169... for col in tackle_cols:
            home_player_processed[col] = (
                home_player_processed
                    .groupby("POSITION")[col]
                    .transform(lambda x: x.fillna(x.median()))
            )

```

```

In [170... for col in tackle_cols:
            home_player_processed[col].fillna(
                home_player_processed[col].median(),
                inplace=True
            )

```

```

In [171... home_player_processed[tackle_cols].isnull().sum()

```

```

Out[171...  PLAYER_TACKLES_season_sum      0
            PLAYER_TACKLES_season_average  0
            PLAYER_TACKLES_season_std      0
            PLAYER_TACKLES_5_last_match_sum  0
            PLAYER_TACKLES_5_last_match_average  0
            PLAYER_TACKLES_5_last_match_std  0
dtype: int64

```

```

In [172... feature_groups = {
            "aerial": "AERIAL",
            "assist": "ASSIST",
            "big_created": "BIG_CHANCES_CREATED",
            "big_missed": "BIG_CHANCES_MISSED",
            "clearance": "CLEARANCE",
            "clearances": "CLEARANCES",
            "dispossessed": "DISPOSSESSED",
            "duels": "DUELS",
            "interceptions": "INTERCEPTION",

```

```

    "fouls": "FOUL",
    "goals": "GOAL",
    "saves": "SAVE",
    "penalties": "PENALT",
    "own_goals": "OWN_GOAL",
    "woodwork": "WOODWORK",
    "offsides": "OFFSIDE",
    "yellow_cards": "YELLOWCARD",
    "red_cards": "REDCARD",
    "errors": "ERROR",
    "minutes": "MINUTES_PLAYED",
    "starting": "STARTING_LINEUP",
    "rating": "RATING"
}

```

```

In [173... for keyword in feature_groups.values():

    cols = [
        col for col in home_player_processed.columns
        if keyword in col.upper()
    ]

    if len(cols) == 0:
        continue

    # Position-wise median
    for col in cols:
        home_player_processed[col] = (
            home_player_processed
            .groupby("POSITION")[col]
            .transform(lambda x: x.fillna(x.median()))
        )

    # Overall median fallback
    for col in cols:
        home_player_processed[col].fillna(
            home_player_processed[col].median(),
            inplace=True
        )

```

```

In [174... remaining_missing = (
    home_player_processed
    .select_dtypes(include="number")
    .isnull()
    .sum()
)

remaining_missing = remaining_missing[remaining_missing > 0]

print("Columns with missing values:", len(remaining_missing))
print("Total missing values:", remaining_missing.sum())

remaining_missing

```

```

Columns with missing values: 0
Total missing values: 0

```

```

Out[174... Series([], dtype: int64)

```

```
In [175... missing_cols = home_player_processed.columns[
    home_player_processed.isnull().any()
]

print("Number of columns with missing values:", len(missing_cols))
print(missing_cols.tolist())
```

Number of columns with missing values: 1
['POSITION']

```
In [176... home_player_processed["POSITION"].isnull().sum()
```

Out[176... np.int64(42820)

```
In [177... print(home_player_processed.shape)

home_player_processed["POSITION"].value_counts(dropna=False)
```

(237079, 277)

```
Out[177... POSITION
midfielder    78524
defender      67746
NaN           42820
goalkeeper    24298
attacker      23691
Name: count, dtype: int64
```

```
In [178... home_player_processed[
    home_player_processed["POSITION"].isnull()
].head()
```

```
Out[178...    ID  LEAGUE  TEAM_NAME  POSITION  PLAYER_NAME  PLAYER_ACCURATE_CROSSES
```

4	0	Ligue 1	Toulouse	NaN	Efthymios Koulouris
9	0	Ligue 1	Toulouse	NaN	Aaron Leya Iseka
10	0	Ligue 1	Toulouse	NaN	Yaya Sanogo
17	0	Ligue 1	Toulouse	NaN	Wesley Saïd
21	1	Ligue 2	Brest	NaN	Édouard Butin

```
In [179... home_player_processed.loc[
    home_player_processed["POSITION"].isna()
].iloc[0]
```



```
Out[179...] ID                                0
LEAGUE                                Ligue 1
TEAM_NAME                            Toulouse
POSITION                             NaN
PLAYER_NAME                          Efthymios Koulouris

...
PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_std    12.00
PLAYER_TACKLES_5_last_match_std                22.00
PLAYER_TOTAL_CROSSES_5_last_match_std          7.00
PLAYER_TOTAL_DUELS_5_last_match_std            28.00
PLAYER_YELLOWCARDS_5_last_match_std            0.00
Name: 4, Length: 277, dtype: object
```

```
In [180...] # Remove POSITION column
home_player_processed = home_player_processed.drop(columns=["POSITION"])
```

```
In [181...] print("POSITION" in home_player_processed.columns)
```

False

```
In [182...] home_player_processed.columns
```

```
Out[182...] Index(['ID', 'LEAGUE', 'TEAM_NAME', 'PLAYER_NAME',
                  'PLAYER_ACCURATE_CROSSES_season_sum',
                  'PLAYER_ACCURATE_PASSES_season_sum', 'PLAYER_AERIALS_WON_season_s
um',
                  'PLAYER_ASSISTS_season_sum', 'PLAYER_BIG_CHANCES_CREATED_season_s
um',
                  'PLAYER_BIG_CHANCES_MISSED_season_sum',
                  ...
                  'PLAYER_SAVES_INSIDE_BOX_5_last_match_std',
                  'PLAYER_SHOTS_BLOCKED_5_last_match_std',
                  'PLAYER_SHOTS_ON_TARGET_5_last_match_std',
                  'PLAYER_SHOTS_TOTAL_5_last_match_std',
                  'PLAYER_STARTING_LINEUP_5_last_match_std',
                  'PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_std',
                  'PLAYER_TACKLES_5_last_match_std',
                  'PLAYER_TOTAL_CROSSES_5_last_match_std',
                  'PLAYER_TOTAL_DUELS_5_last_match_std',
                  'PLAYER_YELLOWCARDS_5_last_match_std'],
                  dtype='object', length=276)
```

```
In [183...] print("Shape:", home_player_processed.shape)
print("Total missing values:", home_player_processed.isnull().sum().sum())
```

Shape: (237079, 276)

Total missing values: 0

6.1.4 Away Player Dataset

```
In [184...] missing_numeric = (
    away_player
    .select_dtypes(include="number")
    .isnull()
    .mean()
    .mul(100)
    .sort_values(ascending=False)
)
```

```
missing_numeric[missing_numeric > 0]
```

```
Out[184...] PLAYER_CAPTAIN_5_last_match_sum          92.43
            PLAYER_CAPTAIN_season_sum        89.42
            PLAYER_PUNCHES_season_std        72.03
            PLAYER_PUNCHES_5_last_match_std   72.03
            PLAYER_PUNCHES_5_last_match_sum    71.60
            ...
            PLAYER_ASSISTS_5_last_match_average 1.62
            PLAYER_GOALS_5_last_match_average  1.62
            PLAYER_GOALS_CONCEDED_5_last_match_average 1.62
            PLAYER_MINUTES_PLAYED_5_last_match_average 1.62
            PLAYER_PENALTIES_SCORED_5_last_match_average 1.62
            Length: 280, dtype: float64
```

```
In [185...] away_player.shape
```

```
Out[185...] (236132, 285)
```

```
In [186...] cols_to_drop = [
            "PLAYER_CAPTAIN_season_sum",
            "PLAYER_CAPTAIN_5_last_match_sum",
            "PLAYER_PUNCHES_season_sum",
            "PLAYER_PUNCHES_season_average",
            "PLAYER_PUNCHES_season_std",
            "PLAYER_PUNCHES_5_last_match_sum",
            "PLAYER_PUNCHES_5_last_match_average",
            "PLAYER_PUNCHES_5_last_match_std"
        ]
```

```
In [187...] [col for col in cols_to_drop if col in away_player.columns]
```

```
Out[187...] ['PLAYER_CAPTAIN_season_sum',
            'PLAYER_CAPTAIN_5_last_match_sum',
            'PLAYER_PUNCHES_season_sum',
            'PLAYER_PUNCHES_season_average',
            'PLAYER_PUNCHES_season_std',
            'PLAYER_PUNCHES_5_last_match_sum',
            'PLAYER_PUNCHES_5_last_match_average',
            'PLAYER_PUNCHES_5_last_match_std']
```

```
In [188...] away_player_processed = away_player.drop(columns=cols_to_drop)
```

```
In [189...] away_player_processed.shape
```

```
Out[189...] (236132, 277)
```

```
In [190...] away_player_processed.columns[away_player_processed.columns == "POSITION"]
```

```
Out[190...] Index(['POSITION'], dtype='object')
```

```
In [191...] feature_groups = {
            "accurate_passes": "ACCURATE_PASS",
            "passing": "PASS",
            "crossing": "CROSS",
            "dribbling": "DRIBBLE",
            "shooting": "SHOT",
        }
```

```
"tackles": "TACKLE"  
}
```

```
In [192... for keyword in feature_groups.values():  
  
    cols = [  
        col for col in away_player_processed.columns  
        if keyword in col.upper()  
    ]  
  
    if len(cols) == 0:  
        continue  
  
    # Position-wise median  
    for col in cols:  
        away_player_processed[col] = (  
            away_player_processed  
            .groupby("POSITION")[col]  
            .transform(lambda x: x.fillna(x.median()))  
        )  
  
    # Overall median fallback  
    for col in cols:  
        away_player_processed[col] = (  
            away_player_processed[col]  
            .fillna(away_player_processed[col].median())  
        )
```

```
In [193... remaining_missing = (  
    away_player_processed  
    .select_dtypes(include="number")  
    .isnull()  
    .sum()  
)  
  
remaining_missing = remaining_missing[remaining_missing > 0]  
  
print("Columns with missing values:", len(remaining_missing))  
print("Total missing values:", remaining_missing.sum())
```

Columns with missing values: 190
Total missing values: 4332071

```
In [194... feature_groups = {  
    "aerials": "AERIAL",  
    "assists": "ASSIST",  
    "big_chances_created": "BIG_CHANCES_CREATED",  
    "big_chances_missed": "BIG_CHANCES_MISSED",  
    "clearances": "CLEARANCE",  
    "dispossessed": "DISPOSSESSED",  
    "duels_lost": "DUELS_LOST",  
    "duels_won": "DUELS_WON",  
    "errors": "ERROR",  
    "fouls": "FOUL",  
    "goals": "GOAL",  
    "hit_woodwork": "WOODWORK",  
    "interceptions": "INTERCEPTION",  
    "minutes": "MINUTES_PLAYED",  
    "offsides": "OFFSIDE",  
    "own_goals": "OWN_GOAL",
```

```

    "penalties": "PENALT",
    "rating": "RATING",
    "red_cards": "REDCARD",
    "saves": "SAVE",
    "starting_lineup": "STARTING_LINEUP",
    "total_duels": "TOTAL_DUELS",
    "yellow_cards": "YELLOWCARD"
}

for keyword in feature_groups.values():

    cols = [
        col for col in away_player_processed.columns
        if keyword in col.upper()
    ]

    if len(cols) == 0:
        continue

    # Position-wise median
    for col in cols:
        away_player_processed[col] = (
            away_player_processed
            .groupby("POSITION")[col]
            .transform(lambda x: x.fillna(x.median()))
        )

    # Overall median fallback
    for col in cols:
        away_player_processed[col] = (
            away_player_processed[col]
            .fillna(away_player_processed[col].median())
        )

```

```

In [195...] remaining_missing = (
    away_player_processed
    .select_dtypes(include="number")
    .isnull()
    .sum()
)

remaining_missing = remaining_missing[remaining_missing > 0]

print("Columns with missing values:", len(remaining_missing))
print("Total missing values:", remaining_missing.sum())

```

Columns with missing values: 0
Total missing values: 0

```

In [196...] away_player_processed = away_player_processed.drop(columns=["POSITION"])

```

```

In [197...] print(away_player_processed.shape)

missing_cols = away_player_processed.columns[
    away_player_processed.isnull().any()
]

print("Columns with missing values:", len(missing_cols))
print(missing_cols.tolist())

```

```
(236132, 276)  
Columns with missing values: 0  
[]
```

Merging Dataset

```
In [198... print(home_team_processed.columns.tolist())
```

['ID', 'LEAGUE', 'TEAM_NAME', 'TEAM_SHOTS_TOTAL_season_sum', 'TEAM_SHOTS_INSIDEBOX_season_sum', 'TEAM_SHOTS_OFF_TARGET_season_sum', 'TEAM_SHOTS_ON_TARGET_season_sum', 'TEAM_SHOTS_OUTSIDEBOX_season_sum', 'TEAM_PASSES_season_sum', 'TEAM_SUCCESSFUL_PASSES_season_sum', 'TEAM_SAVES_season_sum', 'TEAM_CORNERS_season_sum', 'TEAM_FOULS_season_sum', 'TEAM_YELLOWCARDS_season_sum', 'TEAM_REDCARDS_season_sum', 'TEAM_OFFSIDES_season_sum', 'TEAM_ATTACKS_season_sum', 'TEAM_PENALTIES_season_sum', 'TEAM_SUBSTITUTIONS_season_sum', 'TEAM_BALL_SAFE_season_sum', 'TEAM_DANGEROUS_ATTACKS_season_sum', 'TEAM_INJURIES_season_sum', 'TEAM_GOALS_season_sum', 'TEAM_GAME_WON_season_sum', 'TEAM_GAME_DRAW_season_sum', 'TEAM_GAME_LOST_season_sum', 'TEAM_SHOTS_TOTAL_season_average', 'TEAM_SHOTS_INSIDEBOX_season_average', 'TEAM_SHOTS_OFF_TARGET_season_average', 'TEAM_SHOTS_ON_TARGET_season_average', 'TEAM_SHOTS_OUTSIDEBOX_season_average', 'TEAM_PASSES_season_average', 'TEAM_SUCCESSFUL_PASSES_season_average', 'TEAM_SUCCESSFUL_PASSES_PERCENTAGE_season_average', 'TEAM_SAVES_season_average', 'TEAM_CORNERS_season_average', 'TEAM_BALL_POSSESSION_season_average', 'TEAM_FOULS_season_average', 'TEAM_YELLOWCARDS_season_average', 'TEAM_REDCARDS_season_average', 'TEAM_OFFSIDES_season_average', 'TEAM_ATTACKS_season_average', 'TEAM_PENALTIES_season_average', 'TEAM_SUBSTITUTIONS_season_average', 'TEAM_BALL_SAFE_season_average', 'TEAM_DANGEROUS_ATTACKS_season_average', 'TEAM_INJURIES_season_average', 'TEAM_GOALS_season_average', 'TEAM_GAME_WON_season_average', 'TEAM_GAME_DRAW_season_average', 'TEAM_GAME_LOST_season_average', 'TEAM_SHOTS_TOTAL_season_std', 'TEAM_SHOTS_INSIDEBOX_season_std', 'TEAM_SHOTS_OFF_TARGET_season_std', 'TEAM_SHOTS_ON_TARGET_season_std', 'TEAM_SHOTS_OUTSIDEBOX_season_std', 'TEAM_PASSES_season_std', 'TEAM_SUCCESSFUL_PASSES_season_std', 'TEAM_SUCCESSFUL_PASSES_PERCENTAGE_season_std', 'TEAM_SAVES_season_std', 'TEAM_CORNERS_season_std', 'TEAM_BALL_POSSESSION_season_std', 'TEAM_FOULS_season_std', 'TEAM_YELLOWCARDS_season_std', 'TEAM_REDCARDS_season_std', 'TEAM_OFFSIDES_season_std', 'TEAM_ATTACKS_season_std', 'TEAM_PENALTIES_season_std', 'TEAM_SUBSTITUTIONS_season_std', 'TEAM_BALL_SAFE_season_std', 'TEAM_DANGEROUS_ATTACKS_season_std', 'TEAM_INJURIES_season_std', 'TEAM_GOALS_season_std', 'TEAM_SHOTS_TOTAL_5_last_match_sum', 'TEAM_SHOTS_INSIDEBOX_5_last_match_sum', 'TEAM_SHOTS_OFF_TARGET_5_last_match_sum', 'TEAM_SHOTS_ON_TARGET_5_last_match_sum', 'TEAM_SHOTS_OUTSIDEBOX_5_last_match_sum', 'TEAM_PASSES_5_last_match_sum', 'TEAM_SUCCESSFUL_PASSES_5_last_match_sum', 'TEAM_SAVES_5_last_match_sum', 'TEAM_CORNERS_5_last_match_sum', 'TEAM_FOULS_5_last_match_sum', 'TEAM_YELLOWCARDS_5_last_match_sum', 'TEAM_REDCARDS_5_last_match_sum', 'TEAM_OFFSIDES_5_last_match_sum', 'TEAM_ATTACKS_5_last_match_sum', 'TEAM_PENALTIES_5_last_match_sum', 'TEAM_SUBSTITUTIONS_5_last_match_sum', 'TEAM_BALL_SAFE_5_last_match_sum', 'TEAM_DANGEROUS_ATTACKS_5_last_match_sum', 'TEAM_INJURIES_5_last_match_sum', 'TEAM_GOALS_5_last_match_sum', 'TEAM_GAME_WON_5_last_match_sum', 'TEAM_GAME_DRAW_5_last_match_sum', 'TEAM_GAME_LOST_5_last_match_sum', 'TEAM_SHOTS_TOTAL_5_last_match_average', 'TEAM_SHOTS_INSIDEBOX_5_last_match_average', 'TEAM_SHOTS_OFF_TARGET_5_last_match_average', 'TEAM_SHOTS_ON_TARGET_5_last_match_average', 'TEAM_SHOTS_OUTSIDEBOX_5_last_match_average', 'TEAM_PASSES_5_last_match_average', 'TEAM_SUCCESSFUL_PASSES_5_last_match_average', 'TEAM_SUCCESSFUL_PASSES_PERCENTAGE_5_last_match_average', 'TEAM_SAVES_5_last_match_average', 'TEAM_CORNERS_5_last_match_average', 'TEAM_BALL_POSSESSION_5_last_match_average', 'TEAM_FOULS_5_last_match_average', 'TEAM_YELLOWCARDS_5_last_match_average', 'TEAM_REDCARDS_5_last_match_average', 'TEAM_OFFSIDES_5_last_match_average', 'TEAM_ATTACKS_5_last_match_average', 'TEAM_PENALTIES_5_last_match_average', 'TEAM_SUBSTITUTIONS_5_last_match_average', 'TEAM_BALL_SAFE_5_last_match_average', 'TEAM_DANGEROUS_ATTACKS_5_last_match_average', 'TEAM_INJURIES_5_last_match_average', 'TEAM_GOALS_5_last_match_average', 'TEAM_GAME_WON_5_last_match_average', 'TEAM_GAME_DRAW_5_last_match_average', 'TEAM_GAME_LOST_5_last_match_average', 'TEAM_SHOTS_TOTAL_5_last_match_std', 'TEAM_SHOTS_INSIDEBOX_5_last_match_std', 'TEAM_SHOTS_OFF_TARGET_5_last_match_std', 'TEAM_SHOTS_ON_TARGET_5_last_match_std', 'TEAM_SHOTS_OUTSIDEBOX_5_last_match_std', 'TEAM_PASSES_5_last_match_std', 'TEAM_SUCCESSFUL_PASSES_5_last_match_std', 'TEAM_S

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In [199... print(away_team_processed.columns.tolist())
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UCCESSFUL_PASSES_PERCENTAGE_5_last_match_std', 'TEAM_SAVES_5_last_match_std', 'TEAM_CORNERS_5_last_match_std', 'TEAM_BALL_POSSESSION_5_last_match_std', 'TEAM_FOULS_5_last_match_std', 'TEAM_YELLOWCARDS_5_last_match_std', 'TEAM_REDCARDS_5_last_match_std', 'TEAM_OFFSIDES_5_last_match_std', 'TEAM_ATTACKS_5_last_match_std', 'TEAM_PENALTIES_5_last_match_std', 'TEAM_SUBSTITUTIONS_5_last_match_std', 'TEAM_BALL_SAFE_5_last_match_std', 'TEAM_DANGEROUS_ATTACKS_5_last_match_std', 'TEAM_INJURIES_5_last_match_std', 'TEAM_GOALS_5_last_match_std']
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In [200... print(home_player_processed.columns.tolist())
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['ID', 'LEAGUE', 'TEAM_NAME', 'PLAYER_NAME', 'PLAYER_ACCURATE_CROSSES_season_sum', 'PLAYER_ACCURATE_PASSES_season_sum', 'PLAYER_AERIALS_WON_season_sum', 'PLAYER_ASSISTS_season_sum', 'PLAYER_BIG_CHANCES_CREATED_season_sum', 'PLAYER_BIG_CHANCES_MISSED_season_sum', 'PLAYER_BLOCKED_SHOTS_season_sum', 'PLAYER_CLEARANCES_season_sum', 'PLAYER_CLEARANCE_OFFLINE_season_sum', 'PLAYER_DISPOSSESSED_season_sum', 'PLAYER_DRIBBLED_ATTEMPTS_season_sum', 'PLAYER_DRIBBLED_PAST_season_sum', 'PLAYER_DUELS_LOST_season_sum', 'PLAYER_DUELS_WON_season_sum', 'PLAYER_ERROR_LEAD_TO_GOAL_season_sum', 'PLAYER_FOULS_season_sum', 'PLAYER_FOULS_DRAWN_season_sum', 'PLAYER_GOALKEEPER_GOALS_CONCEDED_season_sum', 'PLAYER_GOALS_season_sum', 'PLAYER_GOALS_CONCEDED_season_sum', 'PLAYER_HIT_WOODWORK_season_sum', 'PLAYER_INTERCEPTIONS_season_sum', 'PLAYER_KEY_PASSES_season_sum', 'PLAYER_MINUTES_PLAYED_season_sum', 'PLAYER_OFFSIDES_season_sum', 'PLAYER_OWN_GOALS_season_sum', 'PLAYER_PASSES_season_sum', 'PLAYER_PENALTIES_COMMITTED_season_sum', 'PLAYER_PENALTIES_MISSES_season_sum', 'PLAYER_PENALTIES_SAVED_season_sum', 'PLAYER_PENALTIES_SCORED_season_sum', 'PLAYER_PENALTIES_WON_season_sum', 'PLAYER_REDCARDS_season_sum', 'PLAYER_SAVES_season_sum', 'PLAYER_SAVES_INSIDE_BOX_season_sum', 'PLAYER_SHOTS_BLOCKED_season_sum', 'PLAYER_SHOTS_ON_TARGET_season_sum', 'PLAYER_SHOTS_TOTAL_season_sum', 'PLAYER_STARTING_LINEUP_season_sum', 'PLAYER_SUCCESSFUL_DRIBBLES_season_sum', 'PLAYER_TACKLES_season_sum', 'PLAYER_TOTAL_CROSSES_season_sum', 'PLAYER_TOTAL_DUELS_season_sum', 'PLAYER_YELLOWCARDS_season_sum', 'PLAYER_ACCURATE_CROSSES_season_average', 'PLAYER_ACCURATE_PASSES_season_average', 'PLAYER_ACCURATE_PASSES_PERCENTAGE_season_average', 'PLAYER_AERIALS_WON_season_average', 'PLAYER_ASSISTS_season_average', 'PLAYER_BIG_CHANCES_CREATED_season_average', 'PLAYER_BIG_CHANCES_MISSED_season_average', 'PLAYER_BLOCKED_SHOTS_season_average', 'PLAYER_CLEARANCES_season_average', 'PLAYER_CLEARANCE_OFFLINE_season_average', 'PLAYER_DISPOSSESSED_season_average', 'PLAYER_DRIBBLED_ATTEMPTS_season_average', 'PLAYER_DRIBBLED_PAST_season_average', 'PLAYER_DUELS_LOST_season_average', 'PLAYER_DUELS_WON_season_average', 'PLAYER_ERROR_LEAD_TO_GOAL_season_average', 'PLAYER_FOULS_season_average', 'PLAYER_FOULS_DRAWN_season_average', 'PLAYER_GOALKEEPER_GOALS_CONCEDED_season_average', 'PLAYER_GOALS_season_average', 'PLAYER_GOALS_CONCEDED_season_average', 'PLAYER_HIT_WOODWORK_season_average', 'PLAYER_INTERCEPTIONS_season_average', 'PLAYER_KEY_PASSES_season_average', 'PLAYER_MINUTES_PLAYED_season_average', 'PLAYER_OFFSIDES_season_average', 'PLAYER_OWN_GOALS_season_average', 'PLAYER_PASSES_season_average', 'PLAYER_PENALTIES_COMMITTED_season_average', 'PLAYER_PENALTIES_MISSES_season_average', 'PLAYER_PENALTIES_SAVED_season_average', 'PLAYER_PENALTIES_SCORED_season_average', 'PLAYER_PENALTIES_WON_season_average', 'PLAYER_RATING_season_average', 'PLAYER_REDCARDS_season_average', 'PLAYER_SAVES_season_average', 'PLAYER_SAVES_INSIDE_BOX_season_average', 'PLAYER_SHOTS_BLOCKED_season_average', 'PLAYER_SHOTS_ON_TARGET_season_average', 'PLAYER_SHOTS_TOTAL_season_average', 'PLAYER_STARTING_LINEUP_season_average', 'PLAYER_SUCCESSFUL_DRIBBLES_season_average', 'PLAYER_TACKLES_season_average', 'PLAYER_TOTAL_CROSSES_season_average', 'PLAYER_TOTAL_DUELS_season_average', 'PLAYER_YELLOWCARDS_season_average', 'PLAYER_ACCURATE_CROSSES_season_std', 'PLAYER_ACCURATE_PASSES_season_std', 'PLAYER_ACCURATE_PASSES_PERCENTAGE_season_std', 'PLAYER_AERIALS_WON_season_std', 'PLAYER_ASSISTS_season_std', 'PLAYER_BIG_CHANCES_CREATED_season_std', 'PLAYER_BIG_CHANCES_MISSED_season_std', 'PLAYER_BLOCKED_SHOTS_season_std', 'PLAYER_CLEARANCES_season_std', 'PLAYER_CLEARANCE_OFFLINE_season_std', 'PLAYER_DISPOSSESSED_season_std', 'PLAYER_DRIBBLED_ATTEMPTS_season_std', 'PLAYER_DRIBBLED_PAST_season_std', 'PLAYER_DUELS_LOST_season_std', 'PLAYER_DUELS_WON_season_std', 'PLAYER_ERROR_LEAD_TO_GOAL_season_std', 'PLAYER_FOULS_season_std', 'PLAYER_FOULS_DRAWN_season_std', 'PLAYER_GOALKEEPER_GOALS_CONCEDED_season_std', 'PLAYER_GOALS_season_std', 'PLAYER_GOALS_CONCEDED_season_std', 'PLAYER_HIT_WOODWORK_season_std', 'PLAYER_INTERCEPTIONS_season_std', 'PLAYER_KEY_PASSES_season_std', 'PLAYER_MINUTES_PLAYED_season_std', 'PLAYER_OFFSIDES_season_std', 'PLAYER_OWN_GOALS_season_std', 'PLAYER_PASSES_season_std', 'PLAYER_PENALTIES_COMMITTED_season_std', 'PLAYER_PENALTIES_MISSES_season_std', 'PLAYER_PENALTIES_SAVED_season_std', 'PLAYER_PENALTIES_SCORED_season_std', 'PLAYER_PENALTIES_WON_season_std', 'PLAYER_RATING_season_std', 'PLAYER_REDCARDS_season_std', 'PLAYER_SAVES_season_std', 'PLAYER_SAVES_INSIDE_BOX_season_std', 'PLAYER_SHOTS_BLOCKED_season_std', 'PLAYER_SHOTS_ON_TARGET_season_std', 'PLAYER_SHOTS_TOTAL_season_std', 'PLAYER_STARTING_LINEUP_season_std', 'PLAYER_SUCCESSFUL_DRIBBLES_season_std', 'PLAYER_TACKLES_season_std', 'PLAYER_TOTAL_CROSSES_season_std', 'PLAYER_TOTAL_DUELS_season_std', 'PLAYER_YELLOWCARDS_season_std']

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```

URATE_PASSES_5_last_match_std', 'PLAYER_ACCURATE_PASSES_PERCENTAGE_5_last_
match_std', 'PLAYER_AERIALS_WON_5_last_match_std', 'PLAYER_ASSISTS_5_last_
match_std', 'PLAYER_BIG_CHANCES_CREATED_5_last_match_std', 'PLAYER_BIG_CHA
NCES_MISSED_5_last_match_std', 'PLAYER_BLOCKED_SHOTS_5_last_match_std', 'P
PLAYER_CLEARANCES_5_last_match_std', 'PLAYER_CLEARANCE_OFFLINE_5_last_match
_std', 'PLAYER_DISPOSSESSED_5_last_match_std', 'PLAYER_DRIBBLED_ATTEMPTS_5
_last_match_std', 'PLAYER_DRIBBLED_PAST_5_last_match_std', 'PLAYER_DUELS_L
OST_5_last_match_std', 'PLAYER_DUELS_WON_5_last_match_std', 'PLAYER_ERROR_
LEAD_TO_GOAL_5_last_match_std', 'PLAYER_FOULS_5_last_match_std', 'PLAYER_F
OULS_DRAWN_5_last_match_std', 'PLAYER_GOALKEEPER_GOALS_CONCEDED_5_last_mat
ch_std', 'PLAYER_GOALS_5_last_match_std', 'PLAYER_GOALS_CONCEDED_5_last_ma
tch_std', 'PLAYER_HIT_WOODWORK_5_last_match_std', 'PLAYER_INTERCEPTIONS_5_
last_match_std', 'PLAYER_KEY_PASSES_5_last_match_std', 'PLAYER_MINUTES_PLA
YED_5_last_match_std', 'PLAYER_OFFSIDES_5_last_match_std', 'PLAYER_OW_N GOA
LS_5_last_match_std', 'PLAYER_PASSES_5_last_match_std', 'PLAYER_PENALTIES_
COMMITTED_5_last_match_std', 'PLAYER_PENALTIES_MISSES_5_last_match_std',
'PLAYER_PENALTIES_SAVED_5_last_match_std', 'PLAYER_PENALTIES_SCORED_5_last
_match_std', 'PLAYER_PENALTIES_WON_5_last_match_std', 'PLAYER_RATING_5_las
t_match_std', 'PLAYER_REDCARDS_5_last_match_std', 'PLAYER_SAVES_5_last mat
ch_std', 'PLAYER_SAVES_INSIDE_BOX_5_last_match_std', 'PLAYER_SHOTS_BLOCKED
_5_last_match_std', 'PLAYER_SHOTS_ON_TARGET_5_last_match_std', 'PLAYER_SHO
TS_TOTAL_5_last_match_std', 'PLAYER_STARTING_LINEUP_5_last_match_std', 'PL
AYER_SUCCESSFUL_DRIBBLES_5_last_match_std', 'PLAYER_TACKLES_5_last_match_s
td', 'PLAYER_TOTAL_CROSSES_5_last_match_std', 'PLAYER_TOTAL_DUELS_5_last_m
atch_std', 'PLAYER_YELLOWCARDS_5_last_match_std']

```

In [202... `home_team_processed.head(2)`

Out[202...

	ID	LEAGUE	TEAM_NAME	TEAM_SHOTS_TOTAL_season_sum	TEAM_SHOTS_INSIDE
0	0.00	Ligue 1	Toulouse	3.00	
1	1.00	Ligue 2	Brest	6.00	

In [203... `away_team_processed.head(2)`

Out[203...

	ID	LEAGUE	TEAM_NAME	TEAM_SHOTS_TOTAL_season_sum	TEAM_SHOTS_INSIDE
0	0.00	Ligue 1	Olympique Marseille	4.00	
1	1.00	Ligue 2	Red Star	4.00	

In [204... `home_player_processed.head(2)`

Out[204...

	ID	LEAGUE	TEAM_NAME	PLAYER_NAME	PLAYER_ACCURATE_CROSSES_season_sum
0	0	Ligue 1	Toulouse	Agustín Rogel	0.00
1	0	Ligue 1	Toulouse	Mathieu Goncalves	0.00

In [205... `away_player_processed.head(2)`

Out[205...	ID	LEAGUE	TEAM_NAME	PLAYER_NAME	PLAYER_ACCURATE_CROSSES_season_sum
	0	Ligue 1	Olympique Marseille	Lucas Perrin	0.00
	1	Ligue 1	Olympique Marseille	Kevin Strootman	2.00

```
In [206... home_team_processed["ID"].value_counts().head()
```

```
Out[206... ID
0.00      1
8458.00   1
8196.00   1
8197.00   1
8198.00   1
Name: count, dtype: int64
```

```
In [207... away_team_processed["ID"].value_counts().head()
```

```
Out[207... ID
0.00      1
8458.00   1
8196.00   1
8197.00   1
8198.00   1
Name: count, dtype: int64
```

```
In [208... home_player_processed["ID"].value_counts().head()
```

```
Out[208... ID
5515      25
6569      24
7220      24
11697     24
1983      24
Name: count, dtype: int64
```

```
In [209... away_player_processed["ID"].value_counts().head()
```

```
Out[209... ID
1642      23
5824      23
11489     23
4665      23
11491     23
Name: count, dtype: int64
```

```
In [210... home_player_agg = (
    home_player_processed
    .groupby(["ID", "LEAGUE", "TEAM_NAME"], as_index=False)
    .mean(numeric_only=True)
)

home_player_agg.shape
```

```
Out[210... (12303, 275)
```



```
In [211...] away_player_agg = (  
    away_player_processed  
    .groupby(["ID", "LEAGUE", "TEAM_NAME"], as_index=False)  
    .mean(numeric_only=True)  
    )  
  
away_player_agg.shape
```

```
Out[211...] (12303, 275)
```

```
In [212...] home_player_agg.shape  
away_player_agg.shape
```

```
Out[212...] (12303, 275)
```

```
In [213...] home_team_merge = home_team_processed.copy()  
  
home_team_merge = home_team_merge.rename(  
    columns=lambda x: "HOME_" + x  
    if x not in ["ID", "LEAGUE"]  
    else x  
    )
```

```
In [214...] away_team_merge = away_team_processed.copy()  
  
away_team_merge = away_team_merge.rename(  
    columns=lambda x: "AWAY_" + x  
    if x not in ["ID", "LEAGUE"]  
    else x  
    )
```

```
In [215...] home_player_merge = home_player_agg.copy()  
  
home_player_merge = home_player_merge.rename(  
    columns=lambda x: "HOME_" + x  
    if x not in ["ID", "LEAGUE"]  
    else x  
    )
```

```
In [216...] away_player_merge = away_player_agg.copy()  
  
away_player_merge = away_player_merge.rename(  
    columns=lambda x: "AWAY_" + x  
    if x not in ["ID", "LEAGUE"]  
    else x  
    )
```

```
In [217...] print(home_team_merge.columns[:10])  
  
print(away_team_merge.columns[:10])  
  
print(home_player_merge.columns[:10])  
  
print(away_player_merge.columns[:10])
```

```

Index(['ID', 'LEAGUE', 'HOME_TEAM_NAME', 'HOME_TEAM_SHOTS_TOTAL_season_sum',
      'HOME_TEAM_SHOTS_INSIDEBOX_season_sum',
      'HOME_TEAM_SHOTS_OFF_TARGET_season_sum',
      'HOME_TEAM_SHOTS_ON_TARGET_season_sum',
      'HOME_TEAM_SHOTS_OUTSIDEBOX_season_sum', 'HOME_TEAM_PASSES_season_sum',
      'HOME_TEAM_SUCCESSFUL_PASSES_season_sum'],
      dtype='object')
Index(['ID', 'LEAGUE', 'AWAY_TEAM_NAME', 'AWAY_TEAM_SHOTS_TOTAL_season_sum',
      'AWAY_TEAM_SHOTS_INSIDEBOX_season_sum',
      'AWAY_TEAM_SHOTS_OFF_TARGET_season_sum',
      'AWAY_TEAM_SHOTS_ON_TARGET_season_sum',
      'AWAY_TEAM_SHOTS_OUTSIDEBOX_season_sum', 'AWAY_TEAM_PASSES_season_sum',
      'AWAY_TEAM_SUCCESSFUL_PASSES_season_sum'],
      dtype='object')
Index(['ID', 'LEAGUE', 'HOME_TEAM_NAME',
      'HOME_PLAYER_ACCURATE_CROSSES_season_sum',
      'HOME_PLAYER_ACCURATE_PASSES_season_sum',
      'HOME_PLAYER_AERIALS_WON_season_sum', 'HOME_PLAYER_ASSISTS_season_sum',
      'HOME_PLAYER_BIG_CHANCES_CREATED_season_sum',
      'HOME_PLAYER_BIG_CHANCES_MISSED_season_sum',
      'HOME_PLAYER_BLOCKED_SHOTS_season_sum'],
      dtype='object')
Index(['ID', 'LEAGUE', 'AWAY_TEAM_NAME',
      'AWAY_PLAYER_ACCURATE_CROSSES_season_sum',
      'AWAY_PLAYER_ACCURATE_PASSES_season_sum',
      'AWAY_PLAYER_AERIALS_WON_season_sum', 'AWAY_PLAYER_ASSISTS_season_sum',
      'AWAY_PLAYER_BIG_CHANCES_CREATED_season_sum',
      'AWAY_PLAYER_BIG_CHANCES_MISSED_season_sum',
      'AWAY_PLAYER_BLOCKED_SHOTS_season_sum'],
      dtype='object')

```

Merging Home team and Away Team

```

In [218]: match_data = home_team_merge.merge(
          away_team_merge,
          on=["ID", "LEAGUE"],
          how="inner"
        )

print(match_data.shape)

```

(12303, 284)

Merging Home Player

```

In [219]: match_data = match_data.merge(
          home_player_merge,
          on=["ID", "LEAGUE"],
          how="inner"
        )

print(match_data.shape)

```

(12303, 557)

Merging Away Player

```
In [220... match_data = match_data.merge(
    away_player_merge,
    on=["ID", "LEAGUE"],
    how="inner"
)

print(match_data.shape)
```

(12303, 830)

```
In [221... print(match_data.shape)

match_data.head(2)
```

(12303, 830)

```
Out[221...      ID  LEAGUE  HOME_TEAM_NAME_x  HOME_TEAM_SHOTS_TOTAL_season_sum  HOM
```

	ID	LEAGUE	HOME_TEAM_NAME_x	HOME_TEAM_SHOTS_TOTAL_season_sum	HOM
0	0.00	Ligue 1	Toulouse	3.00	
1	1.00	Ligue 2	Brest	6.00	

```
In [222... match_data.shape
```

```
Out[222... (12303, 830)
```

```
In [223... match_data.columns[-10:]
```

```
Out[223... Index(['AWAY_PLAYER_SAVES_INSIDE_BOX_5_last_match_std',
        'AWAY_PLAYER_SHOTS_BLOCKED_5_last_match_std',
        'AWAY_PLAYER_SHOTS_ON_TARGET_5_last_match_std',
        'AWAY_PLAYER_SHOTS_TOTAL_5_last_match_std',
        'AWAY_PLAYER_STARTING_LINEUP_5_last_match_std',
        'AWAY_PLAYER_SUCCESSFUL_DRIBBLES_5_last_match_std',
        'AWAY_PLAYER_TACKLES_5_last_match_std',
        'AWAY_PLAYER_TOTAL_CROSSES_5_last_match_std',
        'AWAY_PLAYER_TOTAL_DUELS_5_last_match_std',
        'AWAY_PLAYER_YELLOWCARDS_5_last_match_std'],
      dtype='object')
```

```
In [224... [c for c in match_data.columns if c.endswith("_x") or c.endswith("_y")]
```

```
Out[224... ['HOME_TEAM_NAME_x',
        'AWAY_TEAM_NAME_x',
        'HOME_TEAM_NAME_y',
        'AWAY_TEAM_NAME_y']
```

```
In [225... match_data = match_data.rename(columns={
    "HOME_TEAM_NAME_x": "HOME_TEAM_NAME",
    "AWAY_TEAM_NAME_x": "AWAY_TEAM_NAME"
})

match_data = match_data.drop(columns=[
    "HOME_TEAM_NAME_y",
    "AWAY_TEAM_NAME_y"
```

```
])  
  
print(match_data.shape)
```

(12303, 828)

```
In [226... missing = match_data.isnull().sum()  
  
missing = missing[missing > 0]  
  
print("Columns with missing values:", len(missing))  
print("Total missing values:", missing.sum())  
  
missing.head(20)
```

Columns with missing values: 0
Total missing values: 0

Out[226... Series([], dtype: int64)

```
In [227... print("Duplicate rows:", match_data.duplicated().sum())
```

Duplicate rows: 0

```
In [228... match_data.dtypes.value_counts()
```

Out[228... float64 825
object 3
Name: count, dtype: int64

```
In [229... numeric = match_data.select_dtypes(include=np.number)  
  
print(np.isinf(numeric).sum().sum())
```

0

```
In [230... print(y_train.shape)  
print(y_train.columns.tolist())  
print(y_train.head())
```

(12303, 4)
['ID', 'HOME_WINS', 'DRAW', 'AWAY_WINS']
 ID HOME_WINS DRAW AWAY_WINS
0 0 0 0 1
1 1 0 1 0
2 2 0 0 1
3 3 1 0 0
4 4 0 1 0

```
In [231... print(y_train.dtypes)
```

ID int64
HOME_WINS int64
DRAW int64
AWAY_WINS int64
dtype: object

```
In [232... # Merging the target into Match data  
match_data = match_data.merge(  
    y_train,  
    on="ID",  
    how="inner"  
)
```

```
print("Shape after merging target:", match_data.shape)
```

Shape after merging target: (12303, 831)

```
In [233...] print(match_data[["ID", "HOME_WINS", "DRAW", "AWAY_WINS"]].head(10))
```

	ID	HOME_WINS	DRAW	AWAY_WINS
0	0.00	0	0	1
1	1.00	0	1	0
2	2.00	0	0	1
3	3.00	1	0	0
4	4.00	0	1	0
5	5.00	1	0	0
6	6.00	1	0	0
7	7.00	0	1	0
8	8.00	0	0	1
9	9.00	1	0	0

```
In [234...] outcome_sum = (  
    match_data[["HOME_WINS", "DRAW", "AWAY_WINS"]]  
    .sum(axis=1)  
)  
  
print("Rows with exactly one outcome:", (outcome_sum == 1).sum())  
print("Rows with invalid outcome:", (outcome_sum != 1).sum())
```

Rows with exactly one outcome: 12303

Rows with invalid outcome: 0

```
In [235...] print("Home wins:", match_data["HOME_WINS"].sum())  
print("Draws:", match_data["DRAW"].sum())  
print("Away wins:", match_data["AWAY_WINS"].sum())
```

Home wins: 5356

Draws: 3186

Away wins: 3761

```
In [236...] # Creating one single target variable  
# Create a single multiclass target  
  
match_data["TARGET"] = np.select(  
    [  
        match_data["HOME_WINS"] == 1,  
        match_data["DRAW"] == 1,  
        match_data["AWAY_WINS"] == 1  
    ],  
    [  
        0,  
        1,  
        2  
    ]  
)  
  
print(match_data["TARGET"].value_counts().sort_index())
```

TARGET

0 5356

1 3186

2 3761

Name: count, dtype: int64

```
In [237... print(match_data[["ID", "HOME_WINS", "DRAW", "AWAY_WINS", "TARGET"]].head
```

	ID	HOME_WINS	DRAW	AWAY_WINS	TARGET
0	0.00	0	0	1	2
1	1.00	0	1	0	1
2	2.00	0	0	1	2
3	3.00	1	0	0	0
4	4.00	0	1	0	1
5	5.00	1	0	0	0
6	6.00	1	0	0	0
7	7.00	0	1	0	1
8	8.00	0	0	1	2
9	9.00	1	0	0	0

```
In [238... match_data = match_data.drop(
    columns=["HOME_WINS", "DRAW", "AWAY_WINS"]
)

print(match_data.shape)
```

(12303, 829)

```
In [239... # Check all non-numeric columns

object_columns = match_data.select_dtypes(include=["object"]).columns.tolist()

print("Number of object columns:", len(object_columns))
print("\nObject columns:")
print(object_columns)
```

Number of object columns: 3

Object columns:
['LEAGUE', 'HOME_TEAM_NAME', 'AWAY_TEAM_NAME']

```
In [240... print("ID unique values:", match_data["ID"].nunique())
print("Total rows:", len(match_data))

print("\nID range:")
print(match_data["ID"].min(), match_data["ID"].max())
```

ID unique values: 12303

Total rows: 12303

ID range:
0.0 12302.0

```
In [241... # Count features by their statistical aggregation type

feature_types = {
    "season_sum": [],
    "season_average": [],
    "season_std": [],
    "last_5_sum": [],
    "last_5_average": [],
    "last_5_std": []
}

for col in match_data.columns:

    if col in ["ID", "LEAGUE", "HOME_TEAM_NAME", "AWAY_TEAM_NAME", "TARGE
```

```

        continue

    if "season_sum" in col:
        feature_types["season_sum"].append(col)

    elif "season_average" in col:
        feature_types["season_average"].append(col)

    elif "season_std" in col:
        feature_types["season_std"].append(col)

    elif "5_last_match_sum" in col:
        feature_types["last_5_sum"].append(col)

    elif "5_last_match_average" in col:
        feature_types["last_5_average"].append(col)

    elif "5_last_match_std" in col:
        feature_types["last_5_std"].append(col)

for name, columns in feature_types.items():
    print(name, ":", len(columns))

```

```

season_sum : 134
season_average : 142
season_std : 136
last_5_sum : 134
last_5_average : 142
last_5_std : 136

```

In [242... *# Look for columns that could directly reveal the match result*

```

keywords = [
    "WIN",
    "DRAW",
    "RESULT",
    "OUTCOME"
]

suspicious_columns = [
    col for col in match_data.columns
    if any(keyword in col.upper() for keyword in keywords)
]

print("Potentially suspicious columns:")
print(suspicious_columns)

```

Potentially suspicious columns:

```
['HOME_TEAM_GAME_DRAW_season_sum', 'HOME_TEAM_GAME_DRAW_season_average',  
'HOME_TEAM_GAME_DRAW_5_last_match_sum', 'HOME_TEAM_GAME_DRAW_5_last_match_  
average', 'AWAY_TEAM_GAME_DRAW_season_sum', 'AWAY_TEAM_GAME_DRAW_season_av  
erage', 'AWAY_TEAM_GAME_DRAW_5_last_match_sum', 'AWAY_TEAM_GAME_DRAW_5_las  
t_match_average', 'HOME_PLAYER_FOULS_DRAWN_season_sum', 'HOME_PLAYER_FOULS  
_DRAWN_season_average', 'HOME_PLAYER_FOULS_DRAWN_season_std', 'HOME_PLAYER  
_FOULS_DRAWN_5_last_match_sum', 'HOME_PLAYER_FOULS_DRAWN_5_last_match_aver  
age', 'HOME_PLAYER_FOULS_DRAWN_5_last_match_std', 'AWAY_PLAYER_FOULS_DRAWN  
_season_sum', 'AWAY_PLAYER_FOULS_DRAWN_season_average', 'AWAY_PLAYER_FOULS  
_DRAWN_season_std', 'AWAY_PLAYER_FOULS_DRAWN_5_last_match_sum', 'AWAY_PLAY  
ER_FOULS_DRAWN_5_last_match_average', 'AWAY_PLAYER_FOULS_DRAWN_5_last_matc  
h_std']
```

```
In [243... X = match_data.drop(columns=["TARGET"])  
y = match_data["TARGET"]  
  
print("X shape:", X.shape)  
print("y shape:", y.shape)
```

X shape: (12303, 828)

y shape: (12303,)

```
In [244... print(  
    match_data[["ID", "TARGET"]]  
    .sort_values("ID")  
    .head(20)  
)
```

	ID	TARGET
0	0.00	2
1	1.00	1
2	2.00	2
3	3.00	0
4	4.00	1
5	5.00	0
6	6.00	0
7	7.00	1
8	8.00	2
9	9.00	0
10	10.00	1
11	11.00	0
12	12.00	2
13	13.00	2
14	14.00	0
15	15.00	1
16	16.00	0
17	17.00	0
18	18.00	0
19	19.00	0

```
In [245... print(  
    match_data[["ID", "TARGET"]]  
    .sort_values("ID")  
    .tail(20)  
)
```


	ID	TARGET
12283	12283.00	0
12284	12284.00	0
12285	12285.00	1
12286	12286.00	1
12287	12287.00	0
12288	12288.00	2
12289	12289.00	1
12290	12290.00	0
12291	12291.00	1
12292	12292.00	0
12293	12293.00	1
12294	12294.00	0
12295	12295.00	1
12296	12296.00	0
12297	12297.00	0
12298	12298.00	2
12299	12299.00	2
12300	12300.00	2
12301	12301.00	0
12302	12302.00	0

```
In [246... id_sorted = match_data["ID"].sort_values().reset_index(drop=True)

print("IDs are sequential:",
      np.array_equal(id_sorted.values, np.arange(len(match_data))))
```

IDs are sequential: True

```
In [247... match_data["ID_GROUP"] = pd.qcut(
    match_data["ID"],
    q=5,
    labels=["Q1", "Q2", "Q3", "Q4", "Q5"]
)

print(
    pd.crosstab(
        match_data["ID_GROUP"],
        match_data["TARGET"],
        normalize="index"
    ).round(3)
)
```

TARGET	0	1	2
ID_GROUP			
Q1	0.45	0.26	0.29
Q2	0.42	0.25	0.34
Q3	0.43	0.27	0.30
Q4	0.44	0.25	0.30
Q5	0.44	0.26	0.30

```
In [248... print("Number of leagues:",
      match_data["LEAGUE"].nunique())

print("\nLeague distribution:")
print(match_data["LEAGUE"].value_counts())
```

Number of leagues: 13

League distribution:

LEAGUE

Serie A	1966
League Two	1376
Superliga	1124
La Liga	973
Ligue 2	884
Ligue 1	851
League One	844
J-League	783
Bundesliga	777
Liga Portugal	772
Eredivisie	674
Pro League	649
Premier League	630

Name: count, dtype: int64

```
In [249... print(  
    pd.crosstab(  
        match_data["LEAGUE"],  
        match_data["TARGET"],  
        normalize="index"  
    ).round(3)  
)
```

TARGET	0	1	2
LEAGUE			
Bundesliga	0.42	0.24	0.34
Eredivisie	0.50	0.21	0.29
J-League	0.42	0.22	0.36
La Liga	0.45	0.28	0.27
League One	0.44	0.27	0.29
League Two	0.43	0.27	0.30
Liga Portugal	0.44	0.23	0.33
Ligue 1	0.42	0.27	0.32
Ligue 2	0.40	0.31	0.28
Premier League	0.43	0.23	0.34
Pro League	0.44	0.23	0.33
Serie A	0.45	0.27	0.28
Superliga	0.42	0.27	0.31

```
In [250... print("Unique home teams:",  
    match_data["HOME_TEAM_NAME"].nunique())  
  
print("Unique away teams:",  
    match_data["AWAY_TEAM_NAME"].nunique())  
  
print("Unique teams overall:",  
    pd.concat([  
        match_data["HOME_TEAM_NAME"],  
        match_data["AWAY_TEAM_NAME"]  
    ]).nunique())
```

Unique home teams: 350
Unique away teams: 350
Unique teams overall: 350

```
In [251... match_data = match_data.drop(columns=["ID_GROUP"])
```

```
print(match_data.shape)
```

(12303, 829)

```
In [252]: # Final modelling-readiness check

print("Dataset shape:", match_data.shape)

print("\nTarget distribution:")
print(match_data["TARGET"].value_counts().sort_index())

print("\nTarget percentages:")
print((match_data["TARGET"].value_counts(normalize=True).sort_index() * 100).round(2))

print("\nObject/categorical columns:")
print(match_data.select_dtypes(include=["object"]).columns.tolist())

print("\nID column:")
print(match_data["ID"].head())
```

Dataset shape: (12303, 829)

Target distribution:

TARGET

0 5356

1 3186

2 3761

Name: count, dtype: int64

Target percentages:

TARGET

0 43.53

1 25.90

2 30.57

Name: proportion, dtype: float64

Object/categorical columns:

['LEAGUE', 'HOME_TEAM_NAME', 'AWAY_TEAM_NAME']

ID column:

0 0.00

1 1.00

2 2.00

3 3.00

4 4.00

Name: ID, dtype: float64

```
In [253]: # Prepare data for modelling

# Remove identifier and raw team-name columns
X = match_data.drop(
    columns=["TARGET", "ID", "HOME_TEAM_NAME", "AWAY_TEAM_NAME"]
)

y = match_data["TARGET"]

print("X shape:", X.shape)
print("y shape:", y.shape)

print("\nRemaining non-numeric columns:")
print(X.select_dtypes(include=["object"]).columns.tolist())
```

```
print("\nMissing values in X:", X.isna().sum().sum())
print("Infinite values in X:", np.isinf(X.select_dtypes(include=np.number
```

X shape: (12303, 825)

y shape: (12303,)

Remaining non-numeric columns:

['LEAGUE']

Missing values in X: 0

Infinite values in X: 0

```
In [254... from sklearn.model_selection import train_test_split

# Stratified 80/20 train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    stratify=y,
    random_state=42
)

print("X_train shape:", X_train.shape)
print("X_test shape:", X_test.shape)
print("y_train shape:", y_train.shape)
print("y_test shape:", y_test.shape)

print("\nTraining target distribution:")
print(y_train.value_counts(normalize=True).sort_index().round(3))

print("\nTest target distribution:")
print(y_test.value_counts(normalize=True).sort_index().round(3))
```

X_train shape: (9842, 825)

X_test shape: (2461, 825)

y_train shape: (9842,)

y_test shape: (2461,)

Training target distribution:

TARGET

0 0.43

1 0.26

2 0.31

Name: proportion, dtype: float64

Test target distribution:

TARGET

0 0.44

1 0.26

2 0.31

Name: proportion, dtype: float64

```
In [255... from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder

# Identify categorical and numerical columns
categorical_features = ["LEAGUE"]

numerical_features = [
```

```

    col for col in X_train.columns
    if col not in categorical_features
]

# Preprocessing
preprocessor = ColumnTransformer(
    transformers=[
        (
            "cat",
            OneHotEncoder(handle_unknown="ignore"),
            categorical_features
        ),
        (
            "num",
            "passthrough",
            numerical_features
        )
    ]
)

print("Categorical features:", categorical_features)
print("Number of numerical features:", len(numerical_features))

```

Categorical features: ['LEAGUE']
 Number of numerical features: 824

Logistic Regression

In [256...

```

from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.linear_model import LogisticRegression

# Preprocessing
preprocessor_lr = ColumnTransformer(
    transformers=[
        (
            "cat",
            OneHotEncoder(handle_unknown="ignore"),
            categorical_features
        ),
        (
            "num",
            StandardScaler(),
            numerical_features
        )
    ]
)

# Multinomial Logistic Regression
logistic_model = Pipeline(
    steps=[
        ("preprocessor", preprocessor_lr),
        (
            "classifier",
            LogisticRegression(
                max_iter=2000,
                solver="lbfgs",
                multi_class="multinomial",
            )
        )
    ]
)

```

```

        random_state=42
    )
)
]
)

# Fit only on the training data
logistic_model.fit(X_train, y_train)

print("Multinomial Logistic Regression fitted successfully.")

```

Multinomial Logistic Regression fitted successfully.

```

In [257... y_pred_lr = logistic_model.predict(X_test)
y_prob_lr = logistic_model.predict_proba(X_test)

print("Predictions generated successfully.")
print("Probability shape:", y_prob_lr.shape)

```

Predictions generated successfully.

Probability shape: (2461, 3)

```

In [258... from sklearn.metrics import (
    accuracy_score,
    balanced_accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    log_loss,
    roc_auc_score,
    confusion_matrix,
    classification_report
)

# Classification metrics
accuracy_lr = accuracy_score(y_test, y_pred_lr)
balanced_accuracy_lr = balanced_accuracy_score(y_test, y_pred_lr)

precision_lr = precision_score(
    y_test, y_pred_lr,
    average="macro",
    zero_division=0
)

recall_lr = recall_score(
    y_test, y_pred_lr,
    average="macro",
    zero_division=0
)

f1_lr = f1_score(
    y_test, y_pred_lr,
    average="macro",
    zero_division=0
)

# Probability-based metrics
log_loss_lr = log_loss(y_test, y_prob_lr)

roc_auc_lr = roc_auc_score(
    y_test,

```

```

    y_prob_lr,
    multi_class="ovr",
    average="macro"
)

# Print results
print("LOGISTIC REGRESSION - TEST SET RESULTS")
print("-" * 50)

print(f"Accuracy:           {accuracy_lr:.4f}")
print(f"Balanced Accuracy: {balanced_accuracy_lr:.4f}")
print(f"Macro Precision:     {precision_lr:.4f}")
print(f"Macro Recall:        {recall_lr:.4f}")
print(f"Macro F1-score:       {f1_lr:.4f}")
print(f"ROC-AUC (Macro):      {roc_auc_lr:.4f}")
print(f"Log Loss:             {log_loss_lr:.4f}")

print("\nClassification Report:")
print(
    classification_report(
        y_test,
        y_pred_lr,
        target_names=["Home Win (0)", "Draw (1)", "Away Win (2)"],
        digits=4,
        zero_division=0
    )
)

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred_lr))

```

LOGISTIC REGRESSION - TEST SET RESULTS

```

-----
Accuracy:           0.4486
Balanced Accuracy:  0.4061
Macro Precision:    0.4009
Macro Recall:       0.4061
Macro F1-score:     0.3975
ROC-AUC (Macro):    0.5871
Log Loss:           1.1228

```

Classification Report:

	precision	recall	f1-score	support
Home Win (0)	0.5130	0.6437	0.5710	1072
Draw (1)	0.2500	0.1586	0.1940	637
Away Win (2)	0.4396	0.4162	0.4276	752
accuracy			0.4486	2461
macro avg	0.4009	0.4061	0.3975	2461
weighted avg	0.4225	0.4486	0.4296	2461

Confusion Matrix:

```

[[690 170 212]
 [349 101 187]
 [306 133 313]]

```

Decision Trees

In [259... *# Decision Tree*

```
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
from sklearn.tree import DecisionTreeClassifier

# Preprocessing
preprocessor_dt = ColumnTransformer(
    transformers=[
        (
            "cat",
            OneHotEncoder(handle_unknown="ignore"),
            categorical_features
        ),
        (
            "num",
            "passthrough",
            numerical_features
        )
    ]
)

# Decision Tree model
decision_tree_model = Pipeline(
    steps=[
        ("preprocessor", preprocessor_dt),
        (
            "classifier",
            DecisionTreeClassifier(
                random_state=42
            )
        )
    ]
)

# Fit ONLY on training data
decision_tree_model.fit(X_train, y_train)

print("Decision Tree fitted successfully.")
```

Decision Tree fitted successfully.

In [260... *# Generate predictions on the unseen test set*

```
y_pred_dt = decision_tree_model.predict(X_test)
y_prob_dt = decision_tree_model.predict_proba(X_test)

print("Decision Tree predictions generated successfully.")
print("Probability shape:", y_prob_dt.shape)
```

Decision Tree predictions generated successfully.
Probability shape: (2461, 3)

In [261... **from** sklearn.metrics **import** (
 accuracy_score,
 balanced_accuracy_score,
 precision_score,
 recall_score,
 f1_score,


```

        log_loss,
        roc_auc_score,
        confusion_matrix,
        classification_report
    )

    # Classification metrics
    accuracy_dt = accuracy_score(y_test, y_pred_dt)

    balanced_accuracy_dt = balanced_accuracy_score(
        y_test,
        y_pred_dt
    )

    precision_dt = precision_score(
        y_test,
        y_pred_dt,
        average="macro",
        zero_division=0
    )

    recall_dt = recall_score(
        y_test,
        y_pred_dt,
        average="macro",
        zero_division=0
    )

    f1_dt = f1_score(
        y_test,
        y_pred_dt,
        average="macro",
        zero_division=0
    )

    # Probability-based metrics
    log_loss_dt = log_loss(
        y_test,
        y_prob_dt
    )

    roc_auc_dt = roc_auc_score(
        y_test,
        y_prob_dt,
        multi_class="ovr",
        average="macro"
    )

    # Print results
    print("DECISION TREE – TEST SET RESULTS")
    print("-" * 50)

    print(f"Accuracy:                {accuracy_dt:.4f}")
    print(f"Balanced Accuracy:          {balanced_accuracy_dt:.4f}")
    print(f"Macro Precision:            {precision_dt:.4f}")
    print(f"Macro Recall:               {recall_dt:.4f}")
    print(f"Macro F1-score:             {f1_dt:.4f}")
    print(f"ROC-AUC (Macro):            {roc_auc_dt:.4f}")
    print(f"Log Loss:                   {log_loss_dt:.4f}")

```

```

print("\nClassification Report:")
print(
    classification_report(
        y_test,
        y_pred_dt,
        target_names=[
            "Home Win (0)",
            "Draw (1)",
            "Away Win (2)"
        ],
        digits=4,
        zero_division=0
    )
)

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred_dt))

```

DECISION TREE – TEST SET RESULTS

```

-----
Accuracy:          0.3807
Balanced Accuracy: 0.3650
Macro Precision:   0.3649
Macro Recall:      0.3650
Macro F1-score:    0.3649
ROC-AUC (Macro):   0.5246
Log Loss:          22.3204

```

Classification Report:

	precision	recall	f1-score	support
Home Win (0)	0.4743	0.4571	0.4656	1072
Draw (1)	0.2772	0.2841	0.2806	637
Away Win (2)	0.3432	0.3537	0.3484	752
accuracy			0.3807	2461
macro avg	0.3649	0.3650	0.3649	2461
weighted avg	0.3832	0.3807	0.3819	2461

Confusion Matrix:

```

[[490 268 314]
 [261 181 195]
 [282 204 266]]

```

Random Forest

```

In [262]: from sklearn.ensemble import RandomForestClassifier
          from sklearn.pipeline import Pipeline
          from sklearn.compose import ColumnTransformer
          from sklearn.preprocessing import OneHotEncoder

          # Preprocessing for Random Forest
          preprocessor_rf = ColumnTransformer(
              transformers=[
                  (
                      "cat",
                      OneHotEncoder(handle_unknown="ignore"),
                      categorical_features

```

```

        ),
        (
            "num",
            "passthrough",
            numerical_features
        )
    ]
)

# Random Forest model
random_forest_model = Pipeline(
    steps=[
        ("preprocessor", preprocessor_rf),
        (
            "classifier",
            RandomForestClassifier(
                n_estimators=300,
                random_state=42,
                n_jobs=-1
            )
        )
    ]
)

# Fit ONLY on training data
random_forest_model.fit(X_train, y_train)

print("Random Forest fitted successfully.")

```

Random Forest fitted successfully.

In [263...

```

# Generate predictions on the unseen test set
y_pred_rf = random_forest_model.predict(X_test)
y_prob_rf = random_forest_model.predict_proba(X_test)

print("Random Forest predictions generated successfully.")
print("Probability shape:", y_prob_rf.shape)

```

Random Forest predictions generated successfully.
Probability shape: (2461, 3)

In [264...

```

from sklearn.metrics import (
    accuracy_score,
    balanced_accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    log_loss,
    roc_auc_score,
    confusion_matrix,
    classification_report
)

# Classification metrics
accuracy_rf = accuracy_score(y_test, y_pred_rf)

balanced_accuracy_rf = balanced_accuracy_score(
    y_test,
    y_pred_rf
)

```

```

precision_rf = precision_score(
    y_test,
    y_pred_rf,
    average="macro",
    zero_division=0
)

recall_rf = recall_score(
    y_test,
    y_pred_rf,
    average="macro",
    zero_division=0
)

f1_rf = f1_score(
    y_test,
    y_pred_rf,
    average="macro",
    zero_division=0
)

# Probability-based metrics
log_loss_rf = log_loss(
    y_test,
    y_prob_rf
)

roc_auc_rf = roc_auc_score(
    y_test,
    y_prob_rf,
    multi_class="ovr",
    average="macro"
)

# Print results
print("RANDOM FOREST – TEST SET RESULTS")
print("-" * 50)

print(f"Accuracy: {accuracy_rf:.4f}")
print(f"Balanced Accuracy: {balanced_accuracy_rf:.4f}")
print(f"Macro Precision: {precision_rf:.4f}")
print(f"Macro Recall: {recall_rf:.4f}")
print(f"Macro F1-score: {f1_rf:.4f}")
print(f"ROC-AUC (Macro): {roc_auc_rf:.4f}")
print(f"Log Loss: {log_loss_rf:.4f}")

print("\nClassification Report:")
print(
    classification_report(
        y_test,
        y_pred_rf,
        target_names=[
            "Home Win (0)",
            "Draw (1)",
            "Away Win (2)"
        ],
        digits=4,
        zero_division=0
    )
)

```

```
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred_rf))
```

RANDOM FOREST – TEST SET RESULTS

```
-----
Accuracy:          0.4892
Balanced Accuracy: 0.4195
Macro Precision:   0.4429
Macro Recall:      0.4195
Macro F1-score:    0.3732
ROC-AUC (Macro):   0.6219
Log Loss:          1.0269
```

Classification Report:

	precision	recall	f1-score	support
Home Win (0)	0.5029	0.8172	0.6226	1072
Draw (1)	0.3607	0.0345	0.0630	637
Away Win (2)	0.4650	0.4069	0.4340	752
accuracy			0.4892	2461
macro avg	0.4429	0.4195	0.3732	2461
weighted avg	0.4545	0.4892	0.4201	2461

Confusion Matrix:

```
[[876  16 180]
 [443  22 172]
 [423  23 306]]
```

Gradient Boosting with XGBoost

In [265... `%pip install xgboost`

```
Requirement already satisfied: xgboost in /opt/anaconda3/lib/python3.13/site-packages (3.4.0)
Requirement already satisfied: numpy in /opt/anaconda3/lib/python3.13/site-packages (from xgboost) (2.1.3)
Requirement already satisfied: scipy in /opt/anaconda3/lib/python3.13/site-packages (from xgboost) (1.15.3)
Note: you may need to restart the kernel to use updated packages.
```

In [266... `from xgboost import XGBClassifier`

```
print("XGBoost imported successfully.")
```

XGBoost imported successfully.

In [267... `# Gradient Boosting with XGBoost`

```
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder

# Preprocessing
preprocessor_xgb = ColumnTransformer(
    transformers=[
        (
            "cat",
```

```

        OneHotEncoder(handle_unknown="ignore"),
        categorical_features
    ),
    (
        "num",
        "passthrough",
        numerical_features
    )
]
)

# XGBoost model
xgb_model = Pipeline(
    steps=[
        ("preprocessor", preprocessor_xgb),
        (
            "classifier",
            XGBClassifier(
                n_estimators=300,
                max_depth=6,
                learning_rate=0.05,
                subsample=0.8,
                colsample_bytree=0.8,
                objective="multi:softprob",
                num_class=3,
                eval_metric="mlogloss",
                random_state=42,
                n_jobs=-1
            )
        )
    ]
)

# Fit ONLY on training data
xgb_model.fit(X_train, y_train)

print("XGBoost fitted successfully.")

```

XGBoost fitted successfully.

```

In [268... # Generate predictions on the unseen test set

y_pred_xgb = xgb_model.predict(X_test)
y_prob_xgb = xgb_model.predict_proba(X_test)

print("XGBoost predictions generated successfully.")
print("Probability shape:", y_prob_xgb.shape)

```

XGBoost predictions generated successfully.
Probability shape: (2461, 3)

```

In [269... # XGBoost evaluation

accuracy_xgb = accuracy_score(y_test, y_pred_xgb)

balanced_accuracy_xgb = balanced_accuracy_score(
    y_test,
    y_pred_xgb
)

precision_xgb = precision_score(

```

```

        y_test,
        y_pred_xgb,
        average="macro",
        zero_division=0
    )

recall_xgb = recall_score(
    y_test,
    y_pred_xgb,
    average="macro",
    zero_division=0
)

f1_xgb = f1_score(
    y_test,
    y_pred_xgb,
    average="macro",
    zero_division=0
)

log_loss_xgb = log_loss(
    y_test,
    y_prob_xgb
)

roc_auc_xgb = roc_auc_score(
    y_test,
    y_prob_xgb,
    multi_class="ovr",
    average="macro"
)

print("XGB00ST – TEST SET RESULTS")
print("-" * 50)

print(f"Accuracy:                {accuracy_xgb:.4f}")
print(f"Balanced Accuracy:        {balanced_accuracy_xgb:.4f}")
print(f"Macro Precision:           {precision_xgb:.4f}")
print(f"Macro Recall:              {recall_xgb:.4f}")
print(f"Macro F1-score:            {f1_xgb:.4f}")
print(f"ROC-AUC (Macro):           {roc_auc_xgb:.4f}")
print(f"Log Loss:                   {log_loss_xgb:.4f}")

print("\nClassification Report:")
print(
    classification_report(
        y_test,
        y_pred_xgb,
        target_names=[
            "Home Win (0)",
            "Draw (1)",
            "Away Win (2)"
        ],
        digits=4,
        zero_division=0
    )
)

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred_xgb))

```

XGB00ST – TEST SET RESULTS

```
-----
Accuracy:          0.4835
Balanced Accuracy: 0.4254
Macro Precision:   0.4269
Macro Recall:      0.4254
Macro F1-score:    0.3973
ROC-AUC (Macro):   0.6221
Log Loss:          1.0421
```

Classification Report:

	precision	recall	f1-score	support
Home Win (0)	0.5201	0.7500	0.6142	1072
Draw (1)	0.3140	0.0848	0.1335	637
Away Win (2)	0.4468	0.4415	0.4441	752
accuracy			0.4835	2461
macro avg	0.4269	0.4254	0.3973	2461
weighted avg	0.4443	0.4835	0.4378	2461

Confusion Matrix:

```
[[804  61 207]
 [379  54 204]
 [363  57 332]]
```

Neural Netowrk

```
In [270]... # Multilayer Perceptron (MLP)

from sklearn.neural_network import MLPClassifier
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder

# Preprocessing
preprocessor_mlp = ColumnTransformer(
    transformers=[
        (
            "cat",
            OneHotEncoder(handle_unknown="ignore"),
            categorical_features
        ),
        (
            "num",
            StandardScaler(),
            numerical_features
        )
    ]
)

# MLP model
mlp_model = Pipeline(
    steps=[
        ("preprocessor", preprocessor_mlp),
        (
            "classifier",
            MLPClassifier(
```



```

        hidden_layer_sizes=(128, 64),
        activation="relu",
        solver="adam",
        alpha=0.0001,
        learning_rate_init=0.001,
        max_iter=100,
        early_stopping=True,
        validation_fraction=0.1,
        random_state=42
    )
)
]
)

# Fit ONLY on training data
mlp_model.fit(X_train, y_train)

print("MLP fitted successfully.")

```

MLP fitted successfully.

```

In [271]... # Generate predictions on the unseen test set

y_pred_mlp = mlp_model.predict(X_test)
y_prob_mlp = mlp_model.predict_proba(X_test)

print("MLP predictions generated successfully.")
print("Probability shape:", y_prob_mlp.shape)

```

MLP predictions generated successfully.

Probability shape: (2461, 3)

```

In [272]... # MLP evaluation

accuracy_mlp = accuracy_score(y_test, y_pred_mlp)

balanced_accuracy_mlp = balanced_accuracy_score(
    y_test,
    y_pred_mlp
)

precision_mlp = precision_score(
    y_test,
    y_pred_mlp,
    average="macro",
    zero_division=0
)

recall_mlp = recall_score(
    y_test,
    y_pred_mlp,
    average="macro",
    zero_division=0
)

f1_mlp = f1_score(
    y_test,
    y_pred_mlp,
    average="macro",
    zero_division=0
)

```

```

log_loss_mlp = log_loss(
    y_test,
    y_prob_mlp
)

roc_auc_mlp = roc_auc_score(
    y_test,
    y_prob_mlp,
    multi_class="ovr",
    average="macro"
)

print("MLP - TEST SET RESULTS")
print("-" * 50)

print(f"Accuracy:           {accuracy_mlp:.4f}")
print(f"Balanced Accuracy:   {balanced_accuracy_mlp:.4f}")
print(f"Macro Precision:      {precision_mlp:.4f}")
print(f"Macro Recall:         {recall_mlp:.4f}")
print(f"Macro F1-score:       {f1_mlp:.4f}")
print(f"ROC-AUC (Macro):      {roc_auc_mlp:.4f}")
print(f"Log Loss:             {log_loss_mlp:.4f}")

print("\nClassification Report:")
print(
    classification_report(
        y_test,
        y_pred_mlp,
        target_names=[
            "Home Win (0)",
            "Draw (1)",
            "Away Win (2)"
        ],
        digits=4,
        zero_division=0
    )
)

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred_mlp))

```

MLP - TEST SET RESULTS

Accuracy: 0.4681
Balanced Accuracy: 0.4149
Macro Precision: 0.4227
Macro Recall: 0.4149
Macro F1-score: 0.4028
ROC-AUC (Macro): 0.6054
Log Loss: 1.0613

Classification Report:

	precision	recall	f1-score	support
Home Win (0)	0.5104	0.7351	0.6024	1072
Draw (1)	0.3020	0.1664	0.2146	637
Away Win (2)	0.4558	0.3431	0.3915	752
accuracy			0.4681	2461
macro avg	0.4227	0.4149	0.4028	2461
weighted avg	0.4398	0.4681	0.4376	2461

Confusion Matrix:

```
[[788 134 150]
 [373 106 158]
 [383 111 258]]
```